Green Technologies and Sustainability 3 (2025) 100202

Contents lists available at ScienceDirect

Green Technologies and Sustainability

journal homepage: https://www.keaipublishing.com/en/journals/green-technologies-and-sustainability/

Review Article

Digital technologies for the Sustainable Development Goals

Dharmendra Hariyani [a,*], Poonam Hariyani [b], Sanjeev Mishra [c]

[a] Department of Mechanical Engineering, Swami Keshvanand Institute of Technology, Management and Gramothan, Jaipur, Rajasthan, India
[b] Department of Chemistry, JECRC University, Jaipur, Rajasthan, India
[c] Department of Mechanical Engineering, University Department of Teaching, Rajasthan Technical University, Kota, Rajasthan, India

ARTICLE INFO

Keywords:
Sustainable Development Goals (SDGs)
Digital technologies (DTs)
Challenges and barriers
Governments
Organizations
Community
And individuals-driven SDGs strategies

ABSTRACT

Digital technologies (DTs) play a crucial role in advancing the United Nations' Sustainable Development Goals (SDGs). It provides innovative solutions to global challenges like poverty alleviation, equitable healthcare, climate action, and responsible consumption. This study is motivated by the growing significance of DTs in addressing sustainability challenges and their critical contribution to achieving the SDGs while identifying barriers to their adoption. Using a systematic review, this review examines the contributions of various DTs, i.e., AI, IoT, blockchain, cloud computing, big data analytics, remote sensing, and GIS in addressing SDGs. The review methodology includes the keywords search in the title, abstract, and keywords "Digital Technologies" and "Sustainable Development" on the Scopus database. A total of 473 peer-reviewed articles are included in the study. The findings highlight the transformative role of DTs in enhancing resource optimization, decision-making, and service delivery and achieving various SDGs. AI facilitates automation and predictive analytics, supporting healthcare (SDG 3) and climate action (SDG 13). Blockchain enhances transparency, security, and ethical governance (SDG 16). IoT enables real-time monitoring for smart cities (SDG 11) and energy efficiency (SDG 7). Big data analytics strengthens supply chains, agricultural productivity (SDG 2), and disaster management (SDG 13). Cloud computing fosters collaboration and scalability (SDG 9, SDG 17), while 5G networks improve connectivity and infrastructure development. Remote sensing and GIS provide essential insights for environmental monitoring (SDG 14, SDG 15) and disaster response.

Despite these benefits, several challenges hinder DT adoption, including the digital divide, high costs, data privacy concerns, and skill shortages. To overcome these barriers, the study emphasizes policy measures such as (i) increased government investments in digital infrastructure, (ii) strategic public–private partnerships, and (iii) capacity-building initiatives for organizations, communities, and individuals. Ensuring equitable access, ethical digital practices, and environmental sustainability is crucial to fully leveraging DTs for SDGs achievement. This comprehensive analysis provides valuable insights for policymakers, industry leaders, businesses, organizations, researchers, communities, and individuals for leveraging and developing inclusive sustainable digital transformation strategies that accelerate progress toward the SDGs.

1. Introduction

The SDGs, established by the United Nations in 2015, serve as a global blueprint to address pressing challenges and achieve a sustainable future by 2030 [1]. Comprising 17 interconnected goals, the SDGs aim to eradicate poverty, reduce inequality, and ensure environmental sustainability while fostering social well-being and economic growth [2]. These goals emphasize the need for inclusive development, promoting access to education, healthcare, clean energy, and decent work opportunities for all [3]. They also highlight the importance of combating climate change, preserving biodiversity, and ensuring sustainable management of natural resources [4,5]. The SDGs are significant because they provide a universal framework that aligns the efforts of governments, businesses, civil society, and individuals toward common objectives [6,7]. They recognize the interdependence of social, economic, and environmental dimensions and advocate for holistic and integrated approaches to sustainable development [8,9]. By fostering partnerships and mobilizing resources, the SDGs encourage innovative solutions to address complex global challenges [10–13]. Moreover, they provide measurable targets and indicators and enable the stakeholders to track progress and identify areas for improvement [11,13]. As a call to action for a sustainable and equitable world, the SDGs inspire collaboration across nations and emphasize that achieving these goals is essential for securing the planet's future and ensuring the well-being of current and future generations.

* Corresponding author.
E-mail address: dhariyani12002@gmail.com (D. Hariyani).

https://doi.org/10.1016/j.grets.2025.100202
Received 23 December 2024; Received in revised form 20 March 2025; Accepted 23 March 2025
Available online 26 March 2025

Fig. 1. United Nations 17 Sustainable Development Goals (SDGs).

Fig. 1 shows United Nations 17 Sustainable Development Goals (SDGs).

Digital technologies (DTs) play a pivotal role in driving global development and transforming economies, societies, and governance [14]. Innovations such as big data analytics, AI, blockchain, and the IoT enhance operational efficiency, foster unprecedented connectivity, and enable scalable solutions across various sectors [15–17]. These technologies (i) improve decision-making by providing real-time data and insights, (ii) enhance resource management [18], and (iii) improve service delivery in sectors like healthcare, education, and agriculture [19]. Digital platforms bridge gaps in access to information and services, empower marginalized communities, and promote inclusivity [20,21]. For instance, mobile technologies support financial inclusion by extending banking services to remote areas [22–24], while AI-driven solutions enhance disaster prediction and response [25,26]. Moreover, digital tools drive sustainability by optimizing energy use, monitoring environmental changes [27], and fostering circular economies [28–30].

As digital technology adoption continues to accelerate, collaboration among governments, businesses, and civil society becomes crucial to fully harness their potential while addressing challenges such as the digital divide, data security, and ethical concerns [31].

1.1. Literature gap and motivation for the study

Despite the growing research on DTs' contribution to specific SDGs, a comprehensive understanding of their integrated role remains limited. Most existing studies focus on individual technologies or examine a single SDG, leaving a gap in understanding how multiple DTs collectively drive sustainable development. Moreover, challenges unique to developing countries, such as limited digital infrastructure, the digital divide, and the absence of ethical frameworks, remain underexplored.

The study is motivated by the need to bridge these gaps by: (i) providing a holistic review of DTs' contributions across multiple SDGs rather than focusing on isolated technologies, (ii) identifying challenges and barriers hindering DT adoption in sustainable development efforts, and (iii) exploring policy strategies to ensure equitable access, ethical governance, and effective digital transformation.

The study follows the PRISMA guidelines [32] and utilizes the Scopus database to identify relevant studies published between May 17th and November 25th, 2024. A systematic search using "Digital Technologies" and "Sustainable Development" in titles, abstracts, and keywords is conducted. Inclusion criteria focus on peer-reviewed articles, conference proceedings, and reports examining DTs' role in SDGs, while studies not directly addressing specified technologies or SDGs-related outcomes are excluded. A total of 473 peer-reviewed articles are analyzed.

1.2. Contribution of the study

The study contributes to the literature by synthesizing evidence on the multifaceted role of DTs in advancing the SDGs, identifying key barriers such as high costs, regulatory challenges, and digital literacy gaps, and providing policy recommendations for fostering inclusive and sustainable digital transformation. By adopting a multidisciplinary approach, it offers a comprehensive perspective on the impacts, challenges, and opportunities at the intersection of DTs and sustainable development. Additionally, the study informs policymakers, industry leaders, businesses, organizations, researchers, communities, and individuals on strategies for effectively leveraging DTs to accelerate progress toward SDGs.

1.3. Policy implications

The findings from the study offer essential policy implications to optimize the benefits of DTs for sustainable development. Governments must invest in digital infrastructure, including broadband expansion, AI capabilities, and cloud-based services, to bridge the digital divide. Clear regulatory and ethical frameworks on data privacy, cybersecurity, and AI governance are crucial for fostering trust and accountability. Public–private partnerships should be strengthened to accelerate DTs' deployment for achieving SDGs. Additionally, capacity building through enhanced education and training programs is vital to ensure effective adoption of DT-driven solutions. Finally, promoting sustainable technology practices, such as energy-efficient digital solutions and responsible e-waste management, is necessary to minimize the environmental footprint of digital transformation.

The novelty of the study lies in its comprehensive and multidisciplinary approach to analyzing the role of DTs in advancing the SDGs. Unlike previous studies that focus on individual technologies or specific SDGs, this research provides a holistic perspective on how multiple DTs including AI, blockchain, IoT, big data analytics, cloud computing, 5G networks, remote sensing, and GIS interact and complement each other in driving sustainability across various sectors. Moreover, the study moves beyond discussing the benefits of DTs by identifying key adoption barriers such as the digital divide, high costs, regulatory constraints, and ethical concerns, which are often overlooked in fragmented studies. The results highlight that while DTs have transformative potential, their full integration into sustainable development requires targeted policy interventions, including investments in digital infrastructure, capacity-building initiatives, and regulatory frameworks that ensure data security and equitable access. By bridging the gap between technological advancements and policy needs, the study provides actionable insights for governments, businesses, and civil society to develop inclusive and effective digital transformation strategies that accelerate progress toward achieving the SDGs.

Despite its contributions, this study has certain limitations. It primarily relies on published literature, which may not fully capture the latest industry advancements in DTs. While offering a broad analysis of DTs' contribution to sustainable development, it does not account for regional variations in adoption and impact, necessitating further empirical validation. Additionally, the study lacks quantitative assessments of DTs' effectiveness, which future research could address through empirical modeling to provide more data-driven insights into their sustainability outcomes.

2. Literature review

Theoretical foundations and frameworks provide essential insights into the interplay between DTs and sustainable development and offer structured approaches to understanding their contributions to the SDGs. This section provides this information.

2.1. Key theories and models linking DTs and sustainable development

2.1.1. Sociotechnical systems theory

This theory emphasizes the interdependence between social and technical elements in any technological system [33,34]. For DTs to effectively support sustainable development, this theory suggests aligning technological innovations with societal needs, organizational structures, and cultural values [35,36]. For instance, the adoption of IoT in urban infrastructure requires community engagement, robust policies, and technological readiness [37]. By considering these interactions, the theory ensures that technological implementations enhance system functionality while promoting inclusivity, resilience, and equitable development outcomes [38].

2.1.2. Triple Bottom Line (TBL) framework

The TBL framework integrates social, economic, and environmental sustainability and provides a comprehensive approach to assess DTs' contributions to sustainable development [39]. Blockchain, for example, ensures supply chain transparency and reduces environmental impacts while promoting ethical labor practices and economic efficiency [40,41]. The TBL approach demonstrates how technologies can balance profit, people, and the planet. By fostering equitable value distribution, resource optimization, and environmental preservation, TBL serves as a vital framework for analyzing the impact of digital innovations on SDGs [42,43].

2.1.3. Diffusion of innovation theory

This theory explains the process by which DTs are adopted and highlights the categories such as innovators, early adopters, and laggards [44,45]. It underscores factors influencing the adoption of DTs, such as perceived benefits, compatibility, and complexity. For instance, the widespread acceptance of renewable energy technologies like solar panels depends on early adopters' influence and awareness [46,47]. This theory aids in strategizing technology dissemination by addressing barriers, fostering inclusivity, and ensuring sustainable adoption across different social and economic groups [44].

2.1.4. Capabilities approach

The capabilities approach focuses on expanding individual and community capabilities through access to technology and emphasizes inclusivity and equity [48]. DTs such as e-learning platforms or mobile health applications enable marginalized populations to access education and healthcare [49]. This approach aligns with SDGs like SDG 1 and SDG 4 by enhancing human development [50]. It underscores the ethical imperative of using digital innovations to provide opportunities, foster empowerment, and reduce social inequalities [51,52].

2.1.5. Circular economy model

The circular economy model integrates DTs to minimize waste and optimize resource use [53]. It is closely aligned with SDG 12 [54]. IoT-enabled smart systems monitor waste and energy efficiency [55], while AI optimizes recycling processes [56]. Blockchain ensures traceability in resource lifecycle management [57,58]. By creating closed-loop systems that prioritize reuse, recycling, and remanufacturing, the circular economy model demonstrates how digital tools contribute to sustainable practices, reduce environmental impact, and foster economic growth [59,60].

2.1.6. Technology acceptance model (TAM)

The TAM explains factors influencing the individuals' or organizations' willingness to adopt DTs and emphasizes the perceived ease of use and usefulness of DTs [61,62]. In sustainable development, this model is particularly relevant for analyzing barriers to adopting technologies like digital payments or renewable energy systems [63,64]. Understanding acceptance behaviors helps policymakers and businesses address resistance to change, design user-friendly solutions, and improve technology adoption rates. This fosters digital transformation for achieving the SDGs [65–67].

2.1.7. Technology-organization-environment (TOE) framework

The TOE framework provides a comprehensive lens for analyzing the adoption of DTs in advancing the SDGs. The technological dimension examines the capabilities of DTs focusing on their compatibility, scalability, and potential to address complex sustainability challenges [68–70]. The organizational dimension highlights factors like leadership commitment, resource allocation, workforce readiness, and the strategic alignment of DT initiatives with organizational goals [61,62]. The environmental dimension emphasizes the influence of external forces, including regulatory frameworks [71], market demands [72], socio-political factors, and international collaboration, which shape the adoption and deployment of DTs [73,74]. This framework underscores how organizations can (i) leverage technological innovation while navigating internal and external challenges and (ii) ensure that DTs are effectively integrated to drive progress across SDGs [75–77].

2.1.8. Innovation systems theory

Innovation systems theory explores the collaboration between institutions, organizations, and technology in driving technological advancements [78–80]. It highlights the importance of creating ecosystems where DTs can thrive and contribute to sustainable development [81]. For example, partnerships between governments, private sectors, and academia can foster innovation in smart cities [82–85] or renewable energy [86,87]. This theory emphasizes the role of interconnected systems in overcoming challenges, enabling scalable solutions, and accelerating progress toward SDGs through collaborative innovation [88].

2.1.9. Digital impact assessment (DIA) model

The DIA model serves as a structured approach to evaluating the social, economic, and environmental effects of DTs [89]. It assesses the contributions of DTs toward achieving goals like inclusivity, sustainability, and ethical governance [90]. The model emphasizes evaluating the benefits, such as enhanced decision-making and improved service delivery [91], alongside potential risks like digital divides, cybersecurity threats, and environmental impacts [92]. By considering factors such as accessibility, user engagement, and alignment with sustainable development priorities [93], the DIA model provides a comprehensive framework for decision-makers to measure the short and long-term impacts of digital transformations [94,95]. This approach ensures that digital technology implementations are equitable, responsible, and aligned with global objectives like the SDGs [96,97].

2.1.10. Resilience theory

Resilience theory emphasizes the capacity of systems to adapt and recover from disruptions, aligning with sustainable development's focus on long-term viability [98]. DTs enhance resilience by enabling rapid responses to environmental, economic, or social shocks [99–102]. For example, AI and big data analytics improve disaster prediction and management [103–105], while blockchain ensures secure transactions during crises [106]. By fostering adaptability and preparedness, resilience theory highlights how digital innovations strengthen systems to withstand challenges and SDGs [107].

2.1.11. Life cycle assessment (LCA) model

The LCA model evaluates the environmental, social, and economic impacts of DTs across their entire life cycle, from resource extraction and manufacturing to usage and end-of-life disposal [108]. This model identifies both the benefits of DTs in achieving SDGs such as enhancing efficiency, enabling resource optimization, and supporting environmental monitoring and their challenges, including greenhouse gas emissions, high energy consumption, and e-waste generation [109]. By analyzing trade-offs and synergies across SDG targets, the LCA model helps stakeholders adopt sustainable practices, such as energy-efficient design [110], circular economy principles, and robust e-waste management [111]. It provides a framework for aligning the deployment of DTs with goals like SDG 13 and SDG 12, ensuring that their adoption contributes to long-term sustainability while minimizing negative impacts [100,112]. This holistic approach supports policymakers, businesses, and researchers in making informed decisions to maximize the positive contributions of DTs toward achieving global sustainability objectives [108].

Fig. 2 shows various key theories and models linking DTs and sustainable development.

2.2. Frameworks for assessing DTs' impact on SDGs

2.2.1. Digital transformation framework (DTF)

The DTF provides a holistic approach to understanding how DTs drive sustainability across sectors [113–115]. This framework assesses the strategic alignment of DTs with SDGs, examining factors such as technological innovation, organizational change, and societal impact [97,116,117]. It emphasizes the need for integrating DTs with existing systems to address development challenges. DTF assesses the outcomes of digital transformation on economic performance, social inclusion, and environmental sustainability [118] and enables organizations to evaluate their contributions to SDGs like poverty reduction, sustainable production, and climate action [97,101,119].

2.2.2. Sustainable development technology assessment (SDTA) framework

The SDTA framework focuses on evaluating the potential and actual impacts of DTs on sustainable development [120–122]. It considers social, economic, and environmental dimensions and applies a life cycle assessment approach to measure how technologies such as renewable energy solutions [123], smart grids [96,124], or digital health tools [125,126] influence SDG achievement. This framework provides tools to assess technologies' scalability, feasibility, and long-term sustainability and helps decision-makers to identify those with the greatest potential for driving SDG progress while minimizing unintended negative consequences [96,101,127].

2.2.3. SDG digitalization framework (SDF)

The SDF links DTs directly with the specific targets and indicators of the SDGs [96,128]. It evaluates how DTs can be leveraged to accelerate progress toward individual SDGs [129,130]. By focusing on measurable outputs, such as improved access to education (SDG 4) or enhanced healthcare (SDG 3), the SDF offers a data-driven approach to tracking digital interventions' contributions to global development goals [126,131]. The framework emphasizes the role of data analytics, digital tools, and infrastructure in monitoring and achieving SDG targets [129,132,133].

2.2.4. Technology-driven impact assessment framework (TDIAF)

The TDIAF assesses how the implementation of DTs influences sustainable development at multiple levels [134]. The framework uses a mixed-methods approach to measure the direct and indirect impacts of digital tools on SDG progress, considering factors like adoption rates, resource efficiency, and social benefits [120,135]. TDIAF is particularly useful for understanding how digital solutions enhance the capacity of governments, industries, and communities to implement and monitor sustainable development initiatives [75,136]. It helps policymakers make informed decisions about which technologies to prioritize in addressing SDG-related challenges [132].

2.2.5. Integrated assessment modeling (IAM) for DTs

IAM is a framework that combines multiple systems to assess the role of DTs in sustainability [137,138]. IAM integrates environmental, economic, and social modeling to quantify the contributions of technologies such as automation, smart technologies, and digital supply chains toward achieving SDGs [139,140]. IAM provides a detailed, quantitative view of how DTs interact with the broader sustainability ecosystem and helps policymakers to evaluate trade-offs and synergies between technological advancements and SDG outcomes, particularly in areas like SDG 13 and SDG 12 [141].

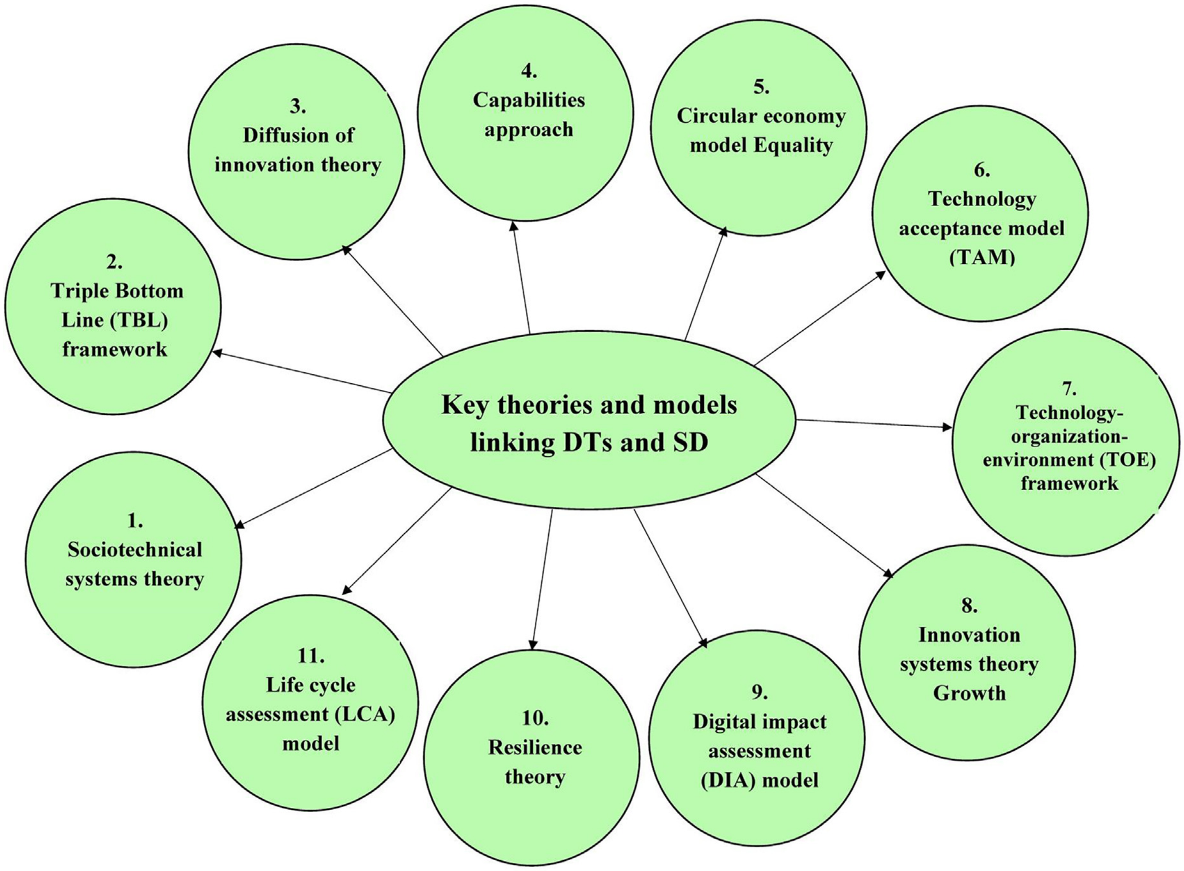

Fig. 2. Key theories and models linking DTs and sustainable development.

2.2.6. Digital ecosystem for SDGs framework (DESF)

The DESF evaluates the interactions between DTs and SDGs through a systems-thinking approach [142,143]. This framework highlights how digital ecosystems, comprising technologies, organizations, and stakeholders, can be optimized to achieve sustainable outcomes [144]. DESF takes into account the dynamic relationships between digital platforms, societal needs, and environmental impact and focuses on collaborative efforts to foster sustainable practices [143]. By assessing the integration of digital solutions within various sectors, DESF provides a comprehensive view of how technological ecosystems can contribute to holistic SDG achievement [145,146].

2.2.7. Impact pathways framework (IPF)

The IPF maps out the pathways through which DTs lead to SDG outcomes. It focuses on tracing the direct and indirect effects of digital tools on societal, economic, and environmental systems [147]. IPF emphasizes feedback loops and interdependencies between technology adoption, policy changes, and sustainable development [148]. By analyzing the effectiveness of different technologies in meeting specific SDG targets, IPF allows researchers and practitioners to understand which digital innovations drive sustainable change and where efforts should be concentrated for maximum impact [149].

2.2.8. Digital sustainability index (DSI)

The DSI is a comprehensive framework that measures the sustainability impact of DTs based on a set of indicators related to environmental, social, and economic performance [150,151]. DSI incorporates factors such as energy consumption, carbon footprint, social equity, and economic inclusivity [152,153]. This index enables stakeholders to assess the sustainability profile of digital innovations across industries and helps them choose technologies that align with SDG principles [154,155]. By providing a standardized measure, the DSI allows for comparisons across sectors and regions and facilitates global tracking of digital contributions to SDGs [156].

Fig. 3 shows various frameworks for assessing DTs' impact on SDGs.

2.3. Theoretical justification for the model for integrating DTs for SDGs

The proposed model for assessing the impact of DTs on SDGs is grounded in well-established theoretical frameworks that offer a structured approach to evaluating digital transformation and its sustainability implications. The DTF framework provides a holistic view of how DTs drive sustainability by aligning technological innovation with economic, social, and environmental goals [113–115]). This framework underscores the need for integrating DTs into existing systems to address development challenges effectively [97,116,117]. Complementing this, the SDTA framework evaluates the potential and actual impacts of digital solutions, incorporating a life cycle perspective to measure their feasibility, scalability, and long-term sustainability [96,101,127]. The SDF framework further strengthens the model by linking DTs directly to SDG targets [96,128], ensuring a data-driven assessment of their contributions [126,131]. Additionally, the TDIAF framework enables a multi-level evaluation of DTs, considering adoption rates, resource efficiency, and social benefits [120,135]. The inclusion of the IAM framework ensures that technological, economic, and environmental dimensions are quantified [139,140], helping policymakers balance trade-offs and synergies in digital interventions [141]. The DESF framework highlights the role of interconnected digital platforms and stakeholders in fostering sustainable practices [145,146], while the IPF framework maps out the direct and indirect effects of DT adoption on SDG progress [147]. Finally, the DSI framework provides a standardized measure of the sustainability performance of DTs [150,151],

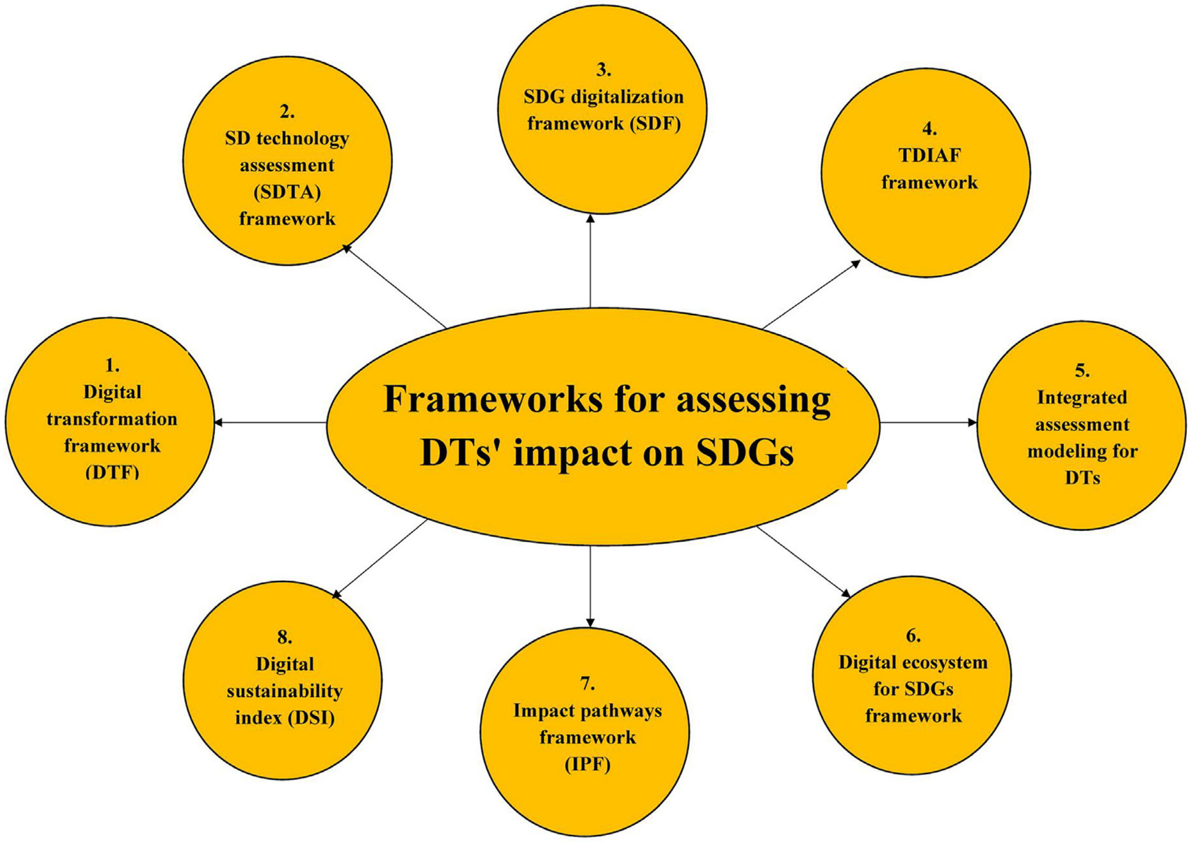

Fig. 3. Frameworks for assessing DTs' impact on SDGs.

facilitating cross-sectoral comparisons [156]. By integrating insights from these frameworks, the proposed model of integrating DTs for SDGs offers a comprehensive, evidence-based approach to evaluating the role of DTs in advancing SDGs, ensuring theoretical rigor and practical applicability.

3. Review questions

The review questions addressed by the study are

RQ1: How do DTs contribute to the achievement of specific SDGs?

RQ2: What are the primary challenges and barriers that hinder the effective adoption of DTs in supporting SDGs and how can these obstacles be mitigated?

4. Review objectives

The review objectives of this literature are to examine the contributions of DTs in advancing specific SDGs and to explore the primary challenges and barriers that impede the effective adoption of these technologies. By analyzing how DTs support the achievement of SDGs, the review aims to provide a comprehensive understanding of their potential impact. Additionally, it seeks to identify the key obstacles hindering their widespread adoption and propose strategies to mitigate these challenges and enable more effective integration of DTs for sustainable development.

5. Review methodology

The review methodology follows the PRISMA guidelines [32], utilizing the Scopus database to identify relevant studies. The search is conducted between May 17th, 2024 to Nov. 25th, 2024. The review employs a systematic search strategy based on the keywords “Digital Technologies” and ”Sustainable Development” in the title, abstract, and keywords. Inclusion criteria focus on peer-reviewed articles, conference proceedings, and reports that examine the role of DTs in supporting SDGs, while exclusion criteria eliminate studies that do not directly address the specified technologies or SDGs-related outcomes. A total of 473 peer-reviewed articles are included in the study.

Fig. 4 shows the review methodology for selecting the articles for the study.

Fig. 5 shows the year-wise distribution of the papers included in the study.

6. DTs supporting SDGs

6.1. Artificial intelligence (AI)

AI refers to the simulation of human intelligence processes by machines, particularly computers [157]. AI technologies such as machine learning, natural language processing, and computer vision are used to analyze large datasets, automate tasks, and enhance decision-making [158]. In sustainable development, AI optimizes resource management [159–161], improves predictive models for climate change [162], enhances healthcare through personalized treatments [163], and drives innovation in clean energy [124,164]. AI also plays a key role in improving efficiency and reducing waste across industries, directly supporting SDGs like SDG 9 and SDG 13 [165].

6.2. Blockchain

Blockchain is “a decentralized digital ledger technology that records transactions across multiple computers and ensures transparency, security, and immutability” [166]. By enabling peer-to-peer transactions

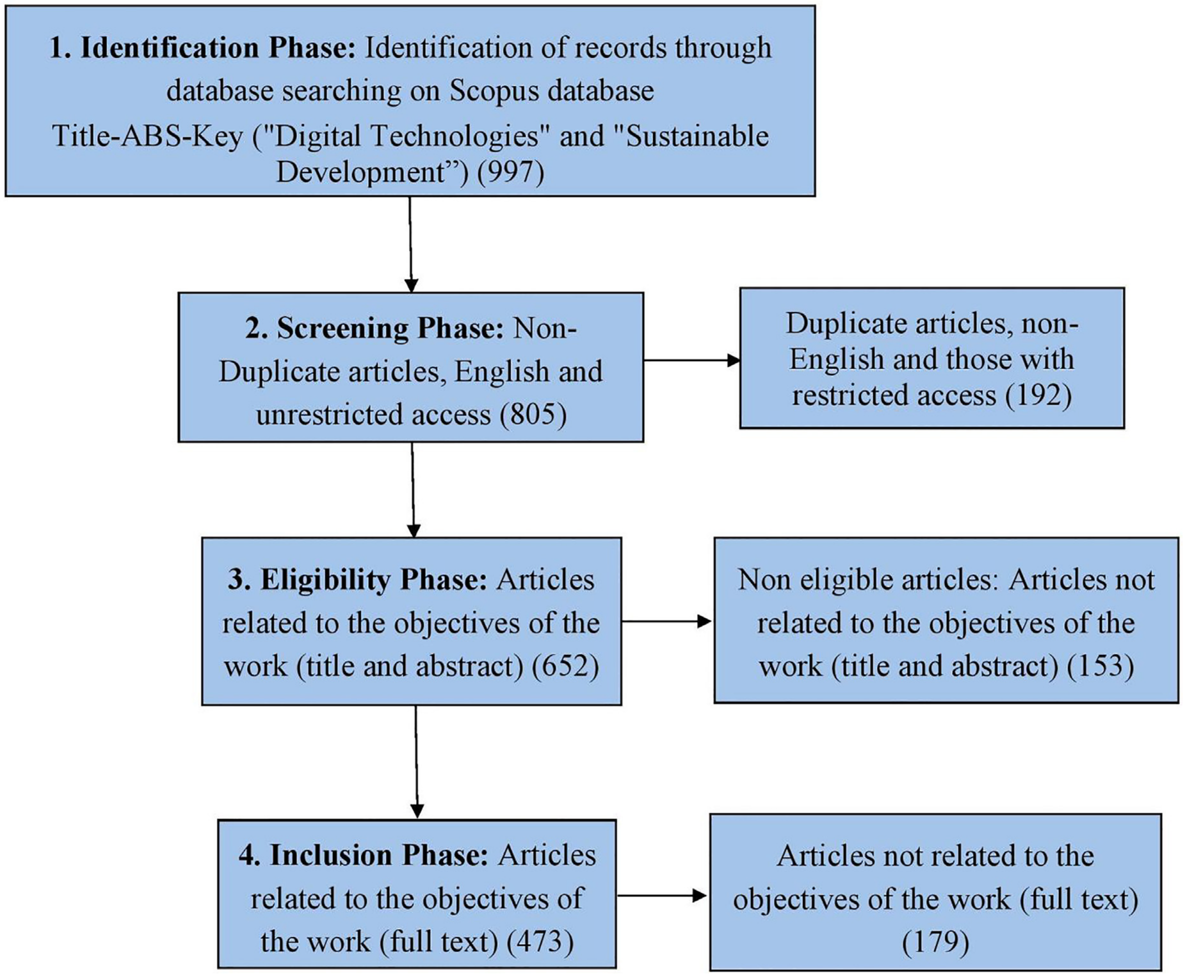

Fig. 4. Review methodology for selecting the articles.

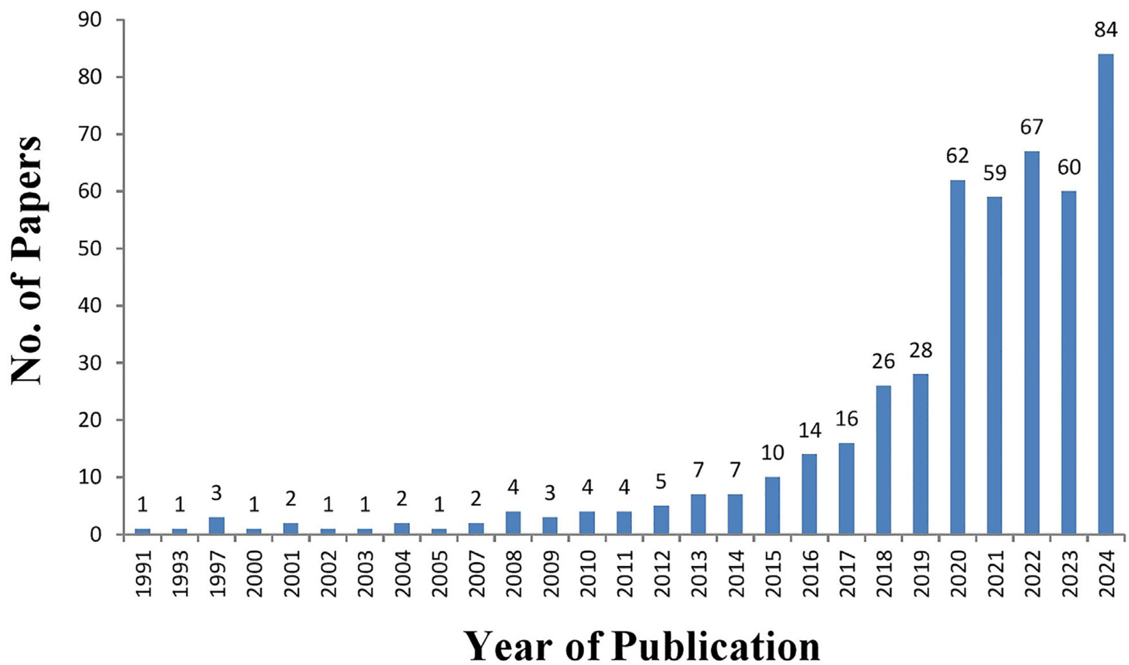

Fig. 5. Year-wise distribution of the papers.

without the need for intermediaries, blockchain facilitates secure, transparent supply chains, enhances data integrity, and supports financial inclusion [167]. In the context of sustainable development, blockchain helps trace the origin of goods, ensures ethical sourcing, and reduces fraud [168]. It also enables transparent, secure transactions for aid distribution, particularly in developing regions, and contributes to SDGs like SDG 8 and SDG 12 [169,170].

6.3. Internet of Things (IoT)

The IoT refers to "a network of interconnected physical devices that collect and exchange data through the internet" [171]. IoT applications in sustainable development range from smart grids and energy-efficient buildings to precision agriculture and environmental monitoring [172]. By gathering real-time data, IoT helps optimize resource usage, reduce waste, and improve operational efficiency [173]. IoT plays a crucial role in achieving SDGs such as SDG 7, SDG 13, and SDG 11 by enabling smarter, more sustainable solutions [174,175].

6.4. Big data analytics

Big data analytics refers to "the collection and analysis of vast amounts of structured and unstructured data to uncover patterns, trends, and associations. It enables organizations to make informed decisions and optimize operations" [176]. In sustainable development, big data analytics is used to analyze environmental data for better resource management, monitor climate change, and improve disaster response strategies [105,177]. It also plays a key role in tracking public health trends, energy consumption, and agricultural productivity and contributes to SDGs like SDG 3 and SDG 12 [178].

6.5. Cloud computing

Cloud computing is "a model that allows users to access and store data and applications over the internet. It provides on-demand computing resources such as storage, processing power, and software" [179]. It enables businesses and governments to scale operations quickly without the need for significant infrastructure investments [180]. Cloud services are essential for enabling collaboration, data sharing, and innovation in areas like healthcare, education, and disaster management [181]. By reducing the need for physical resources, cloud computing also supports SDGs like SDG 7 and SDG 9 [182].

6.6. 5G networks

5G networks are the fifth generation of wireless technology. It offers faster data speeds, higher bandwidth, and more reliable connections compared to previous generations [183]. 5G enables the efficient transmission of large volumes of data, which is essential for the development of smart cities, autonomous vehicles, and IoT applications [184,185]. In sustainable development, 5G supports real-time monitoring of environmental conditions, enhances remote healthcare services, and enables efficient logistics. With its capacity to connect more devices, 5G helps accelerate progress toward SDGs like SDG 11 and SDG 13 [186].

6.7. Remote sensing and geographic information systems (GIS)

These are powerful technologies used to collect, analyze, and interpret spatial data for decision-making [187]. Remote sensing involves the use of satellites, drones, or aircraft to gather data about the Earth's surface [188], while GIS combines this data with mapping and analytical tools to visualize and analyze spatial patterns [189]. These technologies are vital for monitoring environmental changes, managing natural resources, and planning urban development [190–192]. In sustainable development, remote sensing and GIS help (i) track deforestation, monitor climate change, manage water resources, and plan sustainable agriculture and (ii) directly contribute to SDGs like SDG 15, SDG 6, and SDG 13 [193,194].

Fig. 6 shows the various DTs supporting the SDGs.

7. Impact of DTs on individual SDGs

DTs are playing an increasingly vital role in driving progress toward the SDGs. They offer innovative solutions to the various SDGs. From improving healthcare outcomes to enhancing environmental sustainability, digital tools have the potential to reshape industries and improve quality of life globally. DTs provide the foundation for achieving key SDGs by promoting sustainability, inclusivity, and economic growth and fostering innovation and collaboration. Their integration into global development strategies can facilitate transformative changes that are critical to achieving the SDGs by 2030. Table 1 shows the contribution of various DTs to the SDGs. Table 2 shows the mapping of the contribution of various DTs to the SDGs.

8. Challenges and barriers

Challenges and barriers significantly hinder the adoption and integration of DTs that are essential for achieving SDGs. The digital divide remains a critical issue, with disparities in access to technology and infrastructure, particularly in rural and underdeveloped regions. This limits equitable participation in digital solutions [442]. Data privacy and security concerns, including cybersecurity risks and inconsistent regulatory frameworks, further complicate the effective deployment of technologies. This creates uncertainty in sectors such as healthcare and governance [443–446]. Additionally, skill shortages and digital illiteracy in many regions restrict the workforce's ability to fully leverage advanced technologies like AI and the IoT [447–449]. High initial and ongoing costs of implementing and maintaining digital infrastructure pose significant financial constraints, particularly in low-income countries. This makes it difficult to sustain long-term technological investments [96,244,450]. Regulatory and policy barriers, including fragmented global standards and inconsistent enforcement, hinder international collaboration and the seamless integration of technologies across borders [451,452]. Furthermore, challenges related to interoperability, data silos, and environmental impacts, such as electronic waste (e-waste) and excessive energy consumption, underscore the complexities of ensuring sustainable and effective digital solutions [453]. Cultural resistance, social inequities, and public skepticism also contribute to these barriers and restrict the widespread acceptance and utilization of DTs [126,454]. Addressing these challenges requires comprehensive strategies, including robust policy frameworks, capacity-building initiatives, and the adoption of sustainable practices to ensure that DTs contribute effectively to the achievement of the SDGs [455–457].

Fig. 7 shows the various categories of challenges and barriers to the adoption of DTs.

Table 3 shows the various challenges and barriers to the adoption of DTs.

Table 4 shows the general key strategies for the SDGs.

Table 5 outlines the contributions of various stakeholders toward achieving the United Nations SDGs.

9. Conclusion

Digital technologies (DTs) have become indispensable in addressing the multifaceted challenges of sustainable development. They offer innovative and scalable solutions to the complex issues outlined in the United Nations Sustainable Development Goals (SDGs). Technologies such as AI, IoT, blockchain, cloud computing, big data analytics, remote sensing, and GIS play a transformative role in advancing sustainability. By improving decision-making, enhancing access to services, and fostering cross-sectoral innovation, these digital tools facilitate progress toward critical SDG targets, including poverty alleviation (SDG 1), equitable growth (SDG 8), and climate action (SDG 13). This review highlights the transformative potential of DTs in advancing the United Nations SDGs.

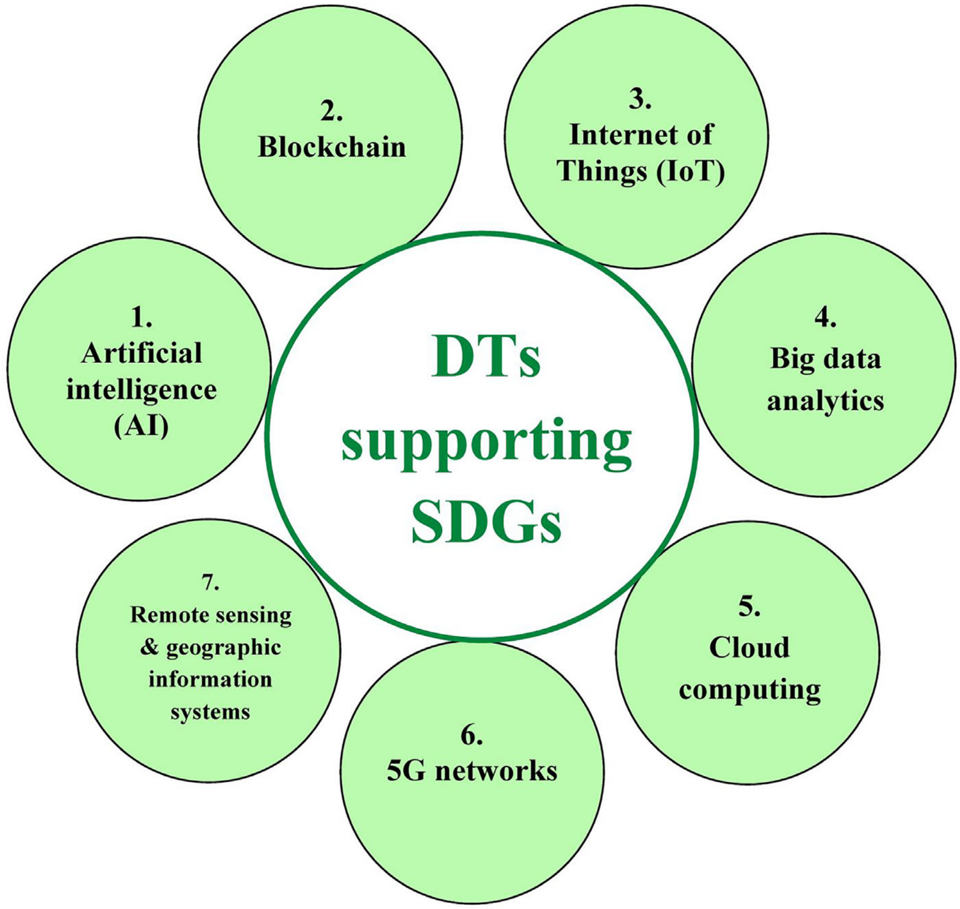

Fig. 6. DTs supporting SDGs.

Artificial intelligence (AI) has emerged as a transformative force across multiple sectors, driving automation, predictive analytics, and optimization while supporting SDGs such as zero hunger (SDG 2), good health and well-being (SDG 3), climate action (SDG 13), and sustainable cities (SDG 11). AI-powered precision agriculture enhances crop monitoring, optimizes irrigation, and minimizes resource wastage, aligning with Altayeb et al. [486], who reported AI-driven agricultural productivity improvements of up to 20%. AI also revolutionizes healthcare by enabling early disease detection, personalized treatments, and advanced diagnostics, consistent with findings by Zeb et al. [487] and Sherani et al. [488], who highlight AI's role in improving healthcare accessibility, particularly in remote areas (SDG 3).

Blockchain technology enhances security, transparency, and ethical governance, making it a crucial tool for promoting responsible consumption and production (SDG 12), financial inclusion (SDG 10), and peace and justice (SDG 16). Blockchain-based supply chain tracking ensures product traceability and regulatory compliance, reinforcing Sarker et al. [489] and Kaplan [490], who emphasized blockchain's role in reducing fraud and corruption. Furthermore, smart contracts automate governance processes, streamlining compliance and minimizing bureaucratic inefficiencies (SDG 16), consistent with Cagigas et al. [491] and Myeong and Jung [492], who highlighted blockchain's impact on reducing bureaucratic bottlenecks in public administration (SDG 16).

The Internet of Things (IoT) facilitates real-time monitoring and automation, driving sustainable cities (SDG 11), clean energy (SDG 7), and climate action (SDG 13). Smart grids optimize energy consumption, reducing electricity wastage by 20%–35%, a trend reported by Mahapatra et al. [493], Mortaji et al. [494], and Saleem et al. [495]. IoT also improves transportation efficiency, mitigating emissions and enhancing urban mobility, aligning with research by Musa et al. [496]. Additionally, IoT-driven health monitoring devices enhance chronic disease management (SDG 3), as noted by Islam et al. [497], Yin et al. [498], and Kelly et al. [309], who emphasized their role in improving patient care and healthcare accessibility.

Big data analytics supports sustainable development by providing actionable insights into environmental, healthcare, and economic systems, addressing disaster management or response planning (SDG 13), improving public health tracking (SDG 3), and enabling predictive analysis in financial markets stability (SDG 8). It enables predictive modeling for climate risk assessment, aligning with Yu et al. [105] and Sarker et al. [102], who demonstrated its effectiveness in disaster response. Additionally, big data enhances precision agriculture and supply chain efficiency, mitigating food insecurity and aligning with zero hunger (SDG 2), as supported by Ding et al. [499] and Rejeb et al. [500], who highlighted its role in optimizing food production and distribution.

Cloud computing fosters industry innovation (SDG 9) and global collaboration and partnerships (SDG 17) by enhancing data accessibility and scalability across healthcare, education, and governance. It reduces infrastructure costs and improves digital inclusivity, particularly in developing economies (SDG 9), findings consistent with Díaz-Arancibia et al. [501] and Sharma et al. [502], who highlighted cloud computing's role in digital transformation. By enabling real-time data sharing, cloud solutions streamline cross-border research and decision-making, accelerating technological progress.

5G networks drive advancements in remote work, telemedicine, and smart manufacturing, significantly improving industrial innovation (SDG 9) and healthcare access (SDG 3). By enabling ultra-fast and low-latency communication, 5G enhances real-time environmental

Table 1
Contribution of various DTs to the SDGs.

SDGs	AI	Blockchain	IoT	Big data analytics	Cloud computing	5G Networks	Remote sensing and GIS
SDG 1: No Poverty	Identifies poverty hotspots and optimizes resource distribution for better targeting of social programs and reducing poverty [195].	Enables secure, transparent financial transactions. This offers financial inclusion, reduces corruption, and lifts people out of poverty [196].	Facilitates affordable access to services like healthcare and education in rural areas. This improves their living standards [197].	Allows governments and organizations to analyze poverty data and design more effective poverty alleviation strategies [129,198,199].	Enables accessible platforms for financial and social services, and helps in reaching underserved communities globally [133,200].	Provide faster, reliable access to digital services and allow remote populations to access banking, healthcare, and education [201–203].	Provide critical data for identifying poverty regions and informing targeted interventions for better resource allocation [204,205].
SDG 2: Zero Hunger	Optimizes agricultural practices, predicts crop yields, and manages food supply chains efficiently to reduce food insecurity [206,207].	Ensures transparency in food supply chains, verifies the origin of food, and promotes sustainable farming practices to address hunger [208,209].	Monitors soil health, crop growth, and environmental conditions to increase agricultural productivity [207,210].	Enables analysis of global food distribution, identifies food shortages and inefficiencies, and improves food security strategies [209,211].	Enables (a) collaborative platforms for farmers to access agricultural resources and share best practices and (b) promote food security [212].	Enhance connectivity for agricultural monitoring systems and enable real-time data collection on crop health and environmental factors [213,214].	(a) aid in monitoring agricultural land, crop health, and environmental factors and (b) facilitate sustainable agriculture and food production [210,214].
SDG 3: Good Health and Well-being	(a) enables personalized healthcare, predictive diagnostics, and treatment recommendations and (b) improves overall health outcomes [215,216].	Provides secure medical data management and promotes trust and data security. This is essential for improving health services and outcomes [217].	IoT devices like wearables monitor health in real time. This facilitates early detection of illnesses and improves chronic disease management [216].	Identifies health trends, tracks disease outbreaks, and optimizes healthcare systems to improve access and efficiency [216].	Provides scalable storage for health data. This enables global collaboration and better healthcare delivery across regions [218,219].	Enable fast, reliable communication for telemedicine, remote patient monitoring, and emergency healthcare services [203,218,220].	Tracks environmental factors that affect health, including air quality, pollution, and vector-borne diseases. This supports the public health efforts [221].
SDG 4: Quality Education	Enables personalized learning experiences, adaptive teaching platforms, and automated assessment tools, enhancing educational outcomes [222,223].	Offers secure credentials, certificates, and diplomas and ensures the credibility of educational qualifications across borders [224].	Provides smart classrooms, connected devices, and remote learning tools that improve access to quality education, especially in remote areas (K. [225,226]).	Analyzes student performance, optimizes learning paths, and addresses gaps in educational resources. Thus, improving the overall education quality [225,227].	Offers remote learning platforms and allows students worldwide to access education, irrespective of geographic location [228].	Support high-quality video streaming for online education and ensure remote and underprivileged areas have access to learning materials [213,229].	Support educational efforts by mapping areas lacking educational resources. This helps governments to allocate resources effectively [230].
SDG 5: Gender Equality	Analyzes gender disparities in various sectors and provides insights to design gender-sensitive policies and programs [231] .	(a) provides secure platforms for women to access financial services and (b) ensures economic empowerment and equal opportunities (Di [232]).	Uses in projects aimed at improving women's health and safety, including wearable safety devices and healthcare monitoring systems [233–235].	Assesses gender-based inequalities, analyzes workforce participation, and designs better policies for gender equality in various sectors [236,237].	Enables online platforms for women's empowerment, including entrepreneurship, education, and networking opportunities. This fosters gender equality [236,238,239].	(a) enable faster and more reliable communication tools for gender-based advocacy and initiatives and (b) promote equality and awareness [240].	Monitor gender-focused initiatives in areas like agriculture and land use and ensure women's inclusion in decision-making processes [240–242].
SDG 6: Clean Water and Sanitation	Optimizes water distribution systems, predicts water demand, and detects leaks. This improves water efficiency [243–245].	Ensures transparency in water distribution and makes sure clean water resources are fairly distributed and managed [246–248].	Monitors water quality, tracks contamination levels, and manages water usage in real-time. This improves access to clean water [244,248–251].	Analyzes water availability, demand, and pollution trends. This aids in better water resource management and policy planning [178,251].	Enables global collaboration on water management, sharing resources, technologies, and data to improve access to clean water [244,252,253].	Enable real-time water quality monitoring. This enhances early detection of contamination and water-related health risks [154,254].	Provide detailed mapping of water resources, help in managing sanitation systems, and support disaster response related to water scarcity [255–257].
SDG 7: Affordable and Clean Energy	Optimizes energy consumption patterns, supports grid management, and aids in the development of renewable energy sources [258–260].	Ensures transparency and fair distribution of energy, promotes decentralized, renewable energy solutions, and reduces energy waste [169,261–263].	Enables the creation of smart grids that optimize energy usage in households, industries, and cities for energy efficiency and sustainability [174,264,265].	Analyzes energy consumption patterns, predicts demand, and optimizes renewable energy deployment. This makes energy cleaner and more accessible [251,266,267].	Facilitates energy management systems, supports innovation in energy efficiency, and enables global collaboration for clean energy solutions [252,265].	Enhance the monitoring of energy systems. This enables real-time adjustments and improvements in energy efficiency and renewable energy integration [186,268,269].	Track renewable energy potential, viz. solar and wind resources, and optimizes energy distribution networks for efficiency [270,271].
SDG 8: Decent Work and Economic Growth	Improves job matching, optimizes workforce allocation, and helps identify emerging skill needs. This fosters job creation and economic growth [272,273].	(a) facilitates secure and transparent business transactions, (b) ensures fair wages, and (c) improves workers' rights in the gig economy [130,274,275].	Creates more efficient and sustainable workplaces, reduces costs, and promotes safe working environments. This leads to better job quality [172,276].	Identifies labor market trends, optimizes workforce allocation, and improves job creation strategies. This fosters economic growth and reduces unemployment [152,277,278].	Fosters global collaboration and provides businesses with scalable solutions. This enables them to grow and create more jobs in various sectors [181,279].	Improve communication in remote or rural areas, support economic activities, enhance access to markets, and provide new job opportunities [280,281].	Identify areas with high unemployment and low economic activity and guides the policymakers to target regions for economic development initiatives [282,283].

(continued on next page)

monitoring and emergency response, findings aligned with Qadir et al. [503] and Naqvi et al. [504], who emphasize its role in disaster preparedness, emergency response, and resilience planning (SDG 13). The technology also supports digital service expansion in rural and underserved regions, bridging connectivity gaps and fostering economic inclusion.

Remote sensing and GIS technologies provide critical insights for environmental sustainability (SDG 14 and SDG 15), disaster management

Table 1 (*continued*).

SDGs	AI	Blockchain	IoT	Big data analytics	Cloud computing	5G Networks	Remote sensing and GIS
SDG 9: Industry, Innovation, and Infrastructure	Fosters innovation in manufacturing, optimizes processes, reduces waste, and helps industries to adopt sustainable practices [284,285].	Promotes innovation by providing secure and transparent solutions for industries to manage supply chains, intellectual property, and financial transactions [286,287].	(a) supports smart manufacturing, predictive maintenance, and automation and (b) drives efficiency and innovation in industries [288–290].	Supports industries in optimizing production, reducing waste, and improving supply chains. This leads to more sustainable and efficient industrial practices [291–295].	Enables innovation in infrastructure by offering scalable computing resources for businesses and government agencies [296,297].	Provide the high-speed connectivity needed for Industry 4.0. This supports real-time data transmission and enhances industrial automation [298–300].	Monitor and optimize infrastructure projects, assist in urban planning, track industrial activity, and ensure sustainability [301,302].
SDG 10: Reduced Inequalities	Identifies gaps in access to services and enables policies to bridge inequalities in education, healthcare, and employment [303,304].	(a) provides equitable access to financial and social services, (b) promotes inclusion and reduces economic inequalities [305–307].	Provides access to digital services in remote or underserved areas. This helps in reducing inequalities in healthcare, education, and resource distribution [308–310].	(a) reveals trends and disparities in income, healthcare, and education and (b) enables more effective policies to reduce inequality [311,312,312,312, 312,312,312].	(a) makes services like education, finance, and healthcare more accessible to underserved populations and (b) reduces digital inequality [308,312–314].	Enable the inclusion of remote or rural populations in digital services like telemedicine and remote education. This reduces the access gaps [201,315].	(a) help target regions suffering from inequality and (b) guide interventions and policies for inclusive growth [242,255,316].
SDG 11: Sustainable Cities and Communities	(a) optimizes urban planning, traffic management, and resource usage and (b) promotes sustainable cities [317–320].	(a) ensures transparency in urban development projects and (b) helps in building sustainable, equitable cities with fair resource distribution [321–323].	Enables smart cities by providing real-time data on traffic, waste, energy, and air quality. This improves sustainability and residents' quality of life [172,324–326].	Aids in urban planning, optimizing resource use, and reducing inefficiencies. Thus, ensures that cities are more sustainable and livable [327–329].	Supports smart city solutions by allowing cities to use digital infrastructure to optimize energy, transportation, and public services [325,330–332].	Enhance connectivity for smart cities, improve traffic management, emergency response, and public services with real-time data transmission [184–186,333].	Support sustainable urban planning by mapping land use, environmental risks, and infrastructure needs. Thus, promoting greener cities [302,334–338].
SDG 12: Responsible Consumption and Production	Promotes sustainable production by optimizing supply chains and reducing waste in manufacturing processes [339–341].	Ensures transparency and traceability in supply chains and makes it easier to track sustainable practices and responsible consumption [40,41,168,342,343].	Supports resource-efficient practices in industries, reduces waste, and promotes circular economy models [344–346].	Monitors resource consumption patterns and identifies inefficiencies in production systems. This aids in sustainable manufacturing practices [347,348].	Fosters resource-sharing platforms. This enables organizations to innovate and adopt sustainable production practices [349–351].	Provide the connectivity for IoT-based systems that track resource use and manage production processes in real-time for better sustainability [186,352,353].	(a) provide data on resource availability, land use, and environmental degradation and (b) aid in the sustainable management of natural resources [193,354,355].
SDG 13: Climate Action	Predicts climate change impacts, optimizes mitigation strategies, and helps monitor carbon footprints in various sectors [356–358].	Ensures transparency in carbon credit markets. This helps businesses and governments track and trade emissions reduction [359–361].	Supports climate action by monitoring environmental parameters, viz. air quality, temperature, and pollution levels [362–365].	Supports climate modeling and helps in predicting environmental changes and designing effective climate action plans [101,102,366–368].	Provides platforms for global collaboration on climate action and facilitates access to climate data for better decision-making [181,369,370].	Enable real-time environmental monitoring systems that help track climate change impacts and support faster responses [100,371].	Offer satellite data and real-time mapping to monitor deforestation, melting ice caps, and greenhouse gas emissions and aid in climate action [100,186,371].
SDG 14: Life Below Water	Analyzes ocean data to predict and mitigate the effects of climate change on marine life. This promotes sustainable marine practices [372–374].	Ensures the traceability of ocean resources. This combats illegal fishing and promotes sustainable fishing practices [375,376].	(a) helps monitor water quality, marine life health, and pollution levels and (b) ensures better management of aquatic ecosystems [362,377–379].	Analyzes oceanic data and supports the design of better conservation strategies for marine life and sustainable fishing practices [380,381].	Supports collaboration between research institutions, NGOs, and governments to conserve marine biodiversity and develop sustainable fisheries [382–384].	Enable faster data transfer for real-time ocean monitoring and support sustainable fisheries and marine conservation efforts [375,379,385].	(a) provide detailed data on ocean pollution, marine biodiversity, and habitat degradation (b) help in monitoring and protecting marine ecosystems [386–391].
SDG 15: Life on Land	(a) monitors deforestation, biodiversity loss, and ecosystem health and (b) supports sustainable land management practices [374,384,392].	(a) supports transparent tracking of land ownership, land use, and conservation efforts and (b) ensures responsible land management [393,394].	Enables real-time monitoring of land use, wildlife health, and agricultural practices. This promotes conservation and sustainable land management [395–398].	Tracks biodiversity, forest cover, and land-use patterns and enables more informed conservation and sustainable land-use decisions [396,399].	Enables data sharing among conservationists, governments, and researchers and helps in improving global land conservation efforts [400].	Enhance real-time monitoring of land ecosystems. This enables faster response to environmental changes and supports conservation efforts [371,401].	Play a crucial role in tracking land use, deforestation, habitat loss, and conservation efforts. This supports the protection of terrestrial ecosystems [193,402–404].
SDG 16: Peace, and Strong Institutions	Detects patterns of corruption, monitors social behavior, and predicts political instability. Thus, promoting peace and justice [405–408].	Enhances transparency in governance, ensures fair distribution of resources and promotes justice by preventing fraud and corruption [409–412].	IoT devices help monitor peace-building activities, track conflicts, and provide data for improving law enforcement and security systems [413].	Analyzes crime, conflict, and justice system data. This helps in designing policies that promote peace, security, and justice [414–416].	Enables global collaboration in peacebuilding efforts and supports the sharing of data for justice reforms [417,418].	Enable real-time communication for conflict resolution, emergency response, and justice-related initiatives.	Provide data on regions of conflict, illegal activity, and environmental degradation. This aids in peacekeeping and justice efforts [419–422].
SDG 17: Partnerships for the Goals	Enhances international cooperation by predicting trends, aligning policies, and optimizing collaboration [285,423].	Provides transparent, secure mechanisms for international agreements and partnerships. This ensures accountability of the partnerships [424,425].	Monitors and tracks the success of global collaborations in areas like healthcare, education, and infrastructure. This fosters better partnerships [426,427].	Enables the assessment of the success of partnerships. This helps in guiding future collaborations and ensuring resource allocation is effective [428–432].	Allows for collaborative platforms that support partnerships across sectors and borders to achieve the SDGs [91,433–435].	Facilitate faster, more reliable communication. This makes international collaborations more efficient and effective in achieving global goals [436–438].	Supports cross-border collaborations by providing global data for sustainable development initiatives and ensuring efficient resource sharing [439–441].

Table 2
Mapping of the contribution of various DTs with the SDGs.

SDGs	AI	Blockchain	IoT	Big data analytics	Cloud computing	5G networks	Remote sensing & GIS
SDG 1: No Poverty	✓	✓	✓	✓	✓	✓	✓
SDG 2: Zero Hunger	✓	✓	✓	✓	✓	✓	✓
SDG 3: Good Health and Well-being	✓	✓	✓	✓	✓	✓	✓
SDG 4: Quality Education	✓	✓	✓	✓	✓	✓	✓
SDG 5: Gender Equality	✓	✓	✓	✓	✓	✓	✓
SDG 6: Clean Water and Sanitation	✓	✓	✓	✓	✓	✓	✓
SDG 7: Affordable and Clean Energy	✓	✓	✓	✓	✓	✓	✓
SDG 8: Decent Work and Economic Growth	✓	✓	✓	✓	✓	✓	✓
SDG 9: Industry, Innovation, and Infrastructure	✓	✓	✓	✓	✓	✓	✓
SDG 10: Reduced Inequalities	✓	✓	✓	✓	✓	✓	✓
SDG 11: Sustainable Cities and Communities	✓	✓	✓	✓	✓	✓	✓
SDG 12: Responsible Consumption and Production	✓	✓	✓	✓	✓	✓	✓
SDG 13: Climate Action	✓	✓	✓	✓	✓	✓	✓
SDG 14: Life Below Water	✓	✓	✓	✓	✓	✓	✓
SDG 15: Life on Land	✓	✓	✓	✓	✓	✓	✓
SDG 16: Peace, and Strong Institutions	✓	✓	✓	✓	✓	✓	✓
SDG 17: Partnerships for the Goals	✓	✓	✓	✓	✓	✓	✓

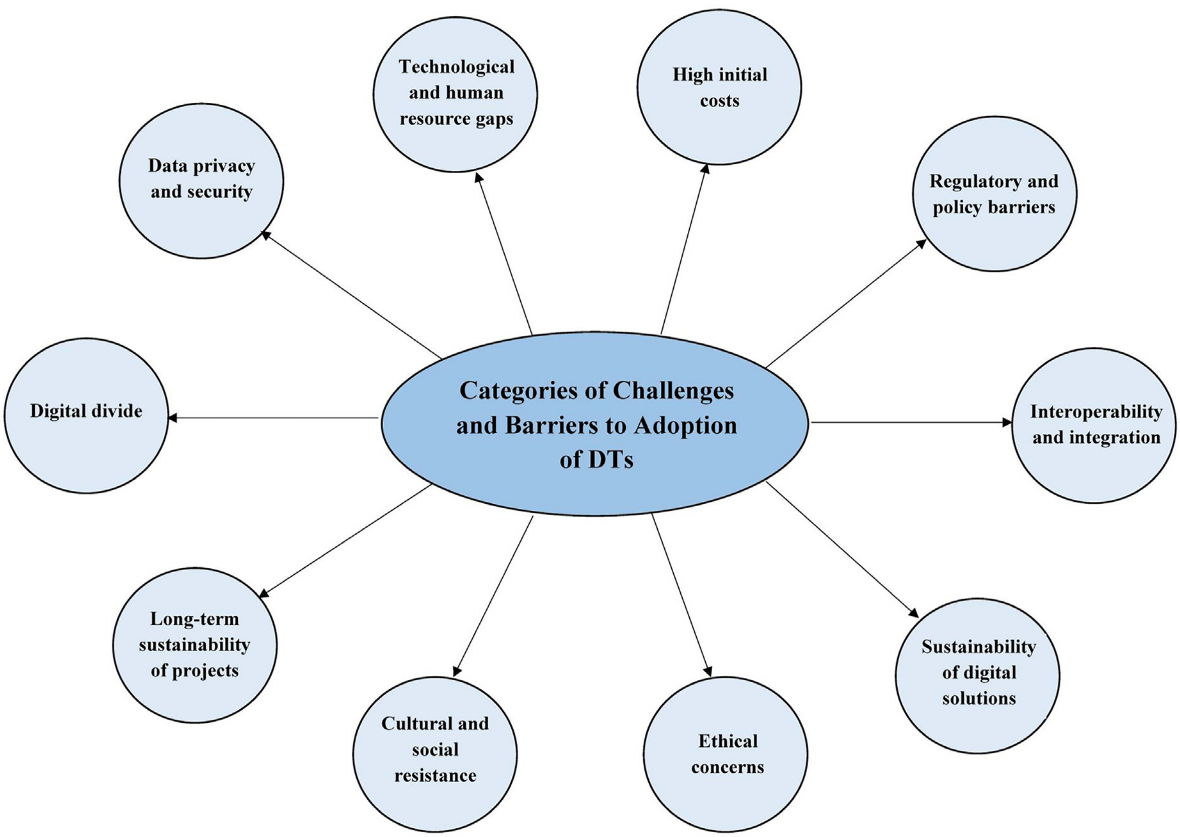

Fig. 7. Categories of challenges and barriers to adoption of DTs.

(SDG 13), and water conservation (SDG 6). GIS-based mapping enables precise monitoring of deforestation, urban expansion, and climate change effects (SDG 15), findings supported by Junge et al. [505] and Mani and Varghese [506], who emphasized its importance in ecological conservation and resource management. These technologies enhance resilience planning and sustainable resource utilization, contributing to long-term environmental sustainability.

DTs provide diverse benefits across domains. Big data and analytics enhance decision-making and resource optimization across sectors, from improving agricultural productivity (SDG 2) to monitoring environmental changes (SDG 13). The IoT supports smart city development (SDG 11), improves healthcare delivery (SDG 3), and optimizes energy use (SDG 7). Blockchain technology fosters transparency and accountability, promoting fair trade (SDG 12), financial inclusion (SDG 10), and ethical governance (SDG 16). AI contributes significantly to automation and predictive analytics, addressing complex challenges such as disease management (SDG 3), personalized education (SDG 4), and resource management for climate action (SDG 13). Cloud computing enhances accessibility and scalability, supporting innovation and collaboration (SDG 9 and SDG 17). Remote sensing and GIS technologies provide critical data for environmental monitoring and disaster response, directly benefiting life below water (SDG 14), life on land (SDG 15), and climate action (SDG 13).

Despite their transformative potential, DTs face several barriers to widespread adoption, particularly in developing regions, where digital divides, high costs, data privacy concerns, and skill gaps hinder equitable access. Limited infrastructure and digital literacy disparities continue to restrict technological adoption, reinforcing findings from

Table 3
Various challenges and barriers to the adoption of DTs.

Category	Challenges and barriers	Description
Digital divide	Access to technology	The unequal distribution of DTs, particularly in rural and developing regions, remains a significant barrier. Many people lack access to high-speed internet, smartphones, or computers, which restricts their participation in digital-driven solutions for SDGs. This digital divide exacerbates inequality and prevents these communities from benefiting from technological advancements that could help in areas like healthcare, education, and economic development [458,459].
	Infrastructure limitations	Insufficient infrastructure, such as unreliable power supplies and limited internet connectivity, impedes the widespread adoption of DTs. Many regions, especially in low-income countries, lack the infrastructure required to implement solutions like AI, IoT, or blockchain, making it difficult to leverage these technologies for SDG progress [460,461].
Data privacy and security	Cybersecurity risks	With the rise of DTs, the volume of data generated has grown exponentially. However, this data is vulnerable to cyberattacks, breaches, and exploitation. Sensitive information, such as personal health data or financial details, can be compromised. This leads to significant risks. This creates hesitance to adopt digital solutions, especially in sectors like healthcare, governance, and education, where privacy is a priority [444,462].
	Regulatory challenges	The lack of consistent, global data privacy standards and regulations complicates cross-border data sharing. Each country has different laws governing data protection, which can create legal barriers to international collaborations that are necessary for addressing global SDGs [443–446]. Additionally, inconsistent regulations may lead to misuse of personal data and hinder the effectiveness of DTs in achieving sustainable outcomes (Adedoyin Tolulope [463–465]).
Technological and human resource gaps	Skill shortages	In many regions, there is a significant shortage of skilled professionals capable of developing, deploying, and maintaining advanced DTs such as AI, blockchain, and IoT. This shortage limits the ability of organizations to leverage these technologies effectively. To overcome this gap, there is a need for focused educational programs and training initiatives that equip the workforce with the necessary skills for sustainable digital transformation [447–449].
	Lack of digital literacy	The success of digital solutions hinges on the literacy and skills of the populations they serve. In many communities, particularly in developing nations, there is insufficient digital literacy. People may struggle to understand and utilize technologies like AI, cloud computing, or remote sensing. This prevents these innovations from having the desired impact on SDGs, especially in areas like education, healthcare, and governance [447–449].
High initial costs	Capital investment	Implementing DTs requires significant financial investment, particularly in low-income countries where the cost of infrastructure, hardware, and software is high. Technologies like IoT, AI, and blockchain require large-scale investment in both physical and human resources, which many organizations and governments find difficult to afford. This upfront cost barrier limits the ability to deploy digital solutions for achieving SDGs, particularly in sectors like agriculture, energy, and healthcare [96,244,450].
	Ongoing costs	Beyond initial investments, maintaining digital infrastructure incurs ongoing costs for software updates, cybersecurity, and data storage. The need for continuous investment in technology and human resources may strain the budgets of both public and private organizations. These ongoing costs are particularly challenging for smaller enterprises and governments in developing nations. This makes it harder for them to scale digital solutions for SDGs effectively [96,244,450].
Regulatory and policy barriers	Lack of supportive policies	In many countries, there is insufficient government support in the form of clear, comprehensive policies for the adoption of emerging DTs. The absence of a digital transformation strategy often leads to fragmented and inefficient efforts in areas such as healthcare, education, and energy. Policymakers must create regulatory frameworks that encourage innovation while ensuring the ethical use of DTs in achieving SDGs.
	Fragmented global policies	Regulatory challenges arise when countries implement differing laws regarding data privacy, technology use, and sustainability practices. This regulatory fragmentation complicates cross-border collaborations and the implementation of global digital solutions for SDGs. Harmonized regulations are crucial to foster international partnerships and ensure the smooth integration of DTs across borders [451,452].
Interoperability and integration	Integration challenges	The integration of new DTs with existing systems is a major challenge. Many technologies do not support seamless integration across various platforms, causing fragmentation. For example, IoT devices may use different communication protocols, while AI systems might not be compatible with legacy software. This lack of interoperability hinders the widespread use of these technologies to drive sustainable outcomes, especially in sectors like healthcare, transportation, and agriculture [466,467].
	Data silos	Despite the proliferation of DTs, data often remains fragmented across different sectors and systems. This results in "data silos," where information from different sources cannot be easily accessed or analyzed together [468,469]. For example, data collected from environmental monitoring systems might not be integrated with health or agricultural data. This prevents the comprehensive analysis and decision-making that could contribute to the SDGs, such as in climate action and responsible consumption.
Sustainability of digital solutions	Environmental impact of technology	Although DTs can drive significant progress towards SDGs, they can also have an environmental footprint. Technologies like blockchain, cloud computing, and AI require substantial energy consumption, which can contribute to carbon emissions if not powered by renewable energy sources. Additionally, the manufacturing of digital devices generates e-waste. This creates challenges for sustainable development. The environmental cost of these technologies must be carefully managed to avoid undermining sustainability efforts [470,471].

(*continued on next page*)

Table 3 (*continued*).

	E-Waste management	The rapid pace of technological innovation leads to an increasing volume of electronic waste (e-waste). E-waste, if not managed properly, poses significant environmental and health risks. Without effective recycling and disposal systems, DTs can become a source of environmental degradation. This limits their positive impact on SDGs. This issue calls for better policies and infrastructure to manage e-waste and ensure DTs remain sustainable throughout their life cycle [472–474].
Ethical concerns	Bias in AI algorithms	AI systems, when not properly designed or trained, can perpetuate or introduce biases. This leads to discrimination or unfair outcomes [475,476]. For example, algorithms used in hiring or criminal justice can inadvertently reinforce existing inequalities. This hinders the efforts to achieve SDGs such as reducing inequalities (SDG 10) [477]. Ensuring that AI systems are fair, transparent, and accountable is essential to avoid undermining sustainable development [475,476].
	Human rights issues	DTs, especially surveillance technologies, can raise serious ethical concerns regarding privacy and human rights. In some cases, digital solutions can be misused by governments or corporations to infringe on people's rights. Striking a balance between using DTs for social good and protecting individual freedoms is crucial to ensure that digital innovations contribute positively to the SDGs without violating fundamental human rights [478].
Cultural and social resistance	Resistance to change	Many societies, particularly in regions with traditional values, may resist the adoption of DTs. This resistance can be driven by fears of job displacement, a lack of trust in technology, or concerns about cultural values being undermined. Overcoming this resistance requires community engagement, education, and the development of technologies that align with local needs and cultural contexts, ensuring that they contribute to sustainable development in a way that is socially accepted [454,479].
	Social equity concerns	The benefits of DTs are often not equally distributed, with wealthier or more developed regions and groups disproportionately benefiting. This can exacerbate social inequalities, particularly if marginalized communities are excluded from the digital revolution. Ensuring equitable access to digital tools and resources is crucial to achieving SDGs related to poverty reduction, education, and social justice [480].
Long-term sustainability of projects	Project longevity	Many DT-driven projects face challenges in ensuring their long-term sustainability. A lack of ongoing financial support, insufficient maintenance, or the failure to continuously engage local communities often leads to the collapse of such initiatives. To ensure the longevity of digital SDG projects, it is essential to implement effective monitoring, maintenance, and funding strategies that can support these initiatives over time [447,481,482].
	Dependency on external support	Many SDG-related digital initiatives depend on foreign investments, external expertise, or aid from international organizations. This dependency can lead to a lack of local ownership and hinders the long-term sustainability of projects. Building local capacity and ensuring that technology adoption is self-sustaining is essential for the success and longevity of these digital solutions [483–485].

Fisk et al. [507] and Molina [508], who identified the digital divide as a key challenge in ensuring inclusive digital transformation (SDG 9, SDG 10). Additionally, the financial burden of implementing DTs remains a major constraint, particularly in low-income economies, where investment in digital infrastructure, cybersecurity, and capacity-building is often inadequate, aligning with Wessels et al. [509] and Rawindaran et al. [510], and Alibekova et al. [511] (SDG 8, SDG 9). Data privacy and cybersecurity concerns further complicate adoption, as regulatory inconsistencies and risks of data breaches undermine trust in digital systems, a concern reported by Toussaint et al. [512], who stress the need for standardized privacy frameworks (SDG 16). Furthermore, skill shortages remain a persistent challenge, with insufficient technical expertise in AI, blockchain, IoT, and cloud computing impeding effective implementation. The study findings align with James [513], Ameen and Gorman [514], and Nedungadi et al. [515], who emphasized that digital literacy deficits hinder the full potential of DTs, particularly in developing nations (SDG 4, SDG 9). Addressing these barriers requires collaborative efforts from policymakers, industry leaders, and civil society to ensure that digital transformation is inclusive, secure, and aligned with sustainability objectives, ultimately contributing to equitable and resilient technological progress (SDG 17).

Policy implications

The integration of DTs into sustainable development efforts requires a structured policy framework that ensures equitable access, responsible governance, and environmental sustainability. A comprehensive approach encompassing infrastructure development, regulatory alignment, capacity building, and cross-sector collaboration is essential to maximize the benefits of DTs while mitigating associated risks.

A foundational priority is the expansion of digital infrastructure to bridge the digital divide, particularly in developing regions. Investments in high-speed internet, cloud computing facilities, and reliable energy supply must be prioritized to support digital transformation. Public–private partnerships (PPPs) can play a key role in financing large-scale infrastructure projects, ensuring affordability and accessibility, and extending digital services to underserved communities. By enabling greater connectivity, governments can facilitate advancements in e-governance, digital education, smart agriculture, and remote healthcare services.

Regulatory frameworks must evolve to address emerging challenges related to data privacy, cybersecurity, and ethical AI deployment. The proliferation of AI, IoT, and blockchain technologies raises concerns about algorithmic bias, data security, and misuse of personal information. Establishing clear guidelines for ethical AI development, algorithmic transparency, and responsible data governance will be crucial in ensuring that digital innovations serve the public good while minimizing risks. Blockchain-enabled smart contracts can further enhance security and transparency in financial transactions, supply chain management, and public service delivery.

Collaboration between governments, private enterprises, and civil society organizations is essential to drive digital innovation for sustainability. Strategic incentives, such as tax benefits, research grants, and sustainability-linked subsidies, can encourage businesses to invest in green and socially responsible digital solutions. Moreover, partnerships between the public and private sectors should focus on advancing AI-driven climate monitoring, IoT-enabled smart cities, and big data analytics for policy optimization. To ensure inclusive benefits, digital literacy programs must be expanded, particularly targeting marginalized communities and small-scale enterprises, enabling them to leverage DTs for economic and social development.

A skilled workforce is central to the successful adoption of DTs. Governments must integrate science, technology, engineering, and mathematics (STEM) education, vocational training, and digital upskilling initiatives into national education policies. Special efforts should be made to promote gender diversity in technology fields, ensuring equal opportunities for women and underrepresented groups in the digital economy. Establishing community-driven digital learning centers will further support knowledge dissemination and skill development, particularly in remote and underserved areas.

As digital transformation accelerates, its environmental impact must be addressed through sustainable digital practices. Data centers, cloud

Table 4
General key strategies for the SDGs.

SDG	Key strategies
SDG 1	- Implement social safety nets and poverty reduction programs. - Provide access to affordable healthcare, education, and housing. - Promote job creation and skill development programs for vulnerable populations.
SDG 2	- Invest in sustainable agriculture and food systems. - Promote food security through subsidies and support for local farmers. - Improve food distribution and reduce food waste across the supply chain.
SDG 3	- Improve access to healthcare services and health insurance. - Strengthen public health infrastructure and services. - Promote health education and preventive care campaigns.
SDG 4	- Ensure universal access to quality education, including rural and marginalized areas. - Invest in teacher training and curriculum development. - Promote lifelong learning and technical skill development programs.
SDG 5	- Implement policies that promote equal pay for equal work. - Support laws and initiatives to prevent gender-based violence. - Promote women's leadership and participation in decision-making processes.
SDG 6	- Improve water and sanitation infrastructure in rural and urban areas. - Promote water conservation and efficient usage policies. - Ensure equitable access to clean and safe drinking water for all communities.
SDG 7	- Invest in renewable energy projects and infrastructure. - Provide incentives and subsidies for clean energy adoption. - Promote energy efficiency policies and green technologies.
SDG 8	- Foster a conducive environment for entrepreneurship and innovation. - Strengthen labor laws and ensure fair working conditions. - Invest in vocational training and skill development programs.
SDG 9	- Develop policies that promote innovation and support startups. - Invest in sustainable infrastructure and smart city projects. - Support research and development in green technologies and industries.
SDG 10	- Implement policies to reduce income inequality and ensure access to resources for marginalized groups. - Strengthen affirmative action and inclusive policies in employment, education, and healthcare. - Promote wealth redistribution through progressive tax systems.
SDG 11	- Promote urban planning policies focused on sustainability and resilience. - Implement green building codes and energy-efficient standards. - Improve public transportation and reduce urban sprawl.
SDG 12	- Enforce regulations that minimize waste and encourage recycling. - Promote sustainable production standards and eco-friendly consumer goods. - Foster a circular economy through regulatory frameworks and incentives.
SDG 13	- Implement climate adaptation and mitigation policies. - Invest in climate-resilient infrastructure and sustainable agriculture. - Commit to international climate agreements and set emission reduction targets.
SDG 14	- Regulate and reduce ocean pollution through stricter waste management and marine conservation laws. - Implement sustainable fisheries management and protect marine biodiversity. - Support international efforts for the preservation of ocean ecosystems.
SDG 15	- Enforce deforestation regulations and support reforestation projects. - Promote sustainable agriculture and conservation efforts. - Protect biodiversity through protected areas and national parks.
SDG 16	- Strengthen judicial systems and law enforcement. - Promote transparency, anti-corruption measures, and good governance. - Implement policies to uphold human rights and reduce violence.
SDG 17	- Foster international and regional cooperation to address global challenges. - Strengthen public–private partnerships to implement SDG-related projects. - Encourage multi-stakeholder collaboration for sustainable development initiatives.

computing, and AI-driven systems consume significant energy, while electronic waste poses growing sustainability concerns. Policies should encourage energy-efficient data management, green computing, and circular economy models for electronic devices. Incentives for low-carbon digital solutions, such as AI-powered climate tools, smart grids, and blockchain-based carbon credit systems, can drive sustainable innovation. Additionally, strict e-waste regulations must be enforced to promote product longevity, repairability, and recyclability.

Ensuring inclusive and equitable access to DTs is critical for reducing socio-economic disparities. Many low-income communities, small-scale farmers, and micro-enterprises remain excluded from AI-driven agricultural solutions, financial technology services, and digital healthcare advancements. Financial support in the form of subsidies, grants, and micro-financing initiatives can enable these groups to adopt DTs for economic growth. Furthermore, digital public services must be designed with accessibility in mind, incorporating local languages and user-friendly interfaces to enhance engagement across diverse populations.

Global cooperation is necessary to harmonize regulations, enhance cross-border data protection, and strengthen international cybersecurity efforts. Participation in multi-stakeholder forums, regional technology alliances, and global governance initiatives can facilitate the exchange of best practices, standardize AI ethics frameworks, and promote transparent data-sharing mechanisms. International partnerships can also enable funding and knowledge-sharing for sustainable digital innovations, ensuring that both developed and developing nations benefit from technological advancements.

Data-driven policymaking can enhance governance efficiency and optimize sustainability efforts. Governments should leverage big data analytics, AI-driven insights, and IoT-enabled monitoring systems for

Table 5
Contributions of various stakeholders toward achieving the United Nations SDGs.

SDG	Government contribution	Business contribution	Civil society contribution	Individual contribution
SDG 1	- Enforce social safety nets and welfare programs. - Create policies that promote inclusive growth.	- Implement fair wages and employee benefits. - Support local businesses and communities.	- Advocate for poverty reduction programs. - Provide support to low-income communities.	- Donate to charities. - Volunteer in local poverty reduction initiatives.
SDG 2	- Implement food security policies and agricultural subsidies. - Invest in sustainable farming practices.	- Promote sustainable food production and reduce food waste. - Provide access to nutritious food.	- Promote community-based agriculture. - Organize food banks and relief efforts.	- Reduce food waste at home. - Support local farmers and food banks.
SDG 3	- Strengthen healthcare infrastructure and systems. - Ensure universal health coverage.	- Provide employee health benefits. - Develop and market health products and services.	- Advocate for health policies and awareness programs. - Organize health camps and community health initiatives.	- Participate in health programs. - Adopt healthy lifestyles and practices.
SDG 4	- Invest in educational infrastructure and teacher training. - Ensure equal access to education for all.	- Provide training and development opportunities for employees. - Support educational programs and scholarships.	- Volunteer in education initiatives. - Raise awareness of the importance of quality education.	- Support education-focused NGOs. - Engage in lifelong learning and skill development.
SDG 5	- Enforce gender equality policies and laws. - Promote equal representation in leadership.	- Implement gender-inclusive hiring and pay policies. - Support women's empowerment initiatives.	- Advocate for gender equality and women's rights. - Raise awareness about gender-based violence.	- Support women-owned businesses. - Challenge gender stereotypes and biases.
SDG 6	- Ensure equitable access to clean water and sanitation. - Implement water management policies.	- Implement water-saving technologies. - Reduce water consumption and pollution.	- Organize water conservation campaigns. - Promote sanitation education and awareness.	- Conserve water at home. - Participate in community water management programs.
SDG 7	- Invest in renewable energy infrastructure. - Provide incentives for clean energy adoption.	- Transition to renewable energy sources. - Innovate in energy-efficient products and services.	- Advocate for clean energy policies. - Organize community renewable energy projects.	- Use energy-efficient appliances. - Adopt renewable energy options like solar panels.
SDG 8	- Create policies that promote decent work and economic growth. - Provide incentives for job creation.	- Ensure fair working conditions and wages. - Invest in innovation and technology for growth.	- Support fair trade and ethical businesses. - Raise awareness of labor rights.	- Support local businesses. - Advocate for workers' rights and fair wages.
SDG 9	- Invest in infrastructure and innovation. - Promote research and development in sustainable technologies.	- Develop sustainable products and services. - Invest in green technologies and innovation.	- Encourage the adoption of sustainable practices. - Participate in innovation challenges and networks.	- Adopt sustainable technologies in daily life. - Support innovative start-ups and businesses.
SDG 10	- Implement policies that promote equality and inclusion. - Ensure equal access to education, healthcare, and employment.	- Promote diversity and inclusion in the workplace. - Support social responsibility initiatives.	- Advocate for policies that reduce inequality. - Organize community-based inclusion programs.	- Volunteer for social equality causes. - Support equal opportunity businesses.
SDG 11	- Promote urban planning focused on sustainability. - Improve public transportation and green spaces.	- Develop sustainable infrastructure and green buildings. - Promote energy-efficient urban solutions.	- Advocate for sustainable urban development. - Organize local green initiatives.	- Reduce personal carbon footprint. - Participate in local urban sustainability projects.
SDG 12	- Enforce regulations to reduce waste and promote recycling. - Promote sustainable procurement policies.	- Adopt sustainable production methods. - Reduce packaging waste and promote recycling.	- Promote responsible consumption and waste reduction. - Support sustainable products and services.	- Reduce, reuse, and recycle. - Make sustainable purchasing decisions.
SDG 13	- Implement climate policies and emission reduction targets. - Invest in climate change adaptation and mitigation projects.	- Reduce carbon emissions through sustainable practices. - Develop green products and services.	- Advocate for climate change policies. - Organize climate action initiatives and protests.	- Reduce personal carbon footprint. - Advocate for climate action and sustainable practices.

(*continued on next page*)

real-time decision-making. Smart urban planning, climate change adaptation, and disaster response mechanisms can be improved through predictive analytics and digital simulations. Additionally, blockchain-based financial management tools can enhance transparency in public

Table 5 (*continued*).

SDG 14	- Enforce regulations to protect marine life and reduce ocean pollution. - Promote sustainable fisheries management.	- Reduce plastic usage and support marine conservation projects. - Adopt sustainable sourcing for seafood products.	- Participate in beach clean-ups. - Advocate for ocean conservation and sustainable fishing.	- Reduce plastic waste. - Support marine conservation efforts.
SDG 15	- Protect natural habitats and biodiversity. - Promote sustainable land use and forest management.	- Support sustainable agricultural practices. - Promote biodiversity-friendly business practices.	- Advocate for the conservation of natural habitats. - Participate in tree planting and reforestation projects.	- Reduce deforestation and support sustainable land use. - Participate in conservation activities.
SDG 16	- Strengthen judicial systems and anti-corruption measures. - Promote transparent and accountable governance.	- Ensure business practices align with ethical standards. - Support human rights and anti-corruption efforts.	- Advocate for justice and accountability. - Monitor and report on human rights violations.	- Promote peace and justice in daily life. - Advocate for strong democratic institutions.
SDG 17	- Foster international and multi-stakeholder partnerships. - Support global collaboration on SDGs.	- Collaborate with other businesses to achieve SDG targets. - Share knowledge and resources for sustainable development.	- Engage in global partnerships for sustainable development. - Mobilize resources for SDG-related initiatives.	- Participate in local and global SDG projects. - Support partnerships that align with SDGs.

spending, ensuring that sustainability-linked investments are allocated efficiently and responsibly.

A holistic policy approach that integrates digital infrastructure expansion, regulatory alignment, ethical AI deployment, workforce development, environmental responsibility, and global cooperation is essential for leveraging DTs in sustainable development. Governments must prioritize investments in digital literacy, establish supportive regulatory mechanisms, and foster cross-sector partnerships to drive responsible and inclusive digital adoption. Businesses must incorporate sustainability into their digital strategies, developing scalable solutions that align with environmental and social objectives. Civil society organizations must advocate for the equitable distribution of digital benefits, monitor the societal impacts of technology adoption, and ensure accountability in governance. Individuals, as key stakeholders, must actively engage in digital literacy programs, adopt sustainable digital practices, and participate in decision-making processes that shape the ethical and responsible use of technology. Through a collaborative effort, DTs can be harnessed effectively to accelerate progress toward global sustainability goals.

Theoretical implications

The theoretical implications of this review on the role of DTs in achieving the SDGs are far-reaching, particularly in the fields of sustainable development, technology adoption, and innovation management. The study expands existing theoretical frameworks by integrating DTs into the discourse of sustainable development, emphasizing their dual role in both environmental and socio-economic sustainability. Traditional sustainability models, which have often focused on ecological, economic, and social factors in isolation, are now increasingly incorporating digital tools as enablers that enhance efficiency, transparency, and inclusivity. This integration suggests that a more holistic view of sustainability must include technological infrastructure and digital literacy as foundational elements for achieving the SDGs.

Furthermore, the paper challenges and refines the technology acceptance model and diffusion of innovations theory by highlighting the barriers to technology adoption, such as the digital divide, lack of technical skills, and high costs, which are particularly relevant in low-income and developing regions. The study also underscores the importance of contextualizing DT adoption within the constraints and opportunities presented by the local socio-economic environment. This necessitates a shift from generic, one-size-fits-all models of technology adoption to more tailored, context-specific approaches.

Additionally, the review calls for the expansion of the Resource-Based View (RBV) theory, incorporating DTs as strategic resources that organizations can leverage to improve their competitive advantage and contribute to the achievement of SDGs. Overall, the study advances the theoretical understanding of the intersection between DTs and sustainable development and proposes new directions for academic inquiry, research, and policy development.

Practical implications

The practical implications of this review emphasize the significant role that DTs can play in achieving the SDGs. The review offers valuable insights for policymakers, businesses, and development organizations. First, the review highlights the need for targeted investments in digital infrastructure, particularly in developing regions where access to technologies is limited. Governments must prioritize policies that reduce the digital divide, such as (i) improving internet connectivity, (ii) providing affordable digital devices, and (iii) investing in digital literacy programs. Ensuring equal access to technology is crucial for maximizing its potential to contribute to SDGs related to poverty reduction, education, and gender equality.

For businesses, the review underscores the importance of integrating DTs into their operational models to enhance sustainability. Technologies like AI, IoT, and Blockchain can improve efficiency, transparency, and resource management, thereby contributing to sustainable business practices. Organizations must consider adopting these technologies to optimize their supply chains, reduce waste, and make data-driven decisions that align with sustainable goals. However, businesses must also be mindful of the high costs associated with adopting these technologies, which may require strategic partnerships, subsidies, or government support to ensure affordability and scalability.

Development organizations and NGOs can (i) use digital tools to improve service delivery in sectors such as healthcare, agriculture, and energy and (ii) ensure more inclusive and equitable access to essential services. By utilizing big data analytics and remote sensing, these organizations can enhance their monitoring, evaluation, and reporting capabilities. This leads to better resource allocation and program outcomes. Lastly, the review suggests that organizations in both the public and private sectors need to invest in capacity-building initiatives, developing the technical expertise required to leverage these technologies effectively.

10. Discussion

The relationship between DTs and sustainable development is grounded in theories and models that emphasize systemic change and innovation. The triple bottom line framework, focusing on balancing social, economic, and environmental sustainability, aligns with DTs' potential to deliver integrated solutions to global challenges. Theories like diffusion of innovations explain how DTs spread across communities and sectors and foster widespread adoption for sustainable outcomes. The technology-organization-environment framework highlights the interplay between technology adoption, organizational capabilities, and environmental contexts. Furthermore, system thinking emphasizes the interconnected nature of SDGs, where DTs enable holistic optimization. These models collectively illustrate the capability of DTs to bridge gaps across various disciplines and sectors and promote integrated strategies to achieve the SDGs.

Frameworks for assessing the impact of DTs on sustainable development continue to evolve. The sustainability assessment of the DTs framework evaluates metrics such as energy efficiency, accessibility, and environmental impact to ensure alignment with SDGs. The digital impact assessment model focuses on identifying positive and negative implications of technology adoption in specific SDG contexts, including SDG 5 and SDG 11. Additionally, life cycle assessment evaluates the cradle-to-grave impacts of digital solutions, particularly regarding resource use and carbon emissions. These frameworks emphasize the importance of data-driven evaluations to maximize benefits while mitigating risks and necessitating continuous refinement to address emerging challenges and improve precision.

DTs significantly advance the SDGs by improving efficiency, optimizing resource use, and expanding access to essential services. For instance, AI and big data analytics are transformative in advancing SDGs by enabling predictive capabilities and optimizing resources across various sectors. In healthcare (SDG 3), these technologies analyze vast health datasets to identify patterns, predict disease outbreaks, and support personalized treatments. This enhances decision-making and improves patient outcomes. In agriculture (SDG 2), AI-driven insights optimize crop yields, resource allocation, and pest management. This ensures more sustainable farming practices. Moreover, AI contributes significantly to addressing climate challenges (SDG 13) by facilitating advanced climate modeling and resource management. This equips the stakeholders with tools to mitigate environmental risks effectively.

The IoT has revolutionized environmental monitoring and infrastructure management and addresses sustainability across various sectors. For energy and the environment (SDG 7 and SDG 13), IoT enables real-time tracking of energy usage, air quality, and water levels. This ensures efficient resource management and rapid responses to crises. In urban development (SDG 11), IoT (a) enhances smart city infrastructure by optimizing waste management, energy consumption, and transportation systems and (b) promotes sustainable urban growth. Additionally, IoT has improved healthcare delivery (SDG 3) through innovations like wearable devices and remote health monitoring. This makes healthcare more accessible and efficient.

Blockchain technology fosters transparency and accountability, which are critical for achieving SDGs. In supply chain management (SDG 12), blockchain creates immutable transaction records. This enables traceability and promotes ethical consumption by providing clear insights into product origins, manufacturing processes, and environmental impacts. Additionally, blockchain supports fair governance and financial inclusion (SDG 10 and SDG 16) by ensuring ethical trade practices, eliminating corruption, and empowering marginalized communities with equitable access to financial systems.

Cloud computing supports sustainable development by providing scalable and accessible platforms for collaboration, innovation, and data management. It drives industry growth and infrastructure development (SDG 9) by enabling enhanced data storage, real-time analysis, and seamless teamwork across geographies. Furthermore, cloud technologies strengthen global partnerships (SDG 17) by facilitating shared research and fostering cross-border cooperation. This creates opportunities for collective problem-solving and data-driven decision-making to address pressing global challenges.

Remote sensing and GIS technologies are indispensable tools for environmental monitoring, disaster management, and biodiversity conservation. They significantly contribute to climate action and biodiversity protection (SDG 13, SDG 14, and SDG 15) by providing real-time data on critical environmental indicators, such as deforestation, marine pollution, and climate patterns. These insights enable precise disaster management, effective land-use planning, and targeted conservation efforts. Moreover, the integration of GIS with predictive analytics enhances disaster preparedness and ensures swift and evidence-based responses to environmental challenges.

Beyond direct contributions, DTs play an indirect role in advancing SDGs by fostering SDG 8 and SDG 10. Digital inclusion initiatives and e-governance tools improve service access. This bridges gaps in underserved communities. However, these contributions must align with ethical and inclusive principles to prevent unintended consequences, such as widening digital divides or increasing resource consumption. Ensuring equity in access and implementation is critical to fully realizing the potential of DTs for sustainable development.

Despite their promise, DTs face challenges that hinder their effective integration into sustainable development strategies. Digital divides persist, particularly in low-income and remote regions, where limited infrastructure, affordability, and low digital literacy constrain access to technology. High energy consumption associated with blockchain and data centers complicates efforts to meet climate-related SDGs. Regulatory hurdles, including inconsistent standards and a lack of global governance, impede cross-border collaboration. Ethical concerns, such as data privacy, algorithmic bias, and cybersecurity risks, further undermine public trust. Without standardized frameworks to evaluate long-term impacts, addressing these challenges remains complex.

Achieving the SDGs through DTs requires a collaborative, multifaceted approach involving innovation, capacity building, and inclusive policies. *Governments* play a central role by investing in digital infrastructure, enhancing digital literacy, and enacting supportive regulations to ensure equitable access to these technologies. By fostering international collaboration, governments can establish global standards addressing critical concerns such as data privacy, interoperability, and ethical use of DTs. *Businesses* are key drivers of sustainable innovation, embedding energy-efficient practices and circular economy models into their strategies while developing scalable digital solutions. Through public–private partnerships, they can accelerate the deployment of technologies that address complex sustainability challenges. *Civil society* organizations contribute by raising awareness, advocating for marginalized communities, and monitoring the social and environmental impacts of digital technology adoption. Their role ensures inclusivity and accountability in how DTs are deployed and leveraged. Meanwhile, *individuals* are empowered through digital literacy to adopt sustainable digital practices, minimize e-waste, and participate in decision-making processes that influence the responsible use of technology.

The collective efforts of these stakeholders must align with shared goals to maximize the impact of DTs on the SDGs. Promoting public–private partnerships and fostering collaboration among nations and industries ensures the effective deployment of innovative solutions while addressing underrepresented areas of the SDGs. Continuous research and development are vital for identifying emerging technologies with the potential to fill gaps and address challenges not currently met by existing DTs.

Additionally, businesses and governments must prioritize policies and strategies that align digital innovation with sustainability goals, ensuring that growth is both inclusive and environmentally responsible. Civil society's engagement ensures that the benefits of digital transformation reach all sections of society, addressing inequalities and promoting ethical practices. With clear strategies, inclusive policies, and active participation from all stakeholders, DTs can effectively drive progress across the SDGs. This strengthens resilience, fosters equity, and promotes global sustainability. The alignment of these efforts with global standards and partnerships enhances the collective capacity to address challenges, optimize resource use, and build a sustainable future for all.

The novelty of the study lies in its comprehensive and multidisciplinary approach to analyzing the role of DTs in advancing the SDGs.

Unlike previous research that tends to focus on individual technologies or specific SDGs, the study presents a holistic perspective by integrating multiple DTs, including AI, blockchain, the IoT, big data analytics, cloud computing, 5G networks, remote sensing, and GIS. This synthesis provides a deeper understanding of how these technologies interact and complement each other to drive sustainability across various sectors.

One of the distinguishing features of this research is its exploration of both the opportunities and challenges associated with DT adoption for sustainable development. While most existing literature emphasizes the benefits of digital transformation, this study identifies key barriers such as the digital divide, high implementation costs, regulatory constraints, cybersecurity risks, and skill shortages. By addressing these challenges, this study offers a more realistic and pragmatic perspective on the adoption and scalability of DTs in different economic and social contexts.

Furthermore, this research extends beyond technological discussions by bridging the gap between digital advancements and policy needs. It highlights the necessity of targeted policy interventions, including investments in digital infrastructure, ethical AI governance, public–private partnerships, and sustainability-driven technology policies. These recommendations provide actionable insights for governments, businesses, and civil society to develop inclusive and effective digital transformation strategies that accelerate progress toward achieving the SDGs.

A key contribution of this study is its comparative analysis with existing literature, demonstrating how emerging digital solutions are evolving to address sustainability challenges. While prior research has established AI's role in precision agriculture and healthcare, this study expands its scope to include AI's applications in climate action, urban planning, and predictive analytics for disaster management. Similarly, blockchain has been widely recognized for enhancing financial transactions and supply chain transparency, but this research highlights its emerging role in ethical governance, anti-corruption efforts, and decentralized decision-making models. By identifying these new applications and trends, this study fills critical gaps in the literature and offers a forward-looking perspective on the evolving role of DTs in sustainable development.

Additionally, the study provides an integrated framework that considers technological, economic, environmental, and policy dimensions, offering a well-rounded analysis of digital transformation for sustainability. It examines how DTs are being implemented at different scales, identifies critical gaps in adoption, and proposes strategic solutions to maximize their impact. The interdisciplinary nature of the study ensures that its findings are relevant to a broad range of stakeholders, including policymakers, researchers, business leaders, and international organizations.

By moving beyond fragmented discussions and providing a novel theoretical synthesis, this study serves as a valuable resource for those seeking to drive responsible and impactful digital transformation. It underscores the importance of aligning technological advancements with regulatory frameworks, ethical considerations, and inclusive access to digital resources. Ultimately, the study insights contribute to shaping a more equitable, resilient, and technology-driven approach to sustainable development, making a significant impact on both academic discourse and real-world policy implementation

11. Limitations of the study

The study primarily relies on published literature, which may not fully capture the latest industry advancements in DTs. While offering a broad analysis of DTs' contribution to sustainable development, it does not account for regional variations in adoption and impact, necessitating further empirical validation. Additionally, the study lacks quantitative assessments of DTs' effectiveness, which future research could address through empirical modeling to provide more data-driven insights into their sustainability outcomes.

12. Future research directions

As DTs continue to evolve, their role in advancing the SDGs will become increasingly vital. Advancing sustainable development requires (i) harnessing emerging technologies and fostering robust digital ecosystems. Innovations like quantum computing and 6G networks can address complex challenges, while AI, IoT, and big data analytics integration enable intelligent systems for resource optimization and climate action. Blockchain technology enhances transparency and inclusivity, driving ethical and efficient resource management. Sustainability-focused technologies such as green computing and remote sensing are critical for environmental conservation. Urban innovations, including digital twins, support sustainable city planning and cross-SDG synergies. Equally vital is education and awareness through digital literacy, AR, and VR, empowering communities and fostering skill development for a sustainable future. Here are the various research directions associated with these areas that require further exploration:

1. Emerging technologies and foundational digital ecosystems

   - Study quantum computing for revolutionizing healthcare (SDG 3) and climate modeling (SDG 13) and explore 6G networks for bridging the digital divide (SDG 10).
   - Assess the role of 5G and future networks in enabling advanced digital solutions like autonomous systems, precision healthcare, and smart cities (SDG 9 and SDG 11).
   - Investigate tailored digital solutions for local and regional contexts, ensuring cultural adaptability and scalability for SDG 1 and SDG 8.

2. AI, IoT, and big data analytics for sustainable systems

   - Create intelligent systems for resource optimization, such as smart grids, precision agriculture, and sustainable urban planning, contributing to SDG 7 and SDG 11.
   - Explore IoT for smart water systems, precision agriculture, and biodiversity monitoring (SDG 12 and SDG 15).
   - Leverage big data analytics for predictive modeling and evidence-based policymaking to address SDG challenges (SDG 13).
   - Optimize data collection and analysis through the integration of AI and machine learning to improve SDG monitoring and evaluation precision.

3. Blockchain technology for transparency and inclusivity

   - Study blockchain's role in ensuring transparency and traceability in supply chains, governance, and climate finance.
   - Blockchain innovations for transparent and inclusive systems, i.e., combine blockchain with IoT for resource management (SDG 12) and utilize decentralized finance (DeFi) for financial inclusion (SDG 1 and SDG 10).

4. Sustainability and environmental conservation technologies

   - Integrate remote sensing, AI, and analytics for climate monitoring, disaster response, and biodiversity conservation (SDG 13 and SDG 15).
   - Develop low-carbon or green computing solutions and sustainable ICT practices to reduce environmental impacts (SDG 12 and SDG 13).

5. Innovations in urban sustainability and design

   - Use digital twins for simulating and optimizing sustainable urban development (SDG 11).

- Integration of DTs for cross-SDG synergies, i.e., analyze interactions among DTs to identify synergies and trade-offs across interconnected SDGs.

6. Education, awareness, and skill development

- Research strategies to improve digital literacy programs among marginalized communities to enhance inclusivity (SDG 4 and SDG 10).
- Leverage augmented reality (AR) and virtual reality (VR) for education and immersive campaigns to promote sustainability literacy and climate action.
- Cross-sector collaborations for scalable solutions, i.e., explore public–private-academic partnerships for workforce skill development in advanced technologies (SDG 4 and SDG 8).

7. Education, awareness, and skill development

- Policy and governance for ethical technology use, i.e., develop frameworks for algorithmic fairness, data privacy, and cybersecurity that align innovations with sustainability and equity (SDG 10 and SDG 13).
- Cybersecurity in digital SDG solutions, i.e., address challenges in ensuring secure and resilient DTs for SDGs.

CRediT authorship contribution statement

Dharmendra Hariyani: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Poonam Hariyani:** Writing – review & editing, Visualization, Validation, Conceptualization. **Sanjeev Mishra:** Writing – review & editing, Visualization, Validation, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] A.A. Kutty, G.M. Abdella, M. Kucukvar, N.C. Onat, M. Bulu, A system thinking approach for harmonizing smart and sustainable city initiatives with united nations sustainable development goals, Sustain. Dev. 28 (2020) 1347–1365, http://dx.doi.org/10.1002/sd.2088.

[2] L.M. Fonseca, J.P. Domingues, A.M. Dima, Mapping the sustainable development goals relationships, Sustainability 12 (2020) 3359, http://dx.doi.org/10.3390/su12083359.

[3] J. Gupta, C. Vegelin, Sustainable development goals and inclusive development, Int. Env. Agreements Polit. Law Econ. 16 (2016) 433–448, http://dx.doi.org/10.1007/s10784-016-9323-z.

[4] A.E. Caglar, S.B. Avci, M. Daştan, M.A. Destek, Investigation of the effect of natural resource dependence on environmental sustainability under the novel load capacity curve hypothesis, Int. J. Sustain. Dev. World Ecol. 31 (2024) 431–446, http://dx.doi.org/10.1080/13504509.2023.2296495.

[5] A. Opoku, Biodiversity and the built environment: Implications for the sustainable development goals (SDGs), Resour. Conserv. Recycl. 141 (2019) 1–7, http://dx.doi.org/10.1016/j.resconrec.2018.10.011.

[6] G. Eweje, A. Sajjad, S.D. Nath, K. Kobayashi, Multi-stakeholder partnerships: a catalyst to achieve sustainable development goals, Mark. Intell. Plan. 39 (2021) 186–212, http://dx.doi.org/10.1108/MIP-04-2020-0135.

[7] I. Montiel, A. Cuervo-Cazurra, J. Park, R. Antolín-López, B.W. Husted, Implementing the United Nations' Sustainable Development Goals in international business, J. Int. Bus. Stud. 52 (2021) 999–1030, http://dx.doi.org/10.1057/s41267-021-00445-y.

[8] N.P. Hariram, K.B. Mekha, V. Suganthan, K. Sudhakar, Sustainalism: An integrated socio-economic-environmental model to address sustainable development and sustainability, Sustainability 15 (2023) 10682, http://dx.doi.org/10.3390/su151310682.

[9] C.A. Adams, The sustainable development goals, integrated thinking and the integrated report, in: IIRC ICAS, 2017, pp. 1–52.

[10] W. Leal Filho, T. Dibbern, M.A. Pimenta Dinis, E. Coggo Cristofoletti, M.F. Mbah, A. Mishra, A. Clarke, N. Samuel, J. Castillo Apraiz, I. Rimi Abubakar, Y.A. Aina, The added value of partnerships in implementing the UN sustainable development goals, J. Clean. Prod. 438 (2024) 140794, http://dx.doi.org/10.1016/j.jclepro.2024.140794.

[11] L. Mariani, B. Trivellato, M. Martini, E. Marafioti, Achieving sustainable development goals through collaborative innovation: Evidence from four European initiatives, J. Bus. Ethics 180 (2022) 1075–1095, http://dx.doi.org/10.1007/s10551-022-05193-z.

[12] L. Stott, D.F. Murphy, An inclusive approach to partnerships for the sdgs: using a relationship lens to explore the potential for transformational collaboration, Sustainability 12 (2020) 7905, http://dx.doi.org/10.3390/su12197905.

[13] A. Florini, M. Pauli, Collaborative governance for the sustainable development goals, Asia Pac. Policy Stud. 5 (2018) 583–598, http://dx.doi.org/10.1002/app5.252.

[14] N.K. Hanna (Ed.), Mastering digital transformation, in: Mastering Digital Transformation (Innovation, Technology, and Education for Growth), Leeds, 2016, pp. i–xxvi, http://dx.doi.org/10.1108/978-1-78560-465-220151009.

[15] N.L. Rane, Ö. Kaya, J. Rane, Integrating internet of things, blockchain, and artificial intelligence techniques for intelligent industry solutions, in: Artificial Intelligence, Machine Learning, and Deep Learning for Sustainable Industry 5.0, Deep Science Publishing, 2024, http://dx.doi.org/10.70593/978-81-981271-8-1_6.

[16] S. Singh, P.K. Sharma, B. Yoon, M. Shojafar, G.H. Cho, I.-H. Ra, Convergence of blockchain and artificial intelligence in IoT network for the sustainable smart city, Sustain. Cities Soc. 63 (2020) 102364, http://dx.doi.org/10.1016/j.scs.2020.102364.

[17] K. Rabah, Convergence of AI, IoT, big data and blockchain: A review, Lake Inst. J. 1 (2018) 1–18.

[18] A.E. Caglar, M.A. Destek, M. Manga, Analyzing the load capacity curve hypothesis for the Turkiye: A perspective for the sustainable environment, J. Clean. Prod. 444 (2024) 141232, http://dx.doi.org/10.1016/j.jclepro.2024.141232.

[19] N. Rane, Enhancing customer loyalty through Artificial Intelligence (AI), Internet of Things (IoT), and big data technologies: Improving customer satisfaction, engagement, relationship, and experience, SSRN Electron. J. (2023) http://dx.doi.org/10.2139/ssrn.4616051.

[20] F. Robards, M. Kang, T. Usherwood, L. Sanci, How marginalized young people access, engage with, and navigate health-care systems in the digital age: systematic review, J. Adolesc. Heal. 62 (2018) 365–381, http://dx.doi.org/10.1016/j.jadohealth.2017.10.018.

[21] A. Powell, A. Bryne, D. Dailey, The essential internet: Digital exclusion in low-income American communities, Policy & Internet 2 (2010) 161–192, http://dx.doi.org/10.2202/1944-2866.1058.

[22] A. Siano, L. Raimi, M. Palazzo, M.C. Panait, Mobile banking: An innovative solution for increasing financial inclusion in Sub-Saharan African countries: Evidence from Nigeria, Sustainability 12 (2020) 10130, http://dx.doi.org/10.3390/su122310130.

[23] S.A. Ouma, T.M. Odongo, M. Were, Mobile financial services and financial inclusion: Is it a boon for savings mobilization? Rev. Dev. Financ. 7 (2017) 29–35, http://dx.doi.org/10.1016/j.rdf.2017.01.001.

[24] S. Bansal, Perspective of technology in achieving financial inclusion in rural India, Procedia Econ. Financ. 11 (2014) 472–480, http://dx.doi.org/10.1016/S2212-5671(14)00213-5.

[25] A.S. Albahri, Y.L. Khaleel, M.A. Habeeb, R.D. Ismael, Q.A. Hameed, M. Deveci, R.Z. Homod, O.S. Albahri, A.H. Alamoodi, L. Alzubaidi, A systematic review of trustworthy artificial intelligence applications in natural disasters, Comput. Electr. Eng. 118 (2024) 109409, http://dx.doi.org/10.1016/j.compeleceng.2024.109409.

[26] S. Gupta, S. Modgil, A. Kumar, U. Sivarajah, Z. Irani, Artificial intelligence and cloud-based Collaborative Platforms for Managing Disaster, extreme weather and emergency operations, Int. J. Prod. Econ. 254 (2022) 108642, http://dx.doi.org/10.1016/j.ijpe.2022.108642.

[27] A.E. Caglar, S.B. Avci, N. Gökçe, M.A. Destek, A sustainable study of competitive industrial performance amidst environmental quality: New insight from novel Fourier perspective, J. Env. Manag. 366 (2024) 121843, http://dx.doi.org/10.1016/j.jenvman.2024.121843.

[28] D. Hariyani, P. Hariyani, S. Mishra, M. Kumar Sharma, Leveraging digital technologies for advancing circular economy practices and enhancing life cycle analysis: A systematic literature review, Waste Manag. Bull. 2 (2024) 69–83, http://dx.doi.org/10.1016/j.wmb.2024.06.007.

[29] M. Rusch, J. Schöggl, R.J. Baumgartner, Application of digital technologies for sustainable product management in a circular economy: A review, Bus. Strat. Env. 32 (2023) 1159–1174, http://dx.doi.org/10.1002/bse.3099.

[30] S. Nižetić, N. Djilali, A. Papadopoulos, J.J.P.C. Rodrigues, Smart technologies for promotion of energy efficiency, Util. Sustain. Resour. Waste Manag. J. Clean. Prod. 231 (2019) 565–591, http://dx.doi.org/10.1016/j.jclepro.2019.04.397.

[31] N. Hanna, A role for the state in the digital age, J. Innov. Entrep. 7 (2018) 5, http://dx.doi.org/10.1186/s13731-018-0086-3.

[32] D. Moher, A. Liberati, J. Tetzlaff, D.G. Altman, Preferred reporting items for systematic reviews and meta-analyses: The PRISMA statement, PLoS Med. 6 (2009) e1000097, http://dx.doi.org/10.1371/journal.pmed.1000097.

[33] G.H. Walker, N.A. Stanton, P.M. Salmon, D.P. Jenkins, A review of sociotechnical systems theory: a classic concept for new command and control paradigms, Theor. Issues Erg. Sci. 9 (2008) 479–499, http://dx.doi.org/10.1080/14639220701635470.

[34] C.W. Clegg, Sociotechnical principles for system design, Appl. Ergon. 31 (2000) 463–477, http://dx.doi.org/10.1016/S0003-6870(00)00009-0.

[35] C. Münch, E. Marx, L. Benz, E. Hartmann, M. Matzner, Capabilities of digital servitization: Evidence from the socio-technical systems theory, Technol. Forecast. Soc. Change 176 (2022) 121361, http://dx.doi.org/10.1016/j.techfore.2021.121361.

[36] M. Sony, S. Naik, Industry 4.0 integration with socio-technical systems theory: A systematic review and proposed theoretical model, Technol. Soc. 61 (2020) 101248, http://dx.doi.org/10.1016/j.techsoc.2020.101248.

[37] A.T. Chatfield, C.G. Reddick, A framework for internet of things-enabled smart government: A case of IoT cybersecurity policies and use cases in U.S. federal government, Gov. Inf. Q. 36 (2019) 346–357, http://dx.doi.org/10.1016/j.giq.2018.09.007.

[38] O.P. Agboola, M. Tunay, Urban resilience in the digital age: The influence of information-communication technology for sustainability, J. Clean. Prod. 428 (2023) 139304, http://dx.doi.org/10.1016/j.jclepro.2023.139304.

[39] D. Hariyani, S. Mishra, Organizational enablers for sustainable manufacturing and industrial ecology, Clean. Eng. Technol. 6 (2022) 100375, http://dx.doi.org/10.1016/j.clet.2021.100375.

[40] A. Upadhyay, S. Mukhuty, V. Kumar, Y. Kazancoglu, Blockchain technology and the circular economy: Implications for sustainability and social responsibility, J. Clean. Prod. 293 (2021) 126130, http://dx.doi.org/10.1016/j.jclepro.2021.126130.

[41] V.G. Venkatesh, K. Kang, B. Wang, R.Y. Zhong, A. Zhang, System architecture for blockchain based transparency of supply chain social sustainability, Robot. Comput.-Integr. Manuf. 63 (2020) 101896, http://dx.doi.org/10.1016/j.rcim.2019.101896.

[42] G. Xu, G. Hou, J. Zhang, Digital sustainable entrepreneurship: A digital capability perspective through digital innovation orientation for social and environmental value creation, Sustainability 14 (2022) 11222, http://dx.doi.org/10.3390/su141811222.

[43] H. Birkel, J.M. Müller, Potentials of industry 4.0 for supply chain management within the triple bottom line of sustainability – A systematic literature review, J. Clean. Prod. 289 (2021) 125612, http://dx.doi.org/10.1016/j.jclepro.2020.125612.

[44] G. Berlingieri, S. Calligaris, C. Criscuolo, R. Verlhac, Laggard Firms, Technology Diffusion and Its Structural and Policy Determinants, OECD Science, Technology and Industry Policy Papers, 2020, http://dx.doi.org/10.1787/281bd7a9-en.

[45] S.F. Jahanmir, J. Cavadas, Factors affecting late adoption of digital innovations, J. Bus. Res. 88 (2018) 337–343, http://dx.doi.org/10.1016/j.jbusres.2018.01.058.

[46] C. Franceschinis, M. Thiene, R. Scarpa, J. Rose, M. Moretto, R. Cavalli, Adoption of renewable heating systems: An empirical test of the diffusion of innovation theory, Energy 125 (2017) 313–326, http://dx.doi.org/10.1016/j.energy.2017.02.060.

[47] K.S. Wolske, P.C. Stern, T. Dietz, Explaining interest in adopting residential solar photovoltaic systems in the United States: Toward an integration of behavioral theories, Energy Res. Soc. Sci. 25 (2017) 134–151, http://dx.doi.org/10.1016/j.erss.2016.12.023.

[48] G. George, A.M. McGahan, J. Prabhu, Innovation for inclusive growth: Towards a theoretical framework and a research agenda, J. Manag. Stud. 49 (2012) 661–683, http://dx.doi.org/10.1111/j.1467-6486.2012.01048.x.

[49] S. Barteit, D. Guzek, A. Jahn, T. Bärnighausen, M.M. Jorge, F. Neuhann, Evaluation of e-learning for medical education in low- and middle-income countries: A systematic review, Comput. Educ. 145 (2020) 103726, http://dx.doi.org/10.1016/j.compedu.2019.103726.

[50] N.N. Ndubuka, E. Rey-Marmonier, Capability approach for realising the sustainable development goals through responsible management education: The case of UK business school academics, Int. J. Manag. Educ. 17 (2019) 100319, http://dx.doi.org/10.1016/j.ijme.2019.100319.

[51] E. Maeckelberghe, K. Zdunek, S. Marceglia, B. Farsides, M. Rigby, The ethical challenges of personalized digital health, Front. Med. 10 (2023) http://dx.doi.org/10.3389/fmed.2023.1123863.

[52] S. Samarakoon, A justice and wellbeing centered framework for analysing energy poverty in the Global South, Ecol. Econ. 165 (2019) 106385, http://dx.doi.org/10.1016/j.ecolecon.2019.106385.

[53] A.E. Caglar, N. Gökçe, F. Şahin, Sustaining environment through municipal solid waste: evidence from European Union economies, Env. Sci. Pollut. Res. 31 (2023) 6040–6053, http://dx.doi.org/10.1007/s11356-023-31494-5.

[54] E. Kristoffersen, F. Blomsma, P. Mikalef, J. Li, The smart circular economy: A digital-enabled circular strategies framework for manufacturing companies, J. Bus. Res. 120 (2020) 241–261, http://dx.doi.org/10.1016/j.jbusres.2020.07.044.

[55] I. Sosunova, J. Porras, IoT-enabled smart waste management systems for smart cities: A systematic review, IEEE Access 10 (2022) 73326–73363, http://dx.doi.org/10.1109/ACCESS.2022.3188308.

[56] K.H. Yu, Y. Zhang, D. Li, C.E. Montenegro-Marin, P.M. Kumar, Environmental planning based on reduce, reuse, recycle and recover using artificial intelligence, Env. Impact Assess. Rev. 86 (2021) 106492, http://dx.doi.org/10.1016/j.eiar.2020.106492.

[57] J. Leng, G. Ruan, P. Jiang, K. Xu, Q. Liu, X. Zhou, C. Liu, Blockchain-empowered sustainable manufacturing and product lifecycle management in industry 4.0: A survey, Renew. Sustain. Energy Rev. 132 (2020) 110112, http://dx.doi.org/10.1016/j.rser.2020.110112.

[58] X.L. Liu, W.M. Wang, H. Guo, A.V. Barenji, Z. Li, G.Q. Huang, Industrial blockchain based framework for product lifecycle management in industry 4.0, Robot. Comput.-Integr. Manuf. 63 (2020) 101897, http://dx.doi.org/10.1016/j.rcim.2019.101897.

[59] Q. Liu, A.H. Trevisan, M. Yang, J. Mascarenhas, A framework of digital technologies for the circular economy: Digital functions and mechanisms, Bus. Strat. Env. 31 (2022) 2171–2192, http://dx.doi.org/10.1002/bse.3015.

[60] G. Bressanelli, F. Adrodegari, M. Perona, N. Saccani, Exploring how usage-focused business models enable circular economy through digital technologies, Sustainability 10 (2018) 639, http://dx.doi.org/10.3390/su10030639.

[61] R. Scherer, F. Siddiq, J. Tondeur, The technology acceptance model (TAM): A meta-analytic structural equation modeling approach to explaining teachers' adoption of digital technology in education, Comput. Educ. 128 (2019) 13–35, http://dx.doi.org/10.1016/j.compedu.2018.09.009.

[62] J.-C. Hong, M.-Y. Hwang, H.-F. Hsu, W.-T. Wong, M.-Y. Chen, Applying the technology acceptance model in a study of the factors affecting usage of the Taiwan digital archives system, Comput. Educ. 57 (2011) 2086–2094, http://dx.doi.org/10.1016/j.compedu.2011.04.011.

[63] K. Gbongli, Y. Xu, K.M. Amedjonekou, Extended technology acceptance model to predict mobile-based money acceptance and sustainability: A multi-analytical structural equation modeling and neural network approach, Sustainability 11 (2019) 3639, http://dx.doi.org/10.3390/su11133639.

[64] B. Chen, N. Sintov, Bridging the gap between sustainable technology adoption and protecting natural resources: Predicting intentions to adopt energy management technologies in California, Energy Res. Soc. Sci. 22 (2016) 210–223, http://dx.doi.org/10.1016/j.erss.2016.10.003.

[65] E.C. Makazhe, T.C. Maramura, The role of fourth industrial revolution (4IR) technologies in achieving sustainable development goals, in: S. Kot, B. Khalid, A.u. Haque (Eds.), Corporate Practices: Policies, Methodologies, and Insights in Organizational Management, Springer, Singapore, 2024, pp. 985–998, http://dx.doi.org/10.1007/978-981-97-0996-0_58.

[66] H.A. Alnemer, Determinants of digital banking adoption in the Kingdom of Saudi Arabia: A technology acceptance model approach, Digit. Bus. 2 (2022) 100037, http://dx.doi.org/10.1016/j.digbus.2022.100037.

[67] I. Park, D. Kim, J. Moon, S. Kim, Y. Kang, S. Bae, Searching for new technology acceptance model under social context: Analyzing the determinants of acceptance of intelligent information technology in digital transformation and implications for the requisites of digital sustainability, Sustainability 14 (2022) 579, http://dx.doi.org/10.3390/su14010579.

[68] S.A. Hendrawan, Afdhal Chatra, Nurul Iman, Soemarno Hidayatullah, Degdo Suprayitno, Digital transformation in MSMEs: Challenges and opportunities in technology management, J. Inf. Dan Teknol. (2024) 141–149, http://dx.doi.org/10.60083/jidt.v6i2.551.

[69] M. Jiang, F. Jia, L. Chen, X. Xing, Technology adoption in socially sustainable supply chain management: Towards an integrated conceptual framework, Technol. Forecast. Soc. Change 206 (2024) 123537, http://dx.doi.org/10.1016/j.techfore.2024.123537.

[70] V.U. Oguanobi, O.T. Joel, O.T. Oguanobi, Scalable business models for startups in renewable energy: Strategies for using GIS technology to enhance SME scaling, Eng. Sci. Technol. J. 5 (2024) 1571–1587, http://dx.doi.org/10.51594/estj.v5i5.1109.

[71] L. Lescrauwaet, H. Wagner, C. Yoon, S. Shukla, Adaptive legal frameworks and economic dynamics in emerging tech-nologies: Navigating the intersection for responsible innovation, Law Econ. 16 (2022) 202–220, http://dx.doi.org/10.35335/laweco.v16i3.61.

[72] F. Berkhout, J. Hertin, De-materialising and re-materialising: digital technologies and the environment, Futures 36 (2004) 903–920, http://dx.doi.org/10.1016/j.futures.2004.01.003.

[73] C. Certomà, Future scenarios of Digital Social Innovation in urban governance. A collective discussion on the socio-political implications in Ghent, Cities 122 (2022) 103542, http://dx.doi.org/10.1016/j.cities.2021.103542.

[74] M.C. Roco, Possibilities for global governance of converging technologies, J. Nanoparticle Res. 10 (2008) 11–29, http://dx.doi.org/10.1007/s11051-007-9269-8.

[75] R. Martínez-Peláez, A. Ochoa-Brust, S. Rivera, V.G. Félix, R. Ostos, H. Brito, R.A. Félix, L.J. Mena, Role of digital transformation for achieving sustainability: Mediated role of stakeholders, Key Capab. Technol. Sustain. 15 (2023) 11221, http://dx.doi.org/10.3390/su151411221.

[76] S. Yadav, A. Samadhiya, A. Kumar, A. Majumdar, J.A. Garza-Reyes, S. Luthra, Achieving the sustainable development goals through net zero emissions: Innovation-driven strategies for transitioning from incremental to radical lean, green and digital technologies, Resour. Conserv. Recycl. 197 (2023) 107094, http://dx.doi.org/10.1016/j.resconrec.2023.107094.
[77] G. Secundo, R. Schena, A. Russo, F. Schiavone, R. Shams, The impact of digital technologies on the achievement of the sustainable development goals: evidence from the agri-food sector, Total. Qual. Manag. Bus. Excel. (2022) 1–17, http://dx.doi.org/10.1080/14783363.2022.2065981.
[78] T. Papaioannou, D. Wield, J. Chataway, Knowledge ecologies and ecosystems? An empirically grounded reflection on recent developments in innovation systems theory, Env. Plan. C Gov. Policy 27 (2009) 319–339, http://dx.doi.org/10.1068/c0832.
[79] F.W. Geels, From sectoral systems of innovation to socio-technical systems: Insights about dynamics and change from sociology and institutional theory, Res. Policy 33 (2004) 897–920, http://dx.doi.org/10.1016/j.respol.2004.01.015.
[80] C. Edquist (Ed.), Systems of Innovation: Technologies, Institutions and Organizations, London, 1997.
[81] L. Satalkina, G. Steiner, Digital entrepreneurship and its role in innovation systems: A systematic literature review as a basis for future research avenues for sustainable transitions, Sustainability 12 (2020) 2764, http://dx.doi.org/10.3390/su12072764.
[82] B.N. Silva, M. Khan, K. Han, Towards sustainable smart cities: A review of trends, architectures, components, and open challenges in smart cities, Sustain. Cities Soc. 38 (2018) 697–713, http://dx.doi.org/10.1016/j.scs.2018.01.053.
[83] S. Zygiaris, Smart city reference model: Assisting planners to conceptualize the building of smart city innovation ecosystems, J. Knowl. Econ. 4 (2013) 217–231, http://dx.doi.org/10.1007/s13132-012-0089-4.
[84] K.A. Paskaleva, The smart city: A nexus for open innovation?, Intell. Build. Int. 3 (2011) 153–171, http://dx.doi.org/10.1080/17508975.2011.586672.
[85] H. Schaffers, N. Komninos, M. Pallot, B. Trousse, M. Nilsson, A. Oliveira, Smart cities and the future internet: Towards cooperation frameworks for open innovation, in: John Domingue, Alex Galis, Anastasius Gavras, Theodore Zahariadis, Dave Lambert, Frances Cleary, Petros Daras, Srdjan Krco, Henning Müller, Man-Sze Li, Hans Schaffers, Volkmar Lotz, Federico Alvarez, Burkhard Stiller, Stamatis Karnouskos, M.N. Susanna Avessta (Eds.), The Future Internet, Springer, 2011, pp. 431–446, http://dx.doi.org/10.1007/978-3-642-20898-0_31.
[86] L.V. Lerman, W. Gerstlberger, M. Ferreira Lima, A.G. Frank, How governments, universities, and companies contribute to renewable energy development? A municipal innovation policy perspective of the triple helix, Energy Res. Soc. Sci. 71 (2021) 101854, http://dx.doi.org/10.1016/j.erss.2020.101854.
[87] G.P. Trencher, M. Yarime, A. Kharrazi, Co-creating sustainability: cross-sector university collaborations for driving sustainable urban transformations, J. Clean. Prod. 50 (2013) 40–55, http://dx.doi.org/10.1016/j.jclepro.2012.11.047.
[88] U. Awan, Steering for sustainable development goals: A typology of sustainable innovation, in: W. Leal Filho, A.M. Azul, L. Brandli, A. Lange Salvia, T. Wall (Eds.), Industry, Innovation and Infrastructure. Encyclopedia of the UN Sustainable Development Goals, Springer, Cham, 2021, pp. 1026–1036, http://dx.doi.org/10.1007/978-3-319-95873-6_64.
[89] Y. Li, J. Dai, L. Cui, The impact of digital technologies on economic and environmental performance in the context of industry 4.0: A moderated mediation model, Int. J. Prod. Econ.. 229 (2020) 107777, http://dx.doi.org/10.1016/j.ijpe.2020.107777.
[90] N. abid, F. Ceci, A. Razzaq, Inclusivity of information and communication technology in ecological governance for sustainable resources management in G10 countries, Resour. Policy 81 (2023) 103378, http://dx.doi.org/10.1016/j.resourpol.2023.103378.
[91] M.A.R.Y. Abeer, The role of the digital economy in sustainable development, Int. J. Humanit. Lang. Res. 5 (2022) 13–25, http://dx.doi.org/10.21608/ijhlr.2023.215933.1013.
[92] S.J. Shackelford, A.N. Craig, Beyond the new digital divide: Analyzing the evolving role of national governments in internet governance and enhancing cybersecurity, Stanf. J. Int. Law 50 (2014) 119–184.
[93] D. Pesce, P. Neirotti, E. Paolucci, When culture meets digital platforms: value creation and stakeholders' alignment in big data use, Curr. Issues Tour. 22 (2019) 1883–1903, http://dx.doi.org/10.1080/13683500.2019.1591354.
[94] M. Skare, M. de las Mercedes de Obesso, S. Ribeiro-Navarrete, Digital transformation and European small and medium enterprises (SMEs): A comparative study using digital economy and society index data, Int. J. Inf. Manage. 68 (2023) 102594, http://dx.doi.org/10.1016/j.ijinfomgt.2022.102594.
[95] J. Dąbrowska, A. Almpanopoulou, A. Brem, H. Chesbrough, V. Cucino, A. Di Minin, F. Giones, H. Hakala, C. Marullo, A. Mention, L. Mortara, S. Nørskov, P.A. Nylund, C.M. Oddo, A. Radziwon, P. Ritala, Digital transformation, for better or worse: a critical multi-level research agenda, R & D Manag. 52 (2022) 930–954, http://dx.doi.org/10.1111/radm.12531.
[96] M.E. Mondejar, R. Avtar, H.L.B. Diaz, R.K. Dubey, J. Esteban, A. Gómez-Morales, B. Hallam, N.T. Mbungu, C.C. Okolo, K.A. Prasad, Q. She, S. Garcia-Segura, Digitalization to achieve sustainable development goals: Steps towards a smart green planet, Sci. Total. Env. 794 (2021) 148539, http://dx.doi.org/10.1016/j.scitotenv.2021.148539.
[97] S. ElMassah, M. Mohieldin, Digital transformation and localizing the Sustainable Development Goals (SDGs), Ecol. Econ. 169 (2020) 106490, http://dx.doi.org/10.1016/j.ecolecon.2019.106490.
[98] U. Pisano, Resilience and Sustainable Development: Theory of Resilience, Systems Thinking and Adaptive Governance, ESDN Quarterly Report, 2012.
[99] Y. Zou, Urban resilience, digital technologies, and the economic recovery of a city from the pandemic, Public Adm. Rev. 84 (2024) 637–650, http://dx.doi.org/10.1111/puar.13678.
[100] S.A. Argyroudis, S.A. Mitoulis, E. Chatzi, J.W. Baker, I. Brilakis, K. Gkoumas, M. Vousdoukas, W. Hynes, S. Carluccio, O. Keou, D.M. Frangopol, I. Linkov, Digital technologies can enhance climate resilience of critical infrastructure, Clim. Risk Manag. 35 (2022) 100387, http://dx.doi.org/10.1016/j.crm.2021.100387.
[101] A.-L. Balogun, D. Marks, R. Sharma, H. Shekhar, C. Balmes, D. Maheng, A. Arshad, P. Salehi, Assessing the potentials of digitalization as a tool for climate change adaptation and sustainable development in urban centres, Sustain. Cities Soc. 53 (2020) 101888, http://dx.doi.org/10.1016/j.scs.2019.101888.
[102] M.N.I. Sarker, Y. Peng, C. Yiran, R.C. Shouse, Disaster resilience through big data: Way to environmental sustainability, Int. J. Disaster Risk Reduct. 51 (2020) 101769, http://dx.doi.org/10.1016/j.ijdrr.2020.101769.
[103] S.K. Abid, N. Sulaiman, S.W. Chan, U. Nazir, M. Abid, H. Han, A. Ariza-Montes, A. Vega-Muñoz, Toward an integrated disaster management approach: How artificial intelligence can boost disaster management, Sustainability 13 (2021) 12560, http://dx.doi.org/10.3390/su132212560.
[104] W. Sun, P. Bocchini, B.D. Davison, Applications of artificial intelligence for disaster management, Nat. Hazards 103 (2020) 2631–2689, http://dx.doi.org/10.1007/s11069-020-04124-3.
[105] M. Yu, C. Yang, Y. Li, Big data in natural disaster management: A review, Geosciences 8 (2018) 165, http://dx.doi.org/10.3390/geosciences8050165.
[106] A. Khurshid, Applying blockchain technology to address the crisis of trust during the COVID-19 pandemic, JMIR Med. Informatics 8 (2020) e20477, http://dx.doi.org/10.2196/20477.
[107] A. Miceli, B. Hagen, M.P. Riccardi, F. Sotti, D. Settembre-Blundo, Thriving, not just surviving in changing times: how sustainability, agility and digitalization intertwine with organizational resilience, Sustainability 13 (2021) 2052, http://dx.doi.org/10.3390/su13042052.
[108] R. Itten, R. Hischier, A.S.G. Andrae, J.C.T. Bieser, L. Cabernard, A. Falke, H. Ferreboeuf, L.M. Hilty, R.L. Keller, E. Lees-Perasso, C. Preist, M. Stucki, Digital transformation—life cycle assessment of digital services, multifunctional devices and cloud computing, Int. J. Life Cycle Assess. 25 (2020) 2093–2098, http://dx.doi.org/10.1007/s11367-020-01801-0.
[109] R. Meinhold, C. Wagner, B.K. Dhar, Digital sustainability and eco-environmental sustainability: A review of emerging technologies, resource challenges, and policy implications, Sustain. Dev. (2024) http://dx.doi.org/10.1002/sd.3240.
[110] A.E. Caglar, M. Daştan, S.B. Avci, Z. Ahmed, S. Gönenç, Modeling the influence of mineral rents and low-carbon energy on environmental quality: New insights from a sustainability perspective, Nat. Resour. Forum 48 (2024) 1456–1476, http://dx.doi.org/10.1111/1477-8947.12472.
[111] M. Ziani, N. Maranzana, A. Feraille, M. Saade, Digital transition and the environment: Towards an approach to assess environmental impacts of digital services using life cycle assessment, in: F. Noël, F. Nyffenegger, L. Rivest, A. Bouras (Eds.), Product Lifecycle Management, in: PLM in Transition Times: The Place of Humans and Transformative Technologies, Springer, Cham, 2023, pp. 577–586, http://dx.doi.org/10.1007/978-3-031-25182-5_56.
[112] V. Varriale, A. Cammarano, F. Michelino, M. Caputo, The role of digital technologies in production systems for achieving sustainable development goals, Sustain. Prod. Consum. 47 (2024) 87–104, http://dx.doi.org/10.1016/j.spc.2024.03.035.
[113] I. Guandalini, Sustainability through digital transformation: A systematic literature review for research guidance, J. Bus. Res. 148 (2022) 456–471, http://dx.doi.org/10.1016/j.jbusres.2022.05.003.
[114] Z. Van Veldhoven, J. Vanthienen, Digital transformation as an interaction-driven perspective between business, society, and technology, Electron. Mark. 32 (2022) 629–644, http://dx.doi.org/10.1007/s12525-021-00464-5.
[115] F. Li, The digital transformation of business models in the creative industries: A holistic framework and emerging trends, Technovation 92–93 (2020) 102012, http://dx.doi.org/10.1016/j.technovation.2017.12.004.
[116] D.E. Ufua, E.T. Emielu, O.J. Olujobi, F. Lakhani, T.T. Borishade, A.S. Ibidunni, E.S. Osabuohien, Digital transformation: a conceptual framing for attaining sustainable development goals 4 and 9 in Nigeria, J. Manag. Organ. 27 (2021) 836–849, http://dx.doi.org/10.1017/jmo.2021.45.
[117] D. Ziadlou, Strategies during digital transformation to make progress in achievement of sustainable development by 2030, Leadersh. Heal. Serv. 34 (2021) 375–391, http://dx.doi.org/10.1108/LHS-08-2020-0056.
[118] L.M. Villar, E. Oliva-Lopez, O. Luis-Pineda, A. Benešová, J. Tupa, J.A. Garza-Reyes, Fostering economic growth, social inclusion & sustainability in Industry 4.0: a systemic approach, Procedia Manuf. 51 (2020) 1755–1762, http://dx.doi.org/10.1016/j.promfg.2020.10.244.
[119] E. Yavuz, E. Kilic, A.E. Caglar, A new hypothesis for the unemployment-environment dilemma: is the environmental phillips curve valid in the framework of load capacity factor in turkiye?, Environ. Dev. Sustain. 26 (2023) 29475–29492, http://dx.doi.org/10.1007/s10668-023-04258-x.

[120] E.G. Popkova, P. De Bernardi, Y.G. Tyurina, B.S. Sergi, A theory of digital technology advancement to address the grand challenges of sustainable development, Technol. Soc. 68 (2022) 101831, http://dx.doi.org/10.1016/j.techsoc.2021.101831.
[121] L. Wang, Y. Chen, T.S. Ramsey, G.J.D. Hewings, Will researching digital technology really empower green development? Technol. Soc. 66 (2021) 101638, http://dx.doi.org/10.1016/j.techsoc.2021.101638.
[122] M.S. Hoosain, B.S. Paul, S. Ramakrishna, The impact of 4IR digital technologies and circular thinking on the united nations sustainable development goals, Sustainability 12 (2020) 10143, http://dx.doi.org/10.3390/su122310143.
[123] A.E. Caglar, M. Daştan, E. Bulut, C. Marangoz, Evaluating a pathway for environmental sustainability: The role of competitive industrial performance and renewable energy consumption in European countries, Sustain. Dev. 32 (2024) 1811–1824, http://dx.doi.org/10.1002/sd.2755.
[124] Z. Fan, Z. Yan, S. Wen, Deep learning and artificial intelligence in sustainability: A review of SDGs, renewable energy, and environmental health, Sustainability 15 (2023) 13493, http://dx.doi.org/10.3390/su151813493.
[125] O. Olu, D. Muneene, J.E. Bataringaya, M.-R. Nahimana, H. Ba, Y. Turgeon, H.C. Karamagi, D. Dovlo, How can digital health technologies contribute to sustainable attainment of universal health coverage in Africa? A perspective, Front. Public Heal. 7 (2019) http://dx.doi.org/10.3389/fpubh.2019.00341.
[126] Y.M. Asi, C. Williams, The role of digital health in making progress toward sustainable development goal (SDG) 3 in conflict-affected populations, Int. J. Med. Inf. 114 (2018) 114–120, http://dx.doi.org/10.1016/j.ijmedinf.2017.11.003.
[127] G. George, R.K. Merrill, S.J.D. Schillebeeckx, Digital sustainability and entrepreneurship: How digital innovations are helping tackle climate change and sustainable development, Entrep. Theory Pr. 45 (2021) 999–1027, http://dx.doi.org/10.1177/1042258719899425.
[128] G. Onyango, J.O. Ondiek, Digitalization and integration of sustainable development goals (SGDs) in public organizations in Kenya, Public Organ. Rev. 21 (2021) 511–526, http://dx.doi.org/10.1007/s11115-020-00504-2.
[129] N. Bachmann, S. Tripathi, M. Brunner, H. Jodlbauer, The contribution of data-driven technologies in achieving the sustainable development goals, Sustainability 14 (2022) 2497, http://dx.doi.org/10.3390/su14052497.
[130] C. de Villiers, S. Kuruppu, D. Dissanayake, A (new) role for business – Promoting the United Nations' Sustainable Development Goals through the internet-of-things and blockchain technology, J. Bus. Res. 131 (2021) 598–609, http://dx.doi.org/10.1016/j.jbusres.2020.11.066.
[131] O. Ahel, K. Lingenau, Opportunities and challenges of digitalization to improve access to education for sustainable development in higher education, in: Walter Leal Filho, Amanda Lange Salvia, Rudi W. Pretorius, Luciana Londero Brandli, Evangelos Manolas, Fatima Alves, Ulisses Azeiteiro, Judy Rogers, A.D.P. Chris Shiel (Eds.), Universities As Living Labs for Sustainable Development, Springer, Cham, 2020, pp. 341–356, http://dx.doi.org/10.1007/978-3-030-15604-6_21.
[132] R. Saner, L. Yiu, M. Nguyen, Monitoring the SDGs: Digital and social technologies to ensure citizen participation, inclusiveness and transparency, Dev. Policy Rev. 38 (2020) 483–500, http://dx.doi.org/10.1111/dpr.12433.
[133] B. Wu, F. Tian, M. Zhang, H. Zeng, Y. Zeng, Cloud services with big data provide a solution for monitoring and tracking sustainable development goals, Geogr. Sustain. 1 (2020) 25–32, http://dx.doi.org/10.1016/j.geosus.2020.03.006.
[134] E. Herman, The interplay between digital entrepreneurship and sustainable development in the context of the EU digital economy: A multivariate analysis, Mathematics 10 (2022) 1682, http://dx.doi.org/10.3390/math10101682.
[135] H. Park, S.O. Choi, Digital innovation adoption and its economic impact focused on path analysis at national level, J. Open Innov. Technol. Mark. Complex 5 (2019) 56, http://dx.doi.org/10.3390/joitmc5030056.
[136] J. Xu, S. She, W. Liu, Role of digitalization in environment, social and governance, and sustainability: Review-based study for implications, Front. Psychol. 13 (2022) http://dx.doi.org/10.3389/fpsyg.2022.961057.
[137] A. Jakeman, R. Letcher, Integrated assessment and modelling: features, principles and examples for catchment management, Env. Model. Softw. 18 (2003) 491–501, http://dx.doi.org/10.1016/S1364-8152(03)00024-0.
[138] S.H. Schneider, Integrated assessment modeling of global climate change: Transparent rational tool for policy making or opaque screen hiding value-laden assumptions, Env. Model. Assess 2 (1997) 229–249, http://dx.doi.org/10.1023/A:1019090117643.
[139] F.M. Tsai, T.-D. Bui, M.-L. Tseng, M.H. Ali, M.K. Lim, A.S. Chiu, Sustainable supply chain management trends in world regions: A data-driven analysis, Resour. Conserv. Recycl. 167 (2021) 105421, http://dx.doi.org/10.1016/j.resconrec.2021.105421.
[140] G. Harris, Integrated assessment and modelling: an essential way of doing science, Env. Model. Softw. 17 (2002) 201– 207, http://dx.doi.org/10.1016/S1364-8152(01)00058-5.
[141] C.L.J. Marcinko, R.J. Nicholls, T.M. Daw, S. Hazra, C.W. Hutton, C.T. Hill, D. Clarke, A. Harfoot, O. Basu, I. Das, S. Giri, S. Pal, P.P. Mondal, The development of a framework for the integrated assessment of SDG trade-offs in the sundarban biosphere reserve, Water 13 (2021) 528, http://dx.doi.org/10.3390/w13040528.
[142] A.S. Gill, S. Germann, Conceptual and normative approaches to AI governance for a global digital ecosystem supportive of the UN Sustainable Development Goals (SDGs), AI Ethics 2 (2022) 293–301, http://dx.doi.org/10.1007/s43681-021-00058-z.
[143] L. Oliveira-Duarte, D.A. Reis, A.L. Fleury, R.A. Vasques, H. Fonseca Filho, M. Koria, J. Baruque-Ramos, Innovation Ecosystem framework directed to Sustainable Development Goal #17 partnerships implementation, Sustain. Dev. 29 (2021) 1018–1036, http://dx.doi.org/10.1002/sd.2191.
[144] B. Sautter, Shaping digital ecosystems for sustainable production: Assessing the policy impact of the 2030 vision for industrie 4.0, Sustainability 13 (2021) 12596, http://dx.doi.org/10.3390/su132212596.
[145] G. Del Río Castro, M.C. González Fernández, Á. Uruburu Colsa, Unleashing the convergence amid digitalization and sustainability towards pursuing the Sustainable Development Goals (SDGs): A holistic review, J. Clean. Prod. 280 (2021) 122204, http://dx.doi.org/10.1016/j.jclepro.2020.122204.
[146] J. Corbett, S. Mellouli, Winning the SDG battle in cities: how an integrated information ecosystem can contribute to the achievement of the 2030 sustainable development goals, Inf. Syst. J. 27 (2017) 427–461, http://dx.doi.org/10.1111/isj.12138.
[147] A. Chaudhuri, B.V. Wæhrens, H. Treiblmaier, S.F. Jensen, Impact pathways: digital product passport for embedding circularity in electronics supply chains, Int. J. Oper. Prod. Manag. (2024) http://dx.doi.org/10.1108/IJOPM-01-2024-0012.
[148] H. Simmonds, A. Gazley, Impact value and sustainable, well-being centred service systems, Eur. J. Mark. 55 (2021) 593–617, http://dx.doi.org/10.1108/EJM-03-2018-0182.
[149] Y. Lyu, Y. Wu, G. Wu, W. Wang, J. Zhang, Digitalization and energy: How could digital economy eliminate energy poverty in China?, Environ. Impact Assess. Rev. 103 (2023) 107243, http://dx.doi.org/10.1016/j.eiar.2023.107243.
[150] M. Imran, X. Liu, R. Wang, S. Saud, Y. Zhao, M.J. Khan, The influence of digital economy and society index on sustainable development indicators: The case of European Union, Sustainability 14 (2022) 11130, http://dx.doi.org/10.3390/su141811130.
[151] L. Cricelli, S. Strazzullo, The economic aspect of digital sustainability: A systematic review, Sustainability 13 (2021) 8241, http://dx.doi.org/10.3390/su13158241.
[152] A. Raihan, A review of the potential opportunities and challenges of the digital economy for sustainability, Innov. Green Dev. 3 (2024) 100174, http://dx.doi.org/10.1016/j.igd.2024.100174.
[153] A.T. Rosário, J.C. Dias, The new digital economy and sustainability: Challenges and opportunities, Sustainability 15 (2023) 10902, http://dx.doi.org/10.3390/su151410902.
[154] S. Gupta, J. Campos Zeballos, G. del Río Castro, A. Tomičić, S. Andrés Morales, M. Mahfouz, I. Osemwegie, V. Phemia Comlan Sessi, M. Schmitz, N. Mahmoud, M. Inyaregh, Operationalizing digitainability: Encouraging mindfulness to harness the power of digitalization for sustainable development, Sustainability 15 (2023) 6844, http://dx.doi.org/10.3390/su15086844.
[155] D. Hidiroğlu, Digital sustainability in businesses, in: Conflict Management in Digital Business, Emerald Publishing Limited, 2022, pp. 241–257, http://dx.doi.org/10.1108/978-1-80262-773-220221022.
[156] A. Ciacci, E. Ivaldi, L. Penco, G. Testa, Measuring digital sustainability paying attention to the economic, social, and environmental dimensions: A European perspective, Soc. Indic. Res. (2024) http://dx.doi.org/10.1007/s11205-024-03459-9.
[157] C. Zhang, Y. Lu, Study on artificial intelligence: The state of the art and future prospects, J. Ind. Inf. Integr. 23 (2021) 100224, http://dx.doi.org/10.1016/j.jii.2021.100224.
[158] I.H. Sarker, AI-based modeling: Techniques, applications and research issues towards automation, intelligent and smart systems, SN Comput. Sci. 3 (2022) 158, http://dx.doi.org/10.1007/s42979-022-01043-x.
[159] A.E. Caglar, S.B. Avci, Z. Ahmed, N. Gökçe, Assessing the role of green investments and green innovation in ecological sustainability: From a climate action perspective on European countries, Sci. Total. Env. 928 (2024) 172527, http://dx.doi.org/10.1016/j.scitotenv.2024.172527.
[160] A.E. Caglar, S. Gönenç, M.A. Destek, The influence of nuclear energy research and development investments on environmental sustainability: evidence from the United States and France, Int. J. Sustain. Dev. World Ecol. 31 (2024) 861–872, http://dx.doi.org/10.1080/13504509.2024.2341797.
[161] M. Lin, Y. Zhao, Artificial intelligence-empowered resource management for future wireless communications: A survey, China Commun. 17 (2020) 58–77, http://dx.doi.org/10.23919/JCC.2020.03.006.
[162] L. Chen, Z. Chen, Y. Zhang, Y. Liu, A.I. Osman, M. Farghali, J. Hua, A. Al-Fatesh, I. Ihara, D.W. Rooney, P.-S. Yap, Artificial intelligence-based solutions for climate change: a review, Env. Chem. Lett. 21 (2023) 2525–2557, http://dx.doi.org/10.1007/s10311-023-01617-y.
[163] N.J. Schork, Artificial intelligence and personalized medicine, in: D. Von Hoff, H. Han (Eds.), Precision Medicine in Cancer Therapy, Springer, Cham, 2019, pp. 265–283, http://dx.doi.org/10.1007/978-3-030-16391-4_11.

[164] A.E. Caglar, M. Daştan, S.B. Avci, Persistence of disaggregate energy RD & D expenditures in top-five economies: Evidence from artificial neural network approach, Appl. Energy 365 (2024) 123216, http://dx.doi.org/10.1016/j.apenergy.2024.123216.
[165] L. Andeobu, S. Wibowo, S. Grandhi, Artificial intelligence applications for sustainable solid waste management practices in Australia: A systematic review, Sci. Total. Env. 834 (2022) 155389, http://dx.doi.org/10.1016/j.scitotenv.2022.155389.
[166] G. Habib, S. Sharma, S. Ibrahim, I. Ahmad, S. Qureshi, M. Ishfaq, Blockchain technology: Benefits, enges, applications, and integration of blockchain technology with cloud computing, Futur. Internet 14 (2022) 341, http://dx.doi.org/10.3390/fi14110341.
[167] D. Mhlanga, Block chain technology for digital financial inclusion in the industry 4.0, towards sustainable development? Front. Blockchain 6 (2023) http://dx.doi.org/10.3389/fbloc.2023.1035405.
[168] N. Kshetri, Blockchain and sustainable supply chain management in developing countries, Int. J. Inf. Manage. 60 (2021) 102376, http://dx.doi.org/10.1016/j.ijinfomgt.2021.102376.
[169] A. Parmentola, A. Petrillo, I. Tutore, F. De Felice, Is blockchain able to enhance environmental sustainability? A systematic review and research agenda from the perspective of Sustainable Development Goals (SDGs), Bus. Strat. Env. 31 (2022) 194–217, http://dx.doi.org/10.1002/bse.2882.
[170] R. Medaglia, J. Damsgaard, Blockchain and the united nations sustainable development goals: Towards an agenda for is research, in: Proc. 24th Pacific Asia Conf. Inf. Syst. Inf. Syst. Futur. PACIS, 2020, 2020, pp. 1–11.
[171] S. Li, L. Da Xu, S. Zhao, The internet of things: a survey, Inf. Syst. Front. 17 (2015) 243–259, http://dx.doi.org/10.1007/s10796-014-9492-7.
[172] S. Nižetić, P. Šolić, D. López-de-Ipiña González-de Artaza, L. Patrono, Internet of Things (IoT): Opportunities, issues and challenges towards a smart and sustainable future, J. Clean. Prod. 274 (2020) 122877, http://dx.doi.org/10.1016/j.jclepro.2020.122877.
[173] M. Maksimovic, Greening the future: Green Internet of Things (G-IoT) as a key technological enabler of sustainable development, in: N. Dey, A. Hassanien, C. Bhatt, A. Ashour, S. Satapathy (Eds.), Internet of Things and Big Data Analytics Toward Next-Generation Intelligence, Springer, Cham, 2018, pp. 283–313, http://dx.doi.org/10.1007/978-3-319-60435-0_12.
[174] N. Hossein Motlagh, M. Mohammadrezaei, J. Hunt, B. Zakeri, Internet of Things (IoT) and the energy sector, Energies 13 (2020) 494, http://dx.doi.org/10.3390/en13020494.
[175] A. Salam, Internet of Things for sustainable community development: Introduction and overview, in: Internet of Things for Sustainable Community Development, Springer, Cham, 2024, pp. 1–31, http://dx.doi.org/10.1007/978-3-031-62162-8_1.
[176] K. Vassakis, E. Petrakis, I. Kopanakis, Big data analytics: Applications, prospects and challenges, in: G. Skourletopoulos, G. Mastorakis, C. Mavromoustakis, C. Dobre, E. Pallis (Eds.), Mobile Big Data, in: Lecture Notes on Data Engineering and Communications Technologies, Springer, Cham, 2018, pp. 3–20, http://dx.doi.org/10.1007/978-3-319-67925-9_1.
[177] A.Y. Sun, B.R. Scanlon, How can big data and machine learning benefit environment and water management: a survey of methods, applications, and future directions, Env. Res. Lett. 14 (2019) 73001, http://dx.doi.org/10.1088/1748-9326/ab1b7d.
[178] H. Hassani, X. Huang, S. MacFeely, M.R. Entezarian, Big data and the united nations sustainable development goals (UN SDGs) at a glance, Big Data Cogn. Comput. 5 (2021) 28, http://dx.doi.org/10.3390/bdcc5030028.
[179] R. Buyya, J. Broberg, A. Goscinski, Cloud Computing: Principles and Paradigms, Wiley, 2011, http://dx.doi.org/10.1002/9780470940105.
[180] T. Muhammad, M.T.Z. Munir, M.T.Z. Munir, M.W. Zafar, Elevating business operations: The transformative power of cloud computing, Int. J. Comput. Sci. Technol. 2 (2018).
[181] C. Yang, Q. Huang, Z. Li, K. Liu, F. Hu, Big data and cloud computing: innovation opportunities and challenges, Int. J. Digit. Earth 10 (2017) 13–53, http://dx.doi.org/10.1080/17538947.2016.1239771.
[182] R. Miskiewicz, Clean and affordable energy within sustainable development goals: The role of governance digitalization, Energies 15 (2022) 9571, http://dx.doi.org/10.3390/en15249571.
[183] L. Vora, Evolution of mobile generation technology: 1G to 5G and review of upcoming wireless technology 5G, Int. J. Mod. Trends Eng. Res. 2 (2015) 281–290.
[184] A. Gohar, G. Nencioni, The role of 5G technologies in a smart city: The case for intelligent transportation system, Sustainability 13 (2021) 5188, http://dx.doi.org/10.3390/su13095188.
[185] L. Guevara, F. Auat Cheein, The role of 5G technologies: Challenges in smart cities and intelligent transportation systems, Sustainability 12 (2020) 6469, http://dx.doi.org/10.3390/su12166469.
[186] M.J. Shehab, I. Kassem, A.A. Kutty, M. Kucukvar, N. Onat, T. Khattab, 5G networks towards smart and sustainable cities: A review of recent developments, applications and future perspectives, IEEE Access 10 (2022) 2987–3006, http://dx.doi.org/10.1109/ACCESS.2021.3139436.
[187] R.S. Lunetta, R.G. Congalton, L.K. Fenstermaker, J.R. Jensen, K.C. McGwire, L.R. Tinney, Remote sensing and geographic information system data integration: error sources and research issues, Photogramm. Eng. Remote Sens. 57 (1991) 677–687.
[188] A.C. Watts, V.G. Ambrosia, E.A. Hinkley, Unmanned aircraft systems in remote sensing and scientific research: Classification and considerations of use, Remote. Sens. 4 (2012) 1671–1692, http://dx.doi.org/10.3390/rs4061671.
[189] P.A. Burrough, GIS and geostatistics: Essential partners for spatial analysis, Environ. Ecol. Stat. 8 (2001) 361–377, http://dx.doi.org/10.1023/A:1012734519752.
[190] A. Emre Caglar, N. Gökçe, D. Balsalobre-Lorente, Towards the vision of going green: The role of different energy research and development investments, Urban. Income Load Capacit. Factor. Sustain. Energy Technol. Assessments 68 (2024) 103888, http://dx.doi.org/10.1016/j.seta.2024.103888.
[191] I. Masser, Managing our urban future: the role of remote sensing and geographic information systems, Habitat. Int. 25 (2001) 503– 512, http://dx.doi.org/10.1016/S0197-3975(01)00021-2.
[192] A.G. Yeh, X. Li, An integrated remote sensing and GIS approach in the monitoring and evaluation of rapid urban growth for sustainable development in the Pearl River Delta, China, Int. Plan. Stud. 2 (1997) 193–210, http://dx.doi.org/10.1080/13563479708721678.
[193] M.A.E. AbdelRahman, An overview of land degradation, desertification and sustainable land management using GIS and remote sensing applications, Rend. Lincei. Sci. Fis. e Nat. 34 (2023) 767–808, http://dx.doi.org/10.1007/s12210-023-01155-3.
[194] P.K. Kingra, D. Majumder, S.P. Singh, Application of remote sensing and gis in agriculture and natural resource management under changing climatic conditions, Agric. Res. J. 53 (2016) 295, http://dx.doi.org/10.5958/2395-146X.2016.00058.2.
[195] D. Mhlanga, Artificial intelligence in the industry 4.0, and its impact on poverty, innovation, infrastructure development, and the sustainable development goals: Lessons from emerging economies? Sustainability 13 (2021) 5788, http://dx.doi.org/10.3390/su13115788.
[196] X. Ning, R. Ramirez, J. Khuntia, Blockchain-enabled government efficiency and impartiality: using blockchain for targeted poverty alleviation in a city in China, Inf. Technol. Dev. 27 (2021) 599–616, http://dx.doi.org/10.1080/02681102.2021.1925619.
[197] A. Roy, A.M.S. Zalzala, A. Kumar, Disruption of things: A model to facilitate adoption of IoT-based innovations by the urban poor, Procedia Eng. 159 (2016) 199–209, http://dx.doi.org/10.1016/j.proeng.2016.08.159.
[198] J. Zhu, H. Xiao, B. Wu, From big data to higher bureaucratic capacity: Poverty alleviation in China, Public Adm. 102 (2024) 61–78, http://dx.doi.org/10.1111/padm.12907.
[199] A. Maaroof, Big Data and the 2030 Agenda for Sustainable Development, UNESCAP, 2015.
[200] M. Firdhous, O. Ghazali, S. Hassan, Cloud computing for rural ICT development: Opportunities and challenges, in: 2013 International Conference on Computing, Electrical and Electronic Engineering, ICCEEE, IEEE, 2013, pp. 680–685, http://dx.doi.org/10.1109/ICCEEE.2013.6634022.
[201] M. Cabanillas-Carbonell, J. Pérez-Martínez, J. Zapata-Paulini, Contributions of the 5G network with respect to poverty (SDG1), Syst. Lit. Rev. Sustain. 15 (2023) 11301, http://dx.doi.org/10.3390/su151411301.
[202] E. Yaacoub, M.-S. Alouini, A key 6G challenge and opportunity—Connecting the base of the pyramid: A survey on rural connectivity, Proc. IEEE 108 (2020) 533–582, http://dx.doi.org/10.1109/JPROC.2020.2976703.
[203] S. Latif, J. Qadir, S. Farooq, M. Imran, How 5G wireless (and concomitant technologies) will revolutionize healthcare?, Futur. Internet 9 (2017) 93, http://dx.doi.org/10.3390/fi9040093.
[204] S. Hu, Y. Ge, M. Liu, Z. Ren, X. Zhang, Village-level poverty identification using machine learning, high-resolution images, and geospatial data, Int. J. Appl. Earth Obs. Geoinf. 107 (2022) 102694, http://dx.doi.org/10.1016/j.jag.2022.102694.
[205] J.E. Steele, P.R. Sundsøy, C. Pezzulo, V.A. Alegana, T.J. Bird, J. Blumenstock, J. Bjelland, K. Engø-Monsen, Y.-A. de Montjoye, A.M. Iqbal, K.N. Hadiuzzaman, X. Lu, E. Wetter, A.J. Tatem, L. Bengtsson, Mapping poverty using mobile phone and satellite data, J. R. Soc. Interface 14 (2017) 20160690, http://dx.doi.org/10.1098/rsif.2016.0690.
[206] Y.J. Amuda, S. Alabdul rahman, Artificial intelligence for food production among smallholder farmers: Towards achieving sustainable development goals (SDGs) in Nigeria, J. Ecohumanism 4 (2024) 175–185, http://dx.doi.org/10.62754/joe.v4i1.4202.
[207] H. Mishra, D. Mishra, Sustainable smart agriculture to ensure zero hunger, in: Sustainable Development Goals: Technologies and Opportunities, CRC Press, 2024.
[208] A. Chandan, M. John, V. Potdar, Achieving UN SDGs in food supply chain using blockchain technology, Sustainability 15 (2023) 2109, http://dx.doi.org/10.3390/su15032109.
[209] M. Kumar, V.K. Choubey, R.D. Raut, S. Jagtap, Enablers to achieve zero hunger through IoT and blockchain technology and transform the green food supply chain systems, J. Clean. Prod. 405 (2023) 136894, http://dx.doi.org/10.1016/j.jclepro.2023.136894.

[210] P.C. Pandey, M. Pandey, Highlighting the role of agriculture and geospatial technology in food security and sustainable development goals, Sustain. Dev. 31 (2023) 3175–3195, http://dx.doi.org/10.1002/sd.2600.
[211] P. Tamasiga, E. houssin Ouassou, H. Onyeaka, M. Bakwena, A. Happonen, M. Molala, Forecasting disruptions in global food value chains to tackle food insecurity: The role of AI and big data analytics – A bibliometric and scientometric analysis, J. Agric. Food Res. 14 (2023) 100819, http://dx.doi.org/10.1016/j.jafr.2023.100819.
[212] S.O. Oruma, S. Misra, L. Fernandez-Sanz, Agriculture 4.0: An implementation framework for food security attainment in Nigeria's post-Covid-19 era, IEEE Access 9 (2021) 83592–83627, http://dx.doi.org/10.1109/ACCESS.2021.3086453.
[213] T.P. Fowdur, M. Indoonundon, M.A. Hosany, D. Milovanovic, Z. Bojkovic, Achieving sustainable development goals through digital infrastructure for intelligent connectivity, in: Z. Boulouard, M. Ouaissa, M. Ouaissa, S. El Himer (Eds.), AI and IoT for Sustainable Development in Emerging Countries, in: Lecture Notes on Data Engineering and Communications Technologies, vol. 105, Springer, Cham, 2022, pp. 3–26, http://dx.doi.org/10.1007/978-3-030-90618-4_1.
[214] V. Martos, A. Ahmad, P. Cartujo, J. Ordoñez, Ensuring agricultural sustainability through remote sensing in the era of agriculture 5.0, Appl. Sci. 11 (2021) 5911, http://dx.doi.org/10.3390/app11135911.
[215] A. Singh, A. Kanaujia, V.K. Singh, R. Vinuesa, Artificial intelligence for Sustainable Development Goals: Bibliometric patterns and concept evolution trajectories, Sustain. Dev. 32 (2024) 724–754, http://dx.doi.org/10.1002/sd.2706.
[216] G. Gabrani, S. Gupta, L. Verma, S. Vyas, The role of health informatics in achieving sustainable development goals, in: G. Tripathi, A. Shakya, S. Kanga, L.T.S. Guite, S.K. Singh (Eds.), Sustainability and Health Informatics, in: Advances in Geographical and Environmental Sciences, Springer, Singapore, 2024, pp. 73–98, http://dx.doi.org/10.1007/978-981-97-6706-9_4.
[217] T. Schoormann, G. Strobel, F. Möller, D. Petrik, P. Zschech, Artificial intelligence for sustainability—A systematic review of information systems literature, Commun. Assoc. Inf. Syst. 52 (2023) 199–237, http://dx.doi.org/10.17705/1CAIS.05209.
[218] K. Hameed, R. Naha, F. Hameed, Digital transformation for sustainable health and well-being: a review and future research directions, Discov. Sustain. 5 (2024) 104, http://dx.doi.org/10.1007/s43621-024-00273-8.
[219] A. Zahid, J.K. Poulsen, R. Sharma, S.C. Wingreen, A systematic review of emerging information technologies for sustainable data-centric health-care, Int. J. Med. Inf. 149 (2021) 104420, http://dx.doi.org/10.1016/j.ijmedinf.2021.104420.
[220] M. Cabanillas-Carbonell, J. Pérez-Martínez, J.A. Yáñez, 5G technology in the digital transformation of healthcare, A Syst. Rev. Sustain. 15 (2023) 3178, http://dx.doi.org/10.3390/su15043178.
[221] S. Singh, S.K. Sharma, A.K. Shukla, The role of geospatial technology in sustainable development, in: C. Sharma, A.K. Shukla, S. Pathak, V.P. Singh (Eds.), Sustainable Development and Geospatial Technology, Springer Nature Switzerland, Cham, 2024, pp. 17–31, http://dx.doi.org/10.1007/978-3-031-65683-5_2.
[222] C.G. Demartini, L. Sciascia, A. Bosso, F. Manuri, Artificial intelligence bringing improvements to adaptive learning in education: A case study, Sustainability 16 (2024) 1347, http://dx.doi.org/10.3390/su16031347.
[223] P. Nedungadi, K.-Y. Tang, R. Raman, The transformative power of generative artificial intelligence for achieving the sustainable development goal of quality education, Sustainability 16 (2024) 9779, http://dx.doi.org/10.3390/su16229779.
[224] M. Caramihai, I. Severin, A blockchain-based solution for diploma management in universities, Sustainability 15 (2023) 15169, http://dx.doi.org/10.3390/su152015169.
[225] K. Joshi, R. Kumar, S. Bharany, D.K.J.B. Saini, R. Kumar, A.O. Ibrahim, A. Abdelmaboud, W. Nagmeldin, M.A. Medani, Exploring the connectivity between education 4.0 and classroom 4.0: Technologies, Stud. Perspect. Engag. Digit. Era. IEEE Access 12 (2024) 24179–24204, http://dx.doi.org/10.1109/ACCESS.2024.3357786.
[226] A. Lopez-Vargas, M. Fuentes, M. Vivar, Challenges and opportunities of the internet of things for global development to achieve the united nations sustainable development goals, IEEE Access 8 (2020) 37202–37213, http://dx.doi.org/10.1109/ACCESS.2020.2975472.
[227] S. Kamarudin, Strategic 4.0 in academia: A comprehensive review of digital transformation and future gaps agenda, in: A. Hamdan (Ed.), Business Development Via AI and Digitalization, in: Studies in Systems, Decision and Control, vol. 538, Springer, Cham, 2024, pp. 31–40, http://dx.doi.org/10.1007/978-3-031-62102-4_3.
[228] M. Airaj, Cloud computing technology and PBL teaching approach for a qualitative education in line with SDG4, Sustainability 14 (2022) 15766, http://dx.doi.org/10.3390/su142315766.
[229] Y. Liu, M.R. Razman, S.Z. Syed Zakaria, L.K. Ern, A. Hussain, V. Chamola, Utilizing ubiquitous learning to foster sustainable development in rural areas: Insights from 6G technology, Comput. Hum. Behav. 161 (2024) 108418, http://dx.doi.org/10.1016/j.chb.2024.108418.
[230] K. Alkaabi, K. Mehmood, P. Bhatacharyya, H. Aldhaheri, Sustainable development goals from theory to practice using spatial data infrastructure: A case study of UAEU undergraduate students, Sustainability 15 (2023) 12394, http://dx.doi.org/10.3390/su151612394.
[231] OECD, Harnessing the Green and Digital Transitions for Gender Equality: Policy Insights from the 2024 OECD Forum on Gender Equality, OECD Public Governance Policy Papers, Paris, 2024, http://dx.doi.org/10.1787/860d0901-en.
[232] A.Di. Vaio, R. Hassan, R. Palladino, Blockchain technology and gender equality: A systematic literature review, Int. J. Inf. Manage. 68 (2023) 102517, http://dx.doi.org/10.1016/j.ijinfomgt.2022.102517.
[233] Á. Verdejo Espinosa, J.L. Lopez Ruiz, F. Mata Mata, M.E. Estevez, Application of IoT in healthcare: Keys to implementation of the sustainable development goals, Sensors 21 (2021) 2330, http://dx.doi.org/10.3390/s21072330.
[234] V. Kulkarni, R. Mathew, IoT: A step towards sustainability, in: R. Das, M. Banerjee, S. De (Eds.), Emerging Trends in Disruptive Technology Management for Sustainable Development, Chapman and Hall/CRC, New York, 2019.
[235] G.C. Harikiran, K. Menasinkai, S. Shirol, Smart security solution for women based on Internet Of Things (IOT), in: 2016 International Conference on Electrical, Electronics, and Optimization Techniques, ICEEOT, IEEE, 2016, pp. 3551–3554, http://dx.doi.org/10.1109/ICEEOT.2016.7755365.
[236] F.A. Ezeugwa, O.O. Olaniyi, J.C. Ugonnia, A.S. Arigbabu, P.C. Joeaneke, Artificial intelligence, big data, and cloud infrastructures: policy recommendations for enhancing women's participation in the tech-driven economy, J. Eng. Res. Rep. 26 (2024) 1–16, http://dx.doi.org/10.9734/jerr/2024/v26i61158.
[237] L. Eden, M.F. Wagstaff, Evidence-based policymaking and the wicked problem of SDG 5 Gender Equality, J. Int. Bus. Policy 4 (2021) 28–57, http://dx.doi.org/10.1057/s42214-020-00054-w.
[238] D. Mhlanga, Empowering african women through industry 4.0 in the 21st century, 2024.
[239] P. Pritiprada, I. Satpathy, B.C.M. Patnaik, P. Thaichon, Fostering women's empowerment in europe case studies of innovation in AI, blockchain, and the metaverse, in: P. Thaichon, P.K. Dutta, P.R. Chelliah, S. Gupta (Eds.), Technology and Luxury Hospitality, Routledge, London, 2024.
[240] E. Lopez-Inesta, C. Botella, S. Rueda, A. Forte, P. Marzal, Towards breaking the gender gap in science, technology, engineering and mathematics, IEEE Rev. Iberoam. Tecnol. Del Aprendiz. 15 (2020) 233–241, http://dx.doi.org/10.1109/RITA.2020.3008114.
[241] T. Ahamed, R. Noguchi, N. Muhsin, R. Ayu Purnamasari, M.A. Islam, F. Tasnim, M.Z. Islam, M.F. Islam, W. Akmam, Sustainable agricultural development: a micro-level GIS-based study on women's perceptions of environmental protection and entrepreneurship in Japan and Bangladesh, GeoJournal 86 (2021) 2071–2103, http://dx.doi.org/10.1007/s10708-020-10169-5.
[242] E. Ozdenerol, The global problem in gender inequality putting gender on the map with GIS, in: E. Ozdenerol (Ed.), Gender Inequalities GIS Approaches to Gender Analysis, CRC Press, Boca Raton, 2021.
[243] M.K. Goyal, S. Kumar, A. Gupta, AI for water treatment, in: AI Innovation for Water Policy and Sustainability, in: SpringerBriefs in Water Science and Technology, Springer, Cham, 2024, pp. 31–40, http://dx.doi.org/10.1007/978-3-031-72014-7_3.
[244] T.A. Kurniawan, A. Mohyuddin, J.C.C. Casila, P.K. Sarangi, H. Al-Hazmi, Y. Wibisono, T.D. Kusworo, M.M.H. Khan, S. Haddout, Digitalization for sustainable wastewater treatment: a way forward for promoting the UN sdg#6 clean water and sanitation towards carbon neutrality goals, Discov. Water 4 (2024) 71, http://dx.doi.org/10.1007/s43832-024-00134-5.
[245] H. Mehmood, D. Liao, K. Mahadeo, A review of artificial intelligence applications to achieve water-related sustainable development goals, in: 2020 IEEE/ITU International Conference on Artificial Intelligence for Good (AI4G), IEEE, 2020, pp. 135–141, http://dx.doi.org/10.1109/AI4G50087.2020.9311018.
[246] T.K. Satilmisoglu, Y. Sermet, M. Kurt, I. Demir, Blockchain opportunities for water resources management: A comprehensive review, Sustainability 16 (2024) 2403, http://dx.doi.org/10.3390/su16062403.
[247] E. Sriyono, Digitizing water management: Toward the innovative use of blockchain technologies to address sustainability, Cogent Eng. 7 (2020) 1769366, http://dx.doi.org/10.1080/23311916.2020.1769366.
[248] E.M. Dogo, A.F. Salami, N.I. Nwulu, C.O. Aigbavboa, Blockchain and internet of things-based technologies for intelligent water management system, in: F. Al-Turjman (Ed.), Artificial Intelligence in IoT. Transactions on Computational Science and Computational Intelligence, Springer, Cham, 2019, pp. 129–150, http://dx.doi.org/10.1007/978-3-030-04110-6_7.
[249] I. Essamlali, H. Nhaila, M. El Khaili, Advances in machine learning and IoT for water quality monitoring: A comprehensive review, Heliyon 10 (2024) e27920, http://dx.doi.org/10.1016/j.heliyon.2024.e27920.
[250] A. Salam, Internet of Things for water sustainability, in: Internet of Things for Sustainable Community Development, Springer, Cham, 2024, pp. 113–145, http://dx.doi.org/10.1007/978-3-031-62162-8_4.
[251] D. Teh, T. Rana, The use of Internet of Things, big data analytics and artificial intelligence for attaining un's SDGs, in: T. Rana, J. Svanberg, P. Öhman, A. Lowe (Eds.), Handbook of Big Data and Analytics in Accounting and Auditing, Springer Nature Singapore, Singapore, 2023, pp. 235–253, http://dx.doi.org/10.1007/978-981-19-4460-4_11.

[252] M. Abaee, M. Saeedi, H. Taherdoost, Examining the role of IoT and cloud computing in achieving sustainable development goals, in: R.K. Gupta, A. Jain, J. Wang, R.K. Pateriya (Eds.), Reshaping Environmental Science Through Machine Learning and IoT, IGI Global, 2024, pp. 74–97, http://dx.doi.org/10.4018/979-8-3693-2351-9.ch005.
[253] W. Li, H. El-Askary, V. Lakshmi, T. Piechota, D. Struppa, Earth observation and cloud computing in support of two sustainable development goals for the river nile watershed countries, Remote. Sens. 12 (2020) 1391, http://dx.doi.org/10.3390/rs12091391.
[254] G. Féry, The Digital Journey of Water and Sanitation Utilities in Latin America and the Caribbean: What Is at Stake and how to Begin, Washington, D. C, 2022, http://dx.doi.org/10.18235/0004562.
[255] S. Qwaider, B. Al-Ramadan, M. Shafiullah, A. Islam, M.Y. Worku, GIS-based progress monitoring of SDGs towards achieving saudi vision 2030, Remote. Sens. 15 (2023) 5770, http://dx.doi.org/10.3390/rs15245770.
[256] A. Shah, K. Kantamaneni, S. Ravan, L.C. Campos, A systematic review investigating the use of earth observation for the assistance of water, sanitation and hygiene in disaster response and recovery, Sustainability 15 (2023) 3290, http://dx.doi.org/10.3390/su15043290.
[257] V. Baskaran, V. R., A systematic review on the role of geographical information systems in monitoring and achieving sustainable development goal 6: Clean water and sanitation, Sustain. Dev. 30 (2022) 1417–1425, http://dx.doi.org/10.1002/sd.2302.
[258] O. Awogbemi, D.V. Von Kallon, K.S. Kumar, Contributions of artificial intelligence and digitization in achieving clean and affordable energy, Intell. Syst. Appl 22 (2024) 200389, http://dx.doi.org/10.1016/j.iswa.2024.200389.
[259] R. Raman, S. Gunasekar, D. Kaliyaperumal, P. Nedungadi, Navigating the nexus of artificial intelligence and renewable energy for the advancement of sustainable development goals, Sustainability 16 (2024) 9144, http://dx.doi.org/10.3390/su16219144.
[260] M. Jayachandran, R.K. Gatla, K.P. Rao, G.S. Rao, S. Mohammed, A.H. Milyani, A.A. Azhari, C. Kalaiarasy, S. Geetha, Challenges in achieving sustainable development goal 7: Affordable and clean energy in light of nascent technologies, Sustain. Energy Technol. Assessments 53 (2022) 102692, http://dx.doi.org/10.1016/j.seta.2022.102692.
[261] J.-L. Hu, M.-Y. Chuang, The importance of energy prosumers for affordable and clean energy development: A review of the literature from the viewpoints of management and policy, Energies 16 (2023) 6270, http://dx.doi.org/10.3390/en16176270.
[262] Y. Lv, Transitioning to sustainable energy: opportunities, challenges, and the potential of blockchain technology, Front. Energy Res. 11 (2023) http://dx.doi.org/10.3389/fenrg.2023.1258044.
[263] M. Vaccargiu, A. Pinna, R. Tonelli, L. Cocco, Blockchain in the energy sector for SDG achievement, Sustainability 15 (2023) 14843, http://dx.doi.org/10.3390/su152014843.
[264] W. Shafik, Smart devices and internet of things for sustainable energy, in: J. Mabrouki, A. Mourade (Eds.), Advanced Technology for Smart Environment and Energy, Springer, Cham, 2024, pp. 67–93, http://dx.doi.org/10.1007/978-3-031-50871-4_5.
[265] P. Mishra, G. Singh, Energy management systems in sustainable smart cities based on the internet of energy: A technical review, Energies 16 (2023) 6903, http://dx.doi.org/10.3390/en16196903.
[266] A.S.I. K.N., M. S., K. Desai, H.K. S., S. Kautish, Optimizing green power and green energy through digital technologies, in: W. Leal Filho, S. Kautish, T. Wall, S. Rewhorn, S.K. Paul (Eds.), Digital Technologies to Implement the UN Sustainable Development Goals, Springer, Cham, 2024, pp. 275–304, http://dx.doi.org/10.1007/978-3-031-68427-2_14.
[267] T. Ahmad, R. Madonski, D. Zhang, C. Huang, A. Mujeeb, Data-driven probabilistic machine learning in sustainable smart energy/smart energy systems: Key developments, challenges, and future research opportunities in the context of smart grid paradigm, Renew. Sustain. Energy Rev. 160 (2022) 112128, http://dx.doi.org/10.1016/j.rser.2022.112128.
[268] K. Obaideen, L. Albasha, U. Iqbal, H. Mir, Wireless power transfer: Applications, challenges, barriers, and the role of AI in achieving sustainable development goals - A bibliometric analysis, Energy Strat. Rev. 53 (2024) 101376, http://dx.doi.org/10.1016/j.esr.2024.101376.
[269] K.Y. Yap, H.H. Chin, J.J. Klemeš, Future outlook on 6G technology for renewable energy sources (RES), Renew. Sustain. Energy Rev. 167 (2022) 112722, http://dx.doi.org/10.1016/j.rser.2022.112722.
[270] M.O. Ukoba, E.O. Diemuodeke, T.A. Briggs, M.M. Ojapah, K. Owebor, K.E. Okedu, K. Akhtar, C. Ilhami, Using remote sensing technique to assess biomass energy potentials and appropriate site locations in Nigeria, Meas. Sensors 33 (2024) 101145, http://dx.doi.org/10.1016/j.measen.2024.101145.
[271] M. Argentiero, P.M. Falcone, The role of earth observation satellites in maximizing renewable energy production: Case studies analysis for renewable power plants, Sustainability 12 (2020) 2062, http://dx.doi.org/10.3390/su12052062.
[272] M.T. Shaker, H. El-Batanouny, M. Abdelsalam, Y. Helmy, Systematic review of machine learning approaches in forecasting economic growth and enhancing decent work opportunities: A comprehensive analysis, in: 2024 6th International Conference on Computing and Informatics, ICCI, IEEE, 2024, pp. 377–392, http://dx.doi.org/10.1109/ICCI61671.2024.10485171.
[273] P. Tripathi, P. Saxena, An assessment of the role of artificial intelligence on sustainable development goals, in: L.V. Sannikova (Ed.), Digital Technologies and Distributed Registries for Sustainable Development, Springer, Cham, 2024, pp. 3–23, http://dx.doi.org/10.1007/978-3-031-51067-0_1.
[274] M. Javaid, A. Haleem, R.P. Singh, R. Suman, S. Khan, A review of blockchain technology applications for financial services, BenchCouncil Trans. Benchmarks, Stand. Eval. 2 (2022) 100073, http://dx.doi.org/10.1016/j.tbench.2022.100073.
[275] A.A.A. Khanfar, M. Iranmanesh, M. Ghobakhloo, M.G. Senali, M. Fathi, Applications of blockchain technology in sustainable manufacturing and supply chain management: A systematic review, Sustainability 13 (2021) 7870, http://dx.doi.org/10.3390/su13147870.
[276] M. Thibaud, H. Chi, W. Zhou, S. Piramuthu, Internet of Things (IoT) in high-risk Environment, Health and Safety (EHS) industries: A comprehensive review, Decis. Support Syst. 108 (2018) 79–95, http://dx.doi.org/10.1016/j.dss.2018.02.005.
[277] I. Rahhal, I. Kassou, M. Ghogho, Data science for job market analysis: A survey on applications and techniques, Expert Syst. Appl. 251 (2024) 124101, http://dx.doi.org/10.1016/j.eswa.2024.124101.
[278] M. Annunziata, H. Bourgeois, The future of work: how G20 countries can leverage digital-industrial innovations into stronger highquality jobs growth, Economics 12 (2018) http://dx.doi.org/10.5018/economics-ejournal.ja.2018-42.
[279] P.K. Ross, M. Blumenstein, Cloud computing as a facilitator of SME entrepreneurship, Technol. Anal. Strat. Manag. 27 (2015) 87–101, http://dx.doi.org/10.1080/09537325.2014.951621.
[280] M. Olokundun, M.E. Ogbari, H. Falola, A.S. Ibidunni, Leveraging 5G network for digital innovation in small and medium enterprises: a conceptual review, J. Innov. Entrep. 11 (2022) 41, http://dx.doi.org/10.1186/s13731-021-00181-5.
[281] K. Campbell, L. Cruz, B. Flanagan, B. Morelli, B. O'neil, S. Téral, J. Watson, The 5G Economy: How 5G Will Contribute to the Global Economy, IHS Economics and IHS Technology, London, 2017.
[282] B. Bhatta, Analysis of Urban Growth and Sprawl from Remote Sensing Data, Springer-Verlag, Berlin, Heidelberg, 2010.
[283] E. Banzhaf, V. Grescho, A. Kindler, Monitoring urban to peri-urban development with integrated remote sensing and GIS information: a Leipzig, Germany case study, Int. J. Remote. Sens. 30 (2009) 1675–1696, http://dx.doi.org/10.1080/01431160802642297.
[284] A. Khang, S. Akhai, Green intelligent and sustainable manufacturing key advancements, benefits, challenges, and applications for transforming industry, in: A. Khang, V.A. Hajimahmud, A. Misra, E. Litvinova (Eds.), Machine Vision and Industrial Robotics in Manufacturing Approaches, Technologies, and Applications, CRC Press, Boca Raton, 2024.
[285] I. Kulkov, J. Kulkova, R. Rohrbeck, L. Menvielle, V. Kaartemo, H. Makkonen, Artificial intelligence - driven sustainable development: Examining organizational, technical, and processing approaches to achieving global goals, Sustain. Dev. 32 (2024) 2253–2267, http://dx.doi.org/10.1002/sd.2773.
[286] P. Dutta, T.-M. Choi, S. Somani, R. Butala, Blockchain technology in supply chain operations: Applications, challenges and research opportunities, Transp. Res. Part E Logist. Transp. Rev. 142 (2020) 102067, http://dx.doi.org/10.1016/j.tre.2020.102067.
[287] S. Saberi, M. Kouhizadeh, J. Sarkis, L. Shen, Blockchain technology and its relationships to sustainable supply chain management, Int. J. Prod. Res. 57 (2019) 2117–2135, http://dx.doi.org/10.1080/00207543.2018.1533261.
[288] E. Costa, Industry 5.0 and SDG 9: a symbiotic dance towards sustainable transformation, Sustain. Earth Rev. 7 (2024) 4, http://dx.doi.org/10.1186/s42055-024-00073-y.
[289] M. Soori, B. Arezoo, R. Dastres, Internet of things for smart factories in industry 4.0, a review, Internet Things Cyber-Phys. Syst. 3 (2023) 192–204, http://dx.doi.org/10.1016/j.iotcps.2023.04.006.
[290] F. Civerchia, S. Bocchino, C. Salvadori, E. Rossi, L. Maggiani, M. Petracca, Industrial Internet of Things monitoring solution for advanced predictive maintenance applications, J. Ind. Inf. Integr. 7 (2017) 4–12, http://dx.doi.org/10.1016/j.jii.2017.02.003.
[291] S. Bag, L.C. Wood, L. Xu, P. Dhamija, Y. Kayikci, Big data analytics as an operational excellence approach to enhance sustainable supply chain performance, Resour. Conserv. Recycl. 153 (2020) 104559, http://dx.doi.org/10.1016/j.resconrec.2019.104559.
[292] S. Ren, Y. Zhang, Y. Liu, T. Sakao, D. Huisingh, C.M.V.B. Almeida, A comprehensive review of big data analytics throughout product lifecycle to support sustainable smart manufacturing: A framework, challenges and future research directions, J. Clean. Prod. 210 (2019) 1343–1365, http://dx.doi.org/10.1016/j.jclepro.2018.11.025.
[293] S. Jeble, R. Dubey, S.J. Childe, T. Papadopoulos, D. Roubaud, A. Prakash, Impact of big data and predictive analytics capability on supply chain sustainability, Int. J. Logist. Manag. 29 (2018) 513–538, http://dx.doi.org/10.1108/IJLM-05-2017-0134.
[294] M.-L. Tseng, R.R. Tan, A.S.F. Chiu, C.-F. Chien, T.C. Kuo, Circular economy meets industry 4.0: Can big data drive industrial symbiosis?, Resour. Conserv. Recycl. 131 (2018) 146–147, http://dx.doi.org/10.1016/j.resconrec.2017.12.028.

[295] G. Wang, A. Gunasekaran, E.W.T. Ngai, T. Papadopoulos, Big data analytics in logistics and supply chain management: Certain investigations for research and applications, Int. J. Prod. Econ.. 176 (2016) 98–110, http://dx.doi.org/10.1016/j.ijpe.2016.03.014.
[296] M.G. Avram, Advantages and challenges of adopting cloud computing from an enterprise perspective, Procedia Technol. 12 (2014) 529–534, http://dx.doi.org/10.1016/j.protcy.2013.12.525.
[297] A. Lin, N.-C. Chen, Cloud computing as an innovation: Percepetion, attitude, and adoption, Int. J. Inf. Manage. 32 (2012) 533–540, http://dx.doi.org/10.1016/j.ijinfomgt.2012.04.001.
[298] M. Attaran, The impact of 5G on the evolution of intelligent automation and industry digitization, J. Ambient. Intell. Humaniz. Comput. 14 (2023) 5977–5993, http://dx.doi.org/10.1007/s12652-020-02521-x.
[299] J. Cheng, W. Chen, F. Tao, C.-L. Lin, Industrial IoT in 5G environment towards smart manufacturing, J. Ind. Inf. Integr. 10 (2018) 10–19, http://dx.doi.org/10.1016/j.jii.2018.04.001.
[300] S.K. Rao, R. Prasad, Impact of 5G technologies on industry 4.0, Wirel. Pers. Commun. 100 (2018) 145–159, http://dx.doi.org/10.1007/s11277-018-5615-7.
[301] R. Avtar, A.A. Komolafe, A. Kouser, D. Singh, A.P. Yunus, J. Dou, P. Kumar, R. Das Gupta, B.A. Johnson, H.V. Thu Minh, A.K. Aggarwal, T.A. Kurniawan, Assessing sustainable development prospects through remote sensing: A review, Remote. Sens. Appl. Soc. Env. 20 (2020) 100402, http://dx.doi.org/10.1016/j.rsase.2020.100402.
[302] W. Tao, Interdisciplinary urban GIS for smart cities: advancements and opportunities, Geo-Spat. Inf. Sci. 16 (2013) 25–34, http://dx.doi.org/10.1080/10095020.2013.774108.
[303] T.M. Holland Brown, M. Bewick, Digital health education: the need for a digitally ready workforce, Arch. Dis. Child. - Educ. Pr. Ed 108 (2023) 214–217, http://dx.doi.org/10.1136/archdischild-2021-322022.
[304] J.M. Tshimula, D.K. Nkashama, K. Kalala, M.V. Dialufuma, M. Mukendi Didier, H. Kanda, J. Tshibangu Muabila, C.N. Mayemba, Redesigning electronic health record systems to support developing countries, in: 2023 the 7th International Conference on Medical and Health Informatics, ICMHI, ACM, New York, NY, USA, 2023, pp. 216–221, http://dx.doi.org/10.1145/3608298.3608338.
[305] D. Mhlanga, Block chain for digital financial inclusion towards reduced inequalities, in: FinTech and Artificial Intelligence for Sustainable Development, Palgrave Macmillan, Cham, 2023, pp. 263–290, http://dx.doi.org/10.1007/978-3-031-37776-1_12.
[306] T. Savelyeva, J. Park, Blockchain technology for sustainable education, Br. J. Educ. Technol. 53 (2022) 1591–1604, http://dx.doi.org/10.1111/bjet.13273.
[307] H. Mora, J.C. Mendoza-Tello, E.G. Varela-Guzmán, J. Szymanski, Blockchain technologies to address smart city and society challenges, Comput. Hum. Behav. 122 (2021) 106854, http://dx.doi.org/10.1016/j.chb.2021.106854.
[308] V. Lakshminarayanan, A. Ravikumar, H. Sriraman, S. Alla, V.K. Chattu, Health care equity through intelligent edge computing and augmented reality/virtual reality: A systematic review, J. Multidiscip. Heal. 16 (2023) 2839–2859, http://dx.doi.org/10.2147/JMDH.S419923.
[309] J.T. Kelly, K.L. Campbell, E. Gong, P. Scuffham, The Internet of Things: Impact and implications for health care delivery, J. Med. Internet Res. 22 (2020) e20135, http://dx.doi.org/10.2196/20135.
[310] A.J.A.M. van Deursen, K. Mossberger, Any thing for anyone? A new digital divide in Internet-of-Things skills, Policy & Internet 10 (2018) 122–140, http://dx.doi.org/10.1002/poi3.171.
[311] H.J. Karyamsetty, S.A. Khan, A. Nayyar, Envisioning toward modernization of society 5.0—A prospective glimpse on status, opportunities, and challenges with XAI, in: XAI Based Intelligent Systems for Society 5.0, Elsevier, 2024, pp. 223–267, http://dx.doi.org/10.1016/B978-0-323-95315-3.00005-X.
[312] R.K. Goel, S. Vishnoi, Urbanization and sustainable development for inclusiveness using ICTs, Telecomm. Policy 46 (2022) 102311, http://dx.doi.org/10.1016/j.telpol.2022.102311.
[313] X. Wang, B. Wong, Bridging knowledge divides utilizing cloud computing learning resources in underfunded schools: Investigating the determinants, J. Educ. Comput. Res. 57 (2019) 591–617, http://dx.doi.org/10.1177/0735633118759455.
[314] S. Goundar, Cloud computing: opportunities and issues for developing countries, in: DiploFoundation Internet Gov. Res. Pap. 1–15, 2010.
[315] D. Featherstone, J. Thomas, I. Holcombe-James, L. Ormond-Parker, Closing the digital gap for remote first nations communities: 5G and beyond?, Media Int. Aust. 190 (2024) 23–38, http://dx.doi.org/10.1177/1329878X231201746.
[316] J. Martínez, The use of GIS and indicators to monitor intra-urban inequalities, A case study in Rosario, Argentina, Habitat. Int. 33 (2009) 387–396, http://dx.doi.org/10.1016/j.habitatint.2008.12.003.
[317] S.E. Bibri, J. Krogstie, A. Kaboli, A. Alahi, Smarter eco-cities and their leading-edge artificial intelligence of things solutions for environmental sustainability: A comprehensive systematic review, Env. Sci. Ecotechnology 19 (2024) 100330, http://dx.doi.org/10.1016/j.ese.2023.100330.
[318] M.E.E. Alahi, A. Sukkuea, F.W. Tina, A. Nag, W. Kurdthongmee, K. Suwannarat, S.C. Mukhopadhyay, Integration of IoT-enabled technologies and artificial intelligence (AI) for smart city scenario: Recent advancements and future trends, Sensors 23 (2023) 5206, http://dx.doi.org/10.3390/s23115206.
[319] T. Yigitcanlar, R. Mehmood, J.M. Corchado, Green artificial intelligence: Towards an efficient, sustainable and equitable technology for smart cities and futures, Sustainability 13 (2021) 8952, http://dx.doi.org/10.3390/su13168952.
[320] A. Ortega-Fernández, R. Martín-Rojas, V.J. García-Morales, Artificial intelligence in the urban environment: Smart cities as models for developing innovation and sustainability, Sustainability 12 (2020) 7860, http://dx.doi.org/10.3390/su12197860.
[321] P. Rani, P. Sharma, I. Gupta, Toward a greener future: A survey on sustainable blockchain applications and impact, J. Env. Manag. 354 (2024) 120273, http://dx.doi.org/10.1016/j.jenvman.2024.120273.
[322] A. Sharifi, Z. Allam, S.E. Bibri, A.R. Khavarian-Garmsir, Smart cities and sustainable development goals (SDGs): A systematic literature review of co-benefits and trade-offs, Cities 146 (2024) 104659, http://dx.doi.org/10.1016/j.cities.2023.104659.
[323] A.S.L. França, J. Amato Neto, R.F. Gonçalves, C.M.V.B. Almeida, Proposing the use of blockchain to improve the solid waste management in small municipalities, J. Clean. Prod. 244 (2020) 118529, http://dx.doi.org/10.1016/j.jclepro.2019.118529.
[324] L. Belli, A. Cilfone, L. Davoli, G. Ferrari, P. Adorni, F. Di Nocera, A. Dall'Olio, C. Pellegrini, M. Mordacci, E. Bertolotti, IoT-enabled smart sustainable cities: Challenges and approaches, Smart Cities 3 (2020) 1039–1071, http://dx.doi.org/10.3390/smartcities3030052.
[325] S.E. Bibri, The IoT for smart sustainable cities of the future: An analytical framework for sensor-based big data applications for environmental sustainability, Sustain. Cities Soc. 38 (2018) 230–253, http://dx.doi.org/10.1016/j.scs.2017.12.034.
[326] M.N. Kamel Boulos, N.M. Al-Shorbaji, On the Internet of Things, smart cities and the WHO healthy cities, Int. J. Heal. Geogr. 13 (2014) 10, http://dx.doi.org/10.1186/1476-072X-13-10.
[327] R. Alsabt, Y.A. Adenle, H.M. Alshuwaikhat, Exploring the roles, future impacts, and strategic integration of artificial intelligence in the optimization of smart city—From systematic literature review to conceptual model, Sustainability 16 (2024) 3389, http://dx.doi.org/10.3390/su16083389.
[328] D.K. Das, Exploring the symbiotic relationship between digital transformation, infrastructure, service delivery, and governance for smart sustainable cities, Smart Cities 7 (2024) 806–835, http://dx.doi.org/10.3390/smartcities7020034.
[329] N. Rane, S. Choudhary, J. Rane, Artificial intelligence for enhancing resilience, J. Appl. Artif. Intell. 5 (2024) 1–33, http://dx.doi.org/10.48185/jaai.v5i2.1053.
[330] T. Alam, M. Tajammul, R. Gupta, Towards the sustainable development of smart cities through cloud computing, in: V. Piuri, R.N. Shaw, A. Ghosh, R. Islam (Eds.), AI and IoT for Smart City Applications. Studies in Computational Intelligence, Springer, Singapore, 2022, pp. 199–222, http://dx.doi.org/10.1007/978-981-16-7498-3_13.
[331] M.A. Ahad, S. Paiva, G. Tripathi, N. Feroz, Enabling technologies and sustainable smart cities, Sustain. Cities Soc. 61 (2020) 102301, http://dx.doi.org/10.1016/j.scs.2020.102301.
[332] K. Nowicka, Smart city logistics on cloud computing model, Procedia-Soc. Behav. Sci. 151 (2014) 266–281, http://dx.doi.org/10.1016/j.sbspro.2014.10.025.
[333] S.K. Rao, R. Prasad, Impact of 5G technologies on smart city implementation, Wirel. Pers. Commun. 100 (2018) 161–176, http://dx.doi.org/10.1007/s11277-018-5618-4.
[334] F. Li, T. Yigitcanlar, M. Nepal, K. Nguyen, F. Dur, Machine learning and remote sensing integration for leveraging urban sustainability: A review and framework, Sustain. Cities Soc. 96 (2023) 104653, http://dx.doi.org/10.1016/j.scs.2023.104653.
[335] T. Wellmann, A. Lausch, E. Andersson, S. Knapp, C. Cortinovis, J. Jache, S. Scheuer, P. Kremer, A. Mascarenhas, R. Kraemer, A. Haase, F. Schug, D. Haase, Remote sensing in urban planning: Contributions towards ecologically sound policies? Landsc, Urban Plan 204 (2020) 103921, http://dx.doi.org/10.1016/j.landurbplan.2020.103921.
[336] M. Netzband, W.L. Stefanov, C.L. Redman, Remote sensing as a tool for urban planning and sustainability, in: M. Netzband, W.L. Stefanov, C. Redman (Eds.), Applied Remote Sensing for Urban Planning, Governance and Sustainability, Springer, Berlin, Heidelberg, 2007, pp. 1–23, http://dx.doi.org/10.1007/978-3-540-68009-3_1.
[337] A. Rahman, Application of remote sensing and GIS technique for urban environmental management and sustainable development of Delhi, India, in: M. Netzband, W.L. Stefanov, C. Redman (Eds.), Applied Remote Sensing for Urban Planning, Governance and Sustainability, Springer, Berlin, Heidelberg, 2007, pp. 165–197, http://dx.doi.org/10.1007/978-3-540-68009-3_8.
[338] C.N. Mundia, M. Aniya, Analysis of land use/cover changes and urban expansion of Nairobi city using remote sensing and GIS, Int. J. Remote Sens. 26 (2005) 2831–2849, http://dx.doi.org/10.1080/01431160500117865.
[339] S.K. Lodhi, A.Y. Gill, H.K. Hussain, Green innovations: Artificial intelligence and sustainable materials in production, BULLET J. Multidisiplin. Ilmu. 3 (2024) 492–507.
[340] M. Chen, Q. Liu, S. Huang, C. Dang, Environmental cost control system of manufacturing enterprises using artificial intelligence based on value chain of circular economy, Enterp. Inf. Syst. 16 (2022) http://dx.doi.org/10.1080/17517575.2020.1856422.

[341] P. Dauvergne, Is artificial intelligence greening global supply chains? Exposing the political economy of environmental costs, Rev. Int. Polit. Econ. 29 (2022) 696–718, http://dx.doi.org/10.1080/09692290.2020.1814381.
[342] P. Centobelli, R. Cerchione, P. Del Vecchio, E. Oropallo, G. Secundo, Blockchain technology for bridging trust, traceability and transparency in circular supply chain, Inf. Manag. 59 (2022) 103508, http://dx.doi.org/10.1016/j.im.2021.103508.
[343] B. Esmaeilian, J. Sarkis, K. Lewis, S. Behdad, Blockchain for the future of sustainable supply chain management in Industry 4.0, Resour. Conserv. Recycl. 163 (2020) 105064, http://dx.doi.org/10.1016/j.resconrec.2020.105064.
[344] Z. Turskis, V. Šniokiene, IoT-driven transformation of circular economy efficiency: An overview, Math. Comput. Appl. 29 (2024) 49, http://dx.doi.org/10.3390/mca29040049.
[345] K. Voulgaridis, T. Lagkas, C.M. Angelopoulos, S.E. Nikoletseas, IoT and digital circular economy: Principles, applications, and challenges, Comput. Netw. 219 (2022) 109456, http://dx.doi.org/10.1016/j.comnet.2022.109456.
[346] C. Magrini, J. Nicolas, H. Berg, A. Bellini, E. Paolini, N. Vincenti, L. Campadello, A. Bonoli, Using Internet of Things and distributed ledger technology for digital circular economy enablement: The case of electronic equipment, Sustainability 13 (2021) 4982, http://dx.doi.org/10.3390/su13094982.
[347] M. Junaid, J. Du, M.S. Mubarik, F. Shahzad, Creating a sustainable future through Industry 4.0 technologies: Untying the role of circular economy practices and supply chain visibility, Bus. Strat. Env. 33 (2024) 5753–5775, http://dx.doi.org/10.1002/bse.3777.
[348] J. Cheng, H.S. Mahinder Singh, Y.-C. Zhang, S.-Y. Wang, The impact of business intelligence, big data analytics capability, and green knowledge management on sustainability performance, J. Clean. Prod. 429 (2023) 139410, http://dx.doi.org/10.1016/j.jclepro.2023.139410.
[349] M. Yenugula, S.K. Sahoo, S.S. Goswami, Cloud computing for sustainable development: An analysis of environmental, economic and social benefits, J. Futur. Sustain. 4 (2024) 59–66, http://dx.doi.org/10.5267/j.jfs.2024.1.005.
[350] V.K. Prasad, D. Dansana, M.D. Bhavsar, B. Acharya, V.C. Gerogiannis, A. Kanavos, Efficient resource utilization in IoT and cloud computing, Information 14 (2023) 619, http://dx.doi.org/10.3390/info14110619.
[351] J. Sun, T. Yang, Z. Xu, Assessing the implementation feasibility of intelligent production systems based on cloud computing, industrial internet of things and business social networks, Kybernetes 51 (2022) 2044–2064, http://dx.doi.org/10.1108/K-04-2021-0272.
[352] S. Mukherjee, S. Gupta, O. Rawlley, S. Jain, Leveraging big data analytics in 5G-enabled IoT and industrial IoT for the development of sustainable smart cities, Trans. Emerg. Telecommun. Technol. 33 (2022) http://dx.doi.org/10.1002/ett.4618.
[353] M.Q. Chau, X.P. Nguyen, T.T. Huynh, V.D. Chu, T.H. Le, T.P. Nguyen, D.T. Nguyen, Prospects of application of IoT-based advanced technologies in remanufacturing process towards sustainable development and energy-efficient use, Energy Sources A Recover. Util. Env. Eff. (2021) 1–25, http://dx.doi.org/10.1080/15567036.2021.1994057.
[354] S. Kumar, R.S. Meena, S. Sheoran, C.K. Jangir, M.K. Jhariya, A. Banerjee, A. Raj, Remote sensing for agriculture and resource management, in: M.K. Jhariya, R.S. Meena, A. Banerjee, S.N. Meena (Eds.), Natural Resources Conservation and Advances for Sustainability, Elsevier, 2022, pp. 91–135, http://dx.doi.org/10.1016/B978-0-12-822976-7.00012-0.
[355] I.E. Olorunfemi, J.T. Fasinmirin, A.A. Olufayo, A.A. Komolafe, GIS and remote sensing-based analysis of the impacts of land use/land cover change (LULCC) on the environmental sustainability of Ekiti State, southwestern Nigeria, Env. Dev. Sustain. 22 (2020) 661–692, http://dx.doi.org/10.1007/s10668-018-0214-z.
[356] J. Cowls, A. Tsamados, M. Taddeo, L. Floridi, The AI gambit: leveraging artificial intelligence to combat climate change—opportunities, challenges, and recommendations, AI Soc. 38 (2023) 283–307, http://dx.doi.org/10.1007/s00146-021-01294-x.
[357] K. Sahil, P. Mehta, S. Kumar Bhardwaj, L.K. Dhaliwal, Development of mitigation strategies for the climate change using artificial intelligence to attain sustainability, in: A. Srivastav, A. Dubey, A. Kumar, S.K. Narang, M.A. Khan (Eds.), Visualization Techniques for Climate Change with Machine Learning and Artificial Intelligence, Elsevier, 2023, pp. 421–448, http://dx.doi.org/10.1016/B978-0-323-99714-0.00021-2.
[358] W. Leal Filho, T. Wall, S.A. Rui Mucova, G.J. Nagy, A.-L. Balogun, J.M. Luetz, A.W. Ng, M. Kovaleva, F.M. Safiul Azam, F. Alves, Z. Guevara, N.R. Matandirotya, A. Skouloudis, A. Tzachor, K. Malakar, O. Gandhi, Deploying artificial intelligence for climate change adaptation, Technol. Forecast. Soc. Change 180 (2022) 121662, http://dx.doi.org/10.1016/j.techfore.2022.121662.
[359] A. Al Sadawi, B. Madani, S. Saboor, M. Ndiaye, G. Abu-Lebdeh, A comprehensive hierarchical blockchain system for carbon emission trading utilizing blockchain of things and smart contract, Technol. Forecast. Soc. Change 173 (2021) 121124, http://dx.doi.org/10.1016/j.techfore.2021.121124.
[360] J. Woo, R. Fatima, C.J. Kibert, R.E. Newman, Y. Tian, R.S. Srinivasan, Applying blockchain technology for building energy performance measurement, reporting, and verification (MRV) and the carbon credit market: A review of the literature, Build. Env. 205 (2021) 108199, http://dx.doi.org/10.1016/j.buildenv.2021.108199.
[361] S.-K. Kim, J.-H. Huh, Blockchain of carbon trading for UN sustainable development goals, Sustainability 12 (2020) 4021, http://dx.doi.org/10.3390/su12104021.
[362] A. Salam, Internet of Things for environmental sustainability and climate change, in: Internet of Things for Sustainable Community Development, Springer, Cham, 2024, pp. 33–69, http://dx.doi.org/10.1007/978-3-031-62162-8_2.
[363] S.L. Ullo, G.R. Sinha, Advances in smart environment monitoring systems using IoT and sensors, Sensors 20 (2020) 3113, http://dx.doi.org/10.3390/s20113113.
[364] S.H. Alsamhi, O. Ma, M.S. Ansari, Q. Meng, Greening internet of things for greener and smarter cities: a survey and future prospects, Telecommun. Syst. 72 (2019) 609–632, http://dx.doi.org/10.1007/s11235-019-00597-1.
[365] S. Fang, L. Da Xu, Y. Zhu, J. Ahati, H. Pei, J. Yan, Z. Liu, An integrated system for regional environmental monitoring and management based on Internet of Things, IEEE Trans. Ind. Informatics 10 (2014) 1596–1605, http://dx.doi.org/10.1109/TII.2014.2302638.
[366] J.D. Ford, S.E. Tilleard, L. Berrang-Ford, M. Araos, R. Biesbroek, A.C. Lesnikowski, G.K. MacDonald, A. Hsu, C. Chen, L. Bizikova, Big data has big potential for applications to climate change adaptation, Proc. Natl. Acad. Sci. 113 (2016) 10729–10732, http://dx.doi.org/10.1073/pnas.1614023113.
[367] R. Lokers, R. Knapen, S. Janssen, Y. van Randen, J. Jansen, Analysis of big data technologies for use in agro-environmental science, Env. Model. Softw. 84 (2016) 494–504, http://dx.doi.org/10.1016/j.envsoft.2016.07.017.
[368] H.-D. Guo, L. Zhang, L.-W. Zhu, Earth observation big data for climate change research, Adv. Clim. Chang. Res. 6 (2015) 108–117, http://dx.doi.org/10.1016/j.accre.2015.09.007.
[369] E. Levin, N. Beisekenov, M. Wilson, M. Sadenova, R. Nabaweesi, L. Nguyen, Empowering climate resilience: Leveraging cloud computing and big data for community climate change impact service (C3IS), Remote. Sens 15 (2023) 5160, http://dx.doi.org/10.3390/rs15215160.
[370] A. Mohamed M., S. Pillutla, Cloud computing: a collaborative green platform for the knowledge society, VINE 44 (2014) 357–374, http://dx.doi.org/10.1108/VINE-07-2013-0038.
[371] G.F. Huseien, K.W. Shah, Potential applications of 5G network technology for climate change control: A scoping review of Singapore, Sustainability 13 (2021) 9720, http://dx.doi.org/10.3390/su13179720.
[372] B.K. Gesami, J. Nunoo, Artificial intelligence in marine ecosystem management: addressing climate threats to Kenya's blue economy, Front. Mar. Sci. 11 (2024) http://dx.doi.org/10.3389/fmars.2024.1404104.
[373] K. Bakker, Smart oceans: Artificial intelligence and marine protected area governance, Earth Syst. Gov. 13 (2022) 100141, http://dx.doi.org/10.1016/j.esg.2022.100141.
[374] D.A. Isabelle, M. Westerlund, A review and categorization of artificial intelligence-based opportunities in wildlife, Ocean. Land Conserv. Sustain. 14 (2022) 1979, http://dx.doi.org/10.3390/su14041979.
[375] J. Leape, M. Abbott, H. Sakaguchi, A. Brett, L. Cao, K. Chand, Y. Golbuu, T. Martin, J. Mayorga, M.S. Myksvoll, Technology, data and new models for sustainably managing ocean resources, in: J. Lubchenco, P.M. Haugan (Eds.), The Blue Compendium, Springer International Publishing, Cham, 2023, pp. 185–211, http://dx.doi.org/10.1007/978-3-031-16277-0_6.
[376] S. Bhattacharya, N. Victor, R. Chengoden, M. Ramalingam, G. Selvi, P. Maddikunta, P. Donta, S. Dustdar, R. Jhaveri, T. Gadekallu, Blockchain for Internet of Underwater Things: State-of-the-art, applications, challenges, and future directions, Sustainability 14 (2022) 15659, http://dx.doi.org/10.3390/su142315659.
[377] M. Manoj, V. Dhilip Kumar, M. Arif, E.-R. Bulai, P. Bulai, O. Geman, State of the art techniques for water quality monitoring systems for fish ponds using IoT and underwater sensors: A review, Sensors 22 (2022) 2088, http://dx.doi.org/10.3390/s22062088.
[378] D.R. Prapti, A.R. Mohamed Shariff, H. Che Man, N.M. Ramli, T. Perumal, M. Shariff, Internet of Things (IoT)-based aquaculture: An overview of IoT application on water quality monitoring, Rev. Aquac. 14 (2022) 979–992, http://dx.doi.org/10.1111/raq.12637.
[379] D.T. Vo, X.P. Nguyen, T.D. Nguyen, R. Hidayat, T.T. Huynh, D.T. Nguyen, A review on the internet of thing (IoT) technologies in controlling ocean environment, Energy Sources A Recover. Util. Env. Eff. (2021) 1–19, http://dx.doi.org/10.1080/15567036.2021.1960932.
[380] M. Jahanbakht, W. Xiang, L. Hanzo, M. Rahimi Azghadi, Internet of underwater things and big marine data analytics—A comprehensive survey, IEEE Commun. Surv. Tutorials 23 (2021) 904–956, http://dx.doi.org/10.1109/COMST.2021.3053118.
[381] A. Trice, C. Robbins, N. Philip, M. Rumsey, Challenges and Opportunities for Ocean Data to Advance Conservation and Management, Ocean Conservancy, 2021.
[382] N. Alsharabi, J. Ktari, T. Frikha, A. Alayba, A.J. Alzahrani, A. Jadi, H. Hamam, Using blockchain and AI technologies for sustainable, biodiverse, and transparent fisheries of the future, J. Cloud Comput. 13 (2024) 135, http://dx.doi.org/10.1186/s13677-024-00696-8.

[383] P. Joshi, A. Bhatt, G. Aggarwal, Fishers 4.0: Revolutionizing contemporary Fisheries management through industry 4.0 integration, in: 2024 International Conference on Healthcare Innovations, Software and Engineering Technologies, HISET, IEEE, 2024, pp. 150–155, http://dx.doi.org/10.1109/HISET61796.2024.00054.

[384] A. Raihan, Artificial intelligence and machine learning applications in forest management and biodiversity conservation, Nat. Resour. Conserv. Res. 6 (2023) 3825, http://dx.doi.org/10.24294/nrcr.v6i2.3825.

[385] A. Kabanov, V. Kramar, Marine Internet of Things platforms for interoperability of marine robotic agents: An overview of concepts and architectures, J. Mar. Sci. Eng. 10 (2022) 1279, http://dx.doi.org/10.3390/jmse10091279.

[386] Z. Yang, X. Yu, S. Dedman, M. Rosso, J. Zhu, J. Yang, Y. Xia, Y. Tian, G. Zhang, J. Wang, UAV remote sensing applications in marine monitoring: Knowledge visualization and review, Sci. Total. Env. 838 (2022) 155939, http://dx.doi.org/10.1016/j.scitotenv.2022.155939.

[387] B. El Mahrad, A. Newton, J. Icely, I. Kacimi, S. Abalansa, M. Snoussi, Contribution of remote sensing technologies to a Holistic Coastal and marine environmental management framework: A review, Remote. Sens. 12 (2020) 2313, http://dx.doi.org/10.3390/rs12142313.

[388] J. Hedley, C. Roelfsema, I. Chollett, A. Harborne, S. Heron, S. Weeks, W. Skirving, A. Strong, C. Eakin, T. Christensen, V. Ticzon, S. Bejarano, P. Mumby, Remote sensing of coral reefs for monitoring and management: A review, Remote. Sens 8 (2016) 118, http://dx.doi.org/10.3390/rs8020118.

[389] W. Ouellette, W. Getinet, Remote sensing for Marine Spatial Planning and Integrated Coastal Areas Management: Achievements, challenges, opportunities and future prospects, Remote. Sens. Appl. Soc. Env. 4 (2016) 138–157, http://dx.doi.org/10.1016/j.rsase.2016.07.003.

[390] H. Nagendra, R. Lucas, J.P. Honrado, R.H.G. Jongman, C. Tarantino, M. Adamo, P. Mairota, Remote sensing for conservation monitoring: assessing protected areas, habitat extent, habitat condition, species diversity, and threats, Ecol. Indic. 33 (2013) 45–59, http://dx.doi.org/10.1016/j.ecolind.2012.09.014.

[391] V. Klemas, Remote sensing techniques for Studying Coastal ecosystems: An overview, J. Coast. Res.. 27 (2010) 2, http://dx.doi.org/10.2112/JCOASTRES-D-10-00103.1.

[392] K.N. Shivaprakash, N. Swami, S. Mysorekar, R. Arora, A. Gangadharan, K. Vohra, M. Jadeyegowda, J.M. Kiesecker, Potential for artificial intelligence (AI) and machine learning (ML) applications in biodiversity conservation, managing forests, and related services in India, Sustainability 14 (2022) 7154, http://dx.doi.org/10.3390/su14127154.

[393] R. Bennett, T. Miller, M. Pickering, A.-K. Kara, Hybrid approaches for smart contracts in land administration: Lessons from three blockchain proofs-of-concept, Land 10 (2021) 220, http://dx.doi.org/10.3390/land10020220.

[394] V.L. Lemieux, Evaluating the use of blockchain in land transactions: An archival science perspective, Eur. Prop. Law J. 6 (2017) 392–440, http://dx.doi.org/10.1515/eplj-2017-0019.

[395] D. Kumar, S. Shanthakumar, M. Banerjee, M.S. Hanspal, IoT based models in healthy natural resource management: Healthy soils for healthy food productions, in: J.A. Parray, A.K. Haghi, G. Meraj (Eds.), IoT-Based Models for Sustainable Environmental Management, Springer, Cham, 2024, pp. 211–242, http://dx.doi.org/10.1007/978-3-031-74374-0_11.

[396] T. Korra, A. Srushtideep, S. Bharati, S. Anand, M. Udayasri, O. Ghosh, P. Thirunarayanan, Future directions of IoT-based models for management of natural resources, in: J.A. Parray, A.K. Haghi, G. Meraj (Eds.), IoT-Based Models for Sustainable Environmental Management, Springer, Cham, 2024, pp. 95–109, http://dx.doi.org/10.1007/978-3-031-74374-0_5.

[397] J.H. Park, M.H. Han, Enhancing livestock management with IoT-based wireless sensor networks: A comprehensive approach for health monitoring, location tracking, behavior analysis, and environmental optimization, J. Sustain. Urban Futur. 13 (2023) 34–46.

[398] B.I. Akhigbe, K. Munir, O. Akinade, L. Akanbi, L.O. Oyedele, IoT technologies for livestock management: A review of present status, opportunities, and future trends, Big Data Cogn. Comput. 5 (2021) 10, http://dx.doi.org/10.3390/bdcc5010010.

[399] P. Bhambri, S. Kautish, Technological advancements in promoting ecosystem health, in: W. Leal Filho, S. Kautish, T. Wall, S. Rewhorn, S.K. Paul (Eds.), Digital Technologies To Implement the UN Sustainable Development Goals, Springer, Cham, 2024, pp. 413–432, http://dx.doi.org/10.1007/978-3-031-68427-2_21.

[400] J.E. Herrick, K.C. Urama, J.W. Karl, J. Boos, M.-V.V. Johnson, K.D. Shepherd, J. Hempel, B.T. Bestelmeyer, J. Davies, J.L. Guerra, C. Kosnik, D.W. Kimiti, A.L. Ekai, K. Muller, L. Norfleet, N. Ozor, T. Reinsch, J. Sarukhan, L.T. West, The global Land-Potential Knowledge System (LandPKS): Supporting evidence-based, site-specific land use and management through cloud computing, mobile applications, and crowdsourcing, J. Soil Water Conserv. 68 (2013) 5A–12A, http://dx.doi.org/10.2489/jswc.68.1.5A.

[401] L. Tomaszewski, R. Kołakowski, Mobile services for smart agriculture and forestry, biodiversity monitoring, and water management: Challenges for 5G/6G networks, Telecom 4 (2023) 67–99, http://dx.doi.org/10.3390/telecom4010006.

[402] M.M. Nizamani, Q. Zhang, G. Muhae-Ud-Din, M. Awais, M. Qayyum, M. Farhan, M. Jabran, Y. Wang, Application of GIS and remote-sensing technology in ecosystem services and biodiversity conservation, in: U.A. Bhatti, H. Mengxing, J. Li, S.U. Bazai, M. Aamir (Eds.), Deep Learning for Multimedia Processing Applications, CRC Press, Boca Raton, 2024.

[403] M. Gabriele, R. Brumana, M. Previtali, A. Cazzani, A combined GIS and remote sensing approach for monitoring climate change-related land degradation to support landscape preservation and planning tools: the Basilicata case study, Appl. Geomatics 15 (2023) 497–532, http://dx.doi.org/10.1007/s12518-022-00437-z.

[404] R.A. Rose, D. Byler, J.R. Eastman, E. Fleishman, G. Geller, S. Goetz, L. Guild, H. Hamilton, M. Hansen, R. Headley, J. Hewson, N. Horning, B.A. Kaplin, N. Laporte, A. Leidner, P. Leimgruber, J. Morisette, J. Musinsky, L. Pintea, A. Prados, V.C. Radeloff, M. Rowen, S. Saatchi, S. Schill, K. Tabor, W. Turner, A. Vodacek, J. Vogelmann, M. Wegmann, D. Wilkie, C. Wilson, Ten ways remote sensing can contribute to conservation, Conserv. Biol. 29 (2015) 350–359, http://dx.doi.org/10.1111/cobi.12397.

[405] M. Giovanardi, AI for peace: mitigating the risks and enhancing opportunities, Data Policy 6 (2024) e41, http://dx.doi.org/10.1017/dap.2024.37.

[406] A.L. Lira Cortes, C. Fuentes Silva, Artificial intelligence models for crime prediction in urban spaces, Mach. Learn. Appl. Int. J. 8 (2021) 1–13, http://dx.doi.org/10.5121/mlaij.2021.8101.

[407] E. Pauwels, Artificial intelligence and data capture technologies in violence and conflict prevention, Glob. Cent. Coop. Secur. (2020) 1–20.

[408] M. Horowitz, G. Allen, E. Saravalle, A. Cho, A. Frederick, P. Charre, Artifcial Intelligence and International Security, Center for a New American Security, Washington, 2018.

[409] M. Arif, Issues and proposed IT solution's by united nations sustainable development goals for peace, justice and strong institutions in relation to Malaysia context, Int. J. Sustain. Dev. Res. 8 (2022) 114–120, http://dx.doi.org/10.11648/j.ijsdr.20220804.11.

[410] A.R. Garcia, AI, IoT, big data, and technologies in digital economy with blockchain at sustainable work satisfaction to smart mankind: Access to 6th dimension of human rights, in: N. Lopes (Ed.), Smart Governance for Cities: Perspectives and Experiences. EAI/Springer Innovations in Communication and Computing, Springer, Cham, 2020, pp. 83–131, http://dx.doi.org/10.1007/978-3-030-22070-9_6.

[411] J.W. Bagby, D. Reitter, P. Chwistek, An emerging political economy of the blockchain: Enhancing regulatory opportunities, UMKC Law Rev. 88 (2019).

[412] K. Hughes, Blockchain, the greater good, and human and civil rights, Metaphilosophy 48 (2017) 654–665, http://dx.doi.org/10.1111/meta.12271.

[413] A. Cheshmehzangi, ICT, Cities, and Reaching Positive Peace, Urban Sustainability, Springer Nature Singapore, Singapore, 2022, http://dx.doi.org/10.1007/978-981-19-3167-3.

[414] A. Završnik, Algorithmic justice: Algorithms and big data in criminal justice settings, Eur. J. Criminol. 18 (2021) 623–642, http://dx.doi.org/10.1177/1477370819876762.

[415] K. Hannah-Moffat, Algorithmic risk governance: Big data analytics, race and information activism in criminal justice debates, Theor. Criminol. 23 (2019) 453–470, http://dx.doi.org/10.1177/1362480618763582.

[416] F. Mancini, New Technology and the Prevention of Violence and Conflict, New Jersey, 2013.

[417] H. Bhatt, R. Bahuguna, S. Swami, R. Singh, A. Gehlot, S.V. Akram, L.R. Gupta, A.K. Thakur, N. Priyadarshi, B. Twala, Integrating industry 4.0 technologies for the administration of courts and justice dispensation—a systematic review, Humanit. Soc. Sci. Commun. 11 (2024) 1076, http://dx.doi.org/10.1057/s41599-024-03587-0.

[418] S. Wang, Y. Li, M.B. Khaskheli, Innovation helps with sustainable business, law, and digital technologies: Economic development and dispute resolution, Sustainability 16 (2024) 3910, http://dx.doi.org/10.3390/su16103910.

[419] M.M. Bennett, J.K. Chen, L.F. Alvarez León, C.J. Gleason, The politics of pixels: A review and agenda for critical remote sensing, Prog. Hum. Geogr. 46 (2022) 729–752, http://dx.doi.org/10.1177/03091325221074691.

[420] R. Avtar, A. Kouser, A. Kumar, D. Singh, P. Misra, A. Gupta, A.P. Yunus, P. Kumar, B.A. Johnson, R. Dasgupta, N. Sahu, A. Besse Rimba, Remote sensing for international peace and security: Its role and implications, Remote. Sens 13 (2021) 439, http://dx.doi.org/10.3390/rs13030439.

[421] D. Rothe, D. Shim, Sensing the ground: On the global politics of satellite-based activism, Rev. Int. Stud. 44 (2018) 414–437, http://dx.doi.org/10.1017/S0260210517000602.

[422] E. Convergne, M.R. Snyder, Making maps to make peace: Geospatial technology as a tool for UN peacekeeping, Int. Peacekeeping 22 (2015) 565–586, http://dx.doi.org/10.1080/13533312.2015.1094193.

[423] J. Bughin, E. Hazan, S. Ramaswamy, M. Chui, T. Allas, P. Dahlström, N. Henke, M. Trench, Artificial Intelligence: The Next Digital Frontier?, McKinsey Global Institute, 2017.

[424] A. Rejeb, J.G. Keogh, S.J. Simske, T. Stafford, H. Treiblmaier, Potentials of blockchain technologies for supply chain collaboration: a conceptual framework, Int. J. Logist. Manag. 32 (2021) 973–994, http://dx.doi.org/10.1108/IJLM-02-2020-0098.

[425] J.-S. Kim, N. Shin, The impact of blockchain technology application on supply chain partnership and performance, Sustainability 11 (2019) 6181, http://dx.doi.org/10.3390/su11216181.

[426] A.K. Srivastav, P. Das, A.K. Srivastava, Future trends, innovations, and global collaboration, in: Biotech and IoT, A Press, Berkeley, CA, CA, 2024, pp. 309–398, http://dx.doi.org/10.1007/979-8-8688-0527-1_10.

[427] P. Friess, O. Vermesan, Building the Hyperconnected Society- Internet of Things Research and Innovation Value Chains, Ecosystems and Markets, River Publishers, New York, 2022, http://dx.doi.org/10.1201/9781003337454.

[428] R.H. Hamilton, W.A. Sodeman, The questions we ask: Opportunities and challenges for using big data analytics to strategically manage human capital resources, Bus. Horiz. 63 (2020) 85–95, http://dx.doi.org/10.1016/j.bushor.2019.10.001.

[429] R. Dubey, A. Gunasekaran, S.J. Childe, D. Roubaud, S.Fosso. Wamba, M. Giannakis, C. Foropon, Big data analytics and organizational culture as complements to swift trust and collaborative performance in the humanitarian supply chain, Int. J. Prod. Econ.. 210 (2019) 120–136, http://dx.doi.org/10.1016/j.ijpe.2019.01.023.

[430] S.K. Singh, A.-N. El-Kassar, Role of big data analytics in developing sustainable capabilities, J. Clean. Prod. 213 (2019) 1264–1273, http://dx.doi.org/10.1016/j.jclepro.2018.12.199.

[431] T. Papadopoulos, A. Gunasekaran, R. Dubey, N. Altay, S.J. Childe, S. Fosso-Wamba, The role of big data in explaining disaster resilience in supply chains for sustainability, J. Clean. Prod. 142 (2017) 1108–1118, http://dx.doi.org/10.1016/j.jclepro.2016.03.059.

[432] V. Morabito, Big Data and Analytics, Springer International Publishing, Cham, 2015, http://dx.doi.org/10.1007/978-3-319-10665-6.

[433] A. Adel, N.H.S. Alani, Human-centric collaboration and industry 5.0 framework in smart cities and communities: Fostering sustainable development goals 3 4, 9, and 11 in society 5.0, Smart Cities 7 (2024) 1723–1775, http://dx.doi.org/10.3390/smartcities7040068.

[434] B. Almadani, S. Alissa, R. Alshareef, F. Aliyu, E. Al-Nahari, A survey of middleware adoption in nonprofit sectors: A sustainable development perspective, Sustainability 16 (2024) 8904, http://dx.doi.org/10.3390/su16208904.

[435] S. Nwokolo, E. Eyime, A. Obiwulu, J. Ogbulezie, Africa's path to sustainability: Harnessing technology, policy, and collaboration, Trends Renew. Energy 10 (2024) 98–131, http://dx.doi.org/10.17737/tre.2024.10.1.00166.

[436] G.A. Akpakwu, B.J. Silva, G.P. Hancke, A.M. Abu-Mahfouz, A survey on 5G networks for the Internet of Things: Communication technologies and challenges, IEEE Access 6 (2018) 3619–3647, http://dx.doi.org/10.1109/ACCESS.2017.2779844.

[437] I.F. Akyildiz, S. Nie, S.-C. Lin, M. Chandrasekaran, 5G roadmap: 10 key enabling technologies, Comput. Netw. 106 (2016) 17–48, http://dx.doi.org/10.1016/j.comnet.2016.06.010.

[438] A. Gupta, R.K. Jha, A survey of 5G network: Architecture and emerging technologies, IEEE Access 3 (2015) 1206–1232, http://dx.doi.org/10.1109/ACCESS.2015.2461602.

[439] Stavros Kalogiannidis, Analyzing the challenges of cross-border disaster response and management: A European perspective, J. Risk Anal. Cris. Response 14 (2024) http://dx.doi.org/10.54560/jracr.v14i2.470.

[440] A. Senthil Kumar, S. Camacho, N.D. Searby, J. Teuben, W. Balogh, Coordinated capacity development to maximize the contributions of space science, technology, and its applications in support of implementing global sustainable development agendas—A conceptual framework, Space Policy 51 (2020) 101346, http://dx.doi.org/10.1016/j.spacepol.2019.101346.

[441] M.A. Brovelli, M. Cannata, M. Rogora, Simile, A geospatial enabler of the monitoring of sustainable development goal 6 (Ensure availability and sustainability of water for all), Int. Arch. Photogramm. Remote. Sens. Spat. Inf. Sci. XLII-4/W20 (2019) 3–10, http://dx.doi.org/10.5194/isprs-archives-XLII-4-W20-3-2019.

[442] P. Perera, S. Selvanathan, J. Bandaralage, J.-J. Su, The impact of digital inequality in achieving sustainable development: a systematic literature review, Equal. Divers. Incl. an Int. J. 42 (2023) 805–825, http://dx.doi.org/10.1108/EDI-08-2022-0224.

[443] A. Salam, Internet of Things for sustainability: Perspectives in privacy, cybersecurity, and future trends, in: Internet of Things for Sustainable Community Development, Springer, Cham, 2024, pp. 299–326, http://dx.doi.org/10.1007/978-3-031-62162-8_10.

[444] M.T. Nguyen, M.Q. Tran, Balancing security and privacy in the digital age: An in-depth analysis of legal and regulatory frameworks impacting cybersecurity practices, Int. J. Intell. Autom. Comput. 6 (2023) 1–12.

[445] A. Holzinger, E. Weippl, A.M. Tjoa, P. Kieseberg, Digital transformation for sustainable development goals (SDGs) - A security, safety and privacy perspective on AI, in: A. Holzinger, P. Kieseberg, A.M. Tjoa, E. Weippl (Eds.), Machine Learning and Knowledge Extraction, Springer, Cham, 2021, pp. 1–20, http://dx.doi.org/10.1007/978-3-030-84060-0_1.

[446] M. Ryan, J. Antoniou, L. Brooks, T. Jiya, K. Macnish, B. Stahl, The ethical balance of using smart information systems for promoting the united nations' sustainable development goals, Sustainability 12 (2020) 4826, http://dx.doi.org/10.3390/su12124826.

[447] S. Sparviero, M. Ragnedda, Towards digital sustainability: the long journey to the sustainable development goals 2030, Digit. Policy, Regul. Gov. 23 (2021) 216–228, http://dx.doi.org/10.1108/DPRG-01-2021-0015.

[448] K. Chetty, L. Qigui, N. Gcora, J. Josie, L. Wenwei, C. Fang, Bridging the digital divide: measuring digital literacy, Economics 12 (2018) http://dx.doi.org/10.5018/economics-ejournal.ja.2018-23.

[449] R. Sharma, A.-R. Fantin, N. Prabhu, C. Guan, A. Dattakumar, Digital literacy and knowledge societies: A grounded theory investigation of sustainable development, Telecomm. Policy 40 (2016) 628–643, http://dx.doi.org/10.1016/j.telpol.2016.05.003.

[450] M. Bhatia, N. Meenakshi, P. Kaur, A. Dhir, Digital technologies and carbon neutrality goals: An in-depth investigation of drivers, barriers, and risk mitigation strategies, J. Clean. Prod. 451 (2024) 141946, http://dx.doi.org/10.1016/j.jclepro.2024.141946.

[451] M. Burri, K. Kugler, Digitization, regulatory barriers and sustainable development, SSRN Electron. J. (2023) http://dx.doi.org/10.2139/ssrn.4424470.

[452] K. Govindan, K.M. Shankar, D. Kannan, Achieving sustainable development goals through identifying and analyzing barriers to industrial sharing economy: A framework development, Int. J. Prod. Econ.. 227 (2020) 107575, http://dx.doi.org/10.1016/j.ijpe.2019.107575.

[453] R. Kumar, S.K. Gupta, H.-C. Wang, C.S. Kumari, S.S.V.P. Korlam, From efficiency to sustainability: Exploring the potential of 6G for a greener future, Sustainability 15 (2023) 16387, http://dx.doi.org/10.3390/su152316387.

[454] S. Milan, Resistance in the data-driven society, Internet Policy Rev. 13 (2024) http://dx.doi.org/10.14763/2024.4.1811.

[455] S. Ibeneme, N. Ukor, M. Ongom, T. Dasa, D. Muneene, J. Okeibunor, Strengthening capacities among digital health leaders for the development and implementation of national digital health programs in Nigeria, BMC Proc. 14 (2020) 9, http://dx.doi.org/10.1186/s12919-020-00193-1.

[456] S. Sharmin, B. Faith, P. Prieto Martín, B. Ramalingam, The Contribution of Digital Technologies To Service Delivery: An Evidence Review, 2017.

[457] E. Roberts, J. Farrington, S. Skerratt, Evaluating new digital technologies through a framework of resilience, Scott. Geogr. J. 131 (2015) 253–264, http://dx.doi.org/10.1080/14702541.2015.1068947.

[458] A. Afzal, S. Khan, S. Daud, Z. Ahmad, A. Butt, Addressing the digital divide: Access and use of technology in education, J. Soc. Sci. Rev. 3 (2023) 883–895, http://dx.doi.org/10.54183/jssr.v3i2.326.

[459] J.A.G.M. Van Dijk, Digital divide: Impact of access, Int. Encycl. Media Eff. (2017) 1–11, http://dx.doi.org/10.1002/9781118783764.wbieme0043.

[460] I.J. Borges do Nascimento, H. Abdulazeem, L.T. Vasanthan, E.Z. Martinez, M.L. Zucoloto, L. Østengaard, N. Azzopardi-Muscat, T. Zapata, D. Novillo-Ortiz, Barriers and facilitators to utilizing digital health technologies by healthcare professionals, Npj Digit. Med. 6 (2023) 161, http://dx.doi.org/10.1038/s41746-023-00899-4.

[461] W. Leal Filho, D.G. Vidal, C. Chen, M. Petrova, M.A.P. Dinis, P. Yang, S. Rogers, L. Álvarez-Castañón, I. Djekic, A. Sharifi, S. Neiva, An assessment of requirements in investments, new technologies, and infrastructures to achieve the SDGs, Env. Sci. Eur. 34 (2022) 58, http://dx.doi.org/10.1186/s12302-022-00629-9.

[462] S. Wang, M. Asif, M.F. Shahzad, M. Ashfaq, Data privacy and cybersecurity challenges in the digital transformation of the banking sector, Comput. Secur. 147 (2024) 104051, http://dx.doi.org/10.1016/j.cose.2024.104051.

[463] Adedoyin Tolulope Oyewole, Bisola Beatrice Oguejiofor, Nkechi Emmanuella Eneh, Chidiogo Uzoamaka Akpuokwe, Seun Solomon Bakare, Data privacy laws and their impact on financial technology companies: a review, Comput. Sci. IT Res. J. 5 (2024) 628–650, http://dx.doi.org/10.51594/csitrj.v5i3.911.

[464] E.A. Gromova, N.S. Koneva, E.V. Titova, Legal barriers to the implementation of digital industry (industry 4.0) components and ways to overcome them, J. World Intellect. Prop. 25 (2022) 186–205, http://dx.doi.org/10.1111/jwip.12215.

[465] D. Grande, X. Luna Marti, R. Feuerstein-Simon, R.M. Merchant, D.A. Asch, A. Lewson, C.C. Cannuscio, Health policy and privacy challenges associated with digital technology, JAMA Netw. Open 3 (2020) e208285, http://dx.doi.org/10.1001/jamanetworkopen.2020.8285.

[466] E.M. dos Santos, N. Reinhard, Electronic government interoperability: Identifying the barriers for frameworks adoption, Soc. Sci. Comput. Rev. 30 (2012) 71–82, http://dx.doi.org/10.1177/0894439310392196.

[467] U. Gasser, J.G. Palfrey, Breaking down digital barriers: When and how ICT interoperability drives innovation, SSRN Electron. J. (2008) http://dx.doi.org/10.2139/ssrn.1033226.

[468] S. Sun, X. Zheng, J. Villalba-Díez, J. Ordieres-Meré, Data handling in industry 4.0: Interoperability based on distributed ledger technology, Sensors 20 (2020) 3046, http://dx.doi.org/10.3390/s20113046.

[469] D. Rasmussen Pennington, L. Cagnazzo, Connecting the silos: Implementations and perceptions of linked data across European libraries, J. Doc. 75 (2019) 643–666, http://dx.doi.org/10.1108/JD-07-2018-0117.

[470] P. Borowski, Digitization, digital twins, blockchain, and industry 4.0 as elements of management process in enterprises in the energy sector, Energies 14 (2021) 1885, http://dx.doi.org/10.3390/en14071885.

[471] P.K. Sharma, N. Kumar, J.H. Park, Blockchain technology toward green IoT: Opportunities and challenges, IEEE Netw. 34 (2020) 263–269, http://dx.doi.org/10.1109/MNET.001.1900526.
[472] D. Sundar, K. Mathiyazhagan, V. Agarwal, M. Janardhanan, A. Appolloni, From linear to a circular economy in the e-waste management sector: Experience from the transition barriers in the United Kingdom, Bus. Strat. Env. 32 (2023) 4282–4298, http://dx.doi.org/10.1002/bse.3365.
[473] J. Jangre, K. Prasad, D. Patel, Analysis of barriers in e-waste management in developing economy: an integrated multiple-criteria decision-making approach, Env. Sci. Pollut. Res. 29 (2022) 72294–72308, http://dx.doi.org/10.1007/s11356-022-21363-y.
[474] J. VanYken, N.J. Boxall, K.Y. Cheng, A.N. Nikoloski, N.R. Moheimani, A.H. Kaksonen, E-waste recycling and resource recovery: A review on technologies, barriers and enablers with a focus on oceania, Met. (Basel) 11 (2021) 1313, http://dx.doi.org/10.3390/met11081313.
[475] N.L. Rane, S.K. Mallick, Ö. Kaya, J. Rane, From challenges to implementation and acceptance: Addressing key barriers in artificial intelligence, machine learning, and deep learning, Appl. Mach. Learn. Deep. Learn.: Archit. Tech. (2024) 153–166, http://dx.doi.org/10.70593/978-81-981271-4-3_8.
[476] S. Kumar, A.K. Verma, A. Mirza, Digital revolution, artificial intelligence, and ethical challenges, in: Digital Transformation, Artificial Intelligence and Society, in: Frontiers of Artificial Intelligence, Ethics and Multidisciplinary Applications, Springer, Singapore, 2024, pp. 161–177, http://dx.doi.org/10.1007/978-981-97-5656-8_11.
[477] A. Kelly-Lyth, Challenging biased hiring algorithms, Oxf. J. Leg. Stud. 41 (2021) 899–928, http://dx.doi.org/10.1093/ojls/gqab006.
[478] J. Coccoli, The challenges of new technologies in the implementation of Human Rights: An analysis of some critical issues in the Digital Era, Peace Hum. Rights Gov. 1 (2017) 223–250, http://dx.doi.org/10.14658/pupj-phrg-2017-2-4.
[479] N. Leesakul, A.-M. Oostveen, I. Eimontaite, M.L. Wilson, R. Hyde, Workplace 4.0: Exploring the implications of technology adoption in digital manufacturing on a sustainable workforce, Sustainability 14 (2022) 3311, http://dx.doi.org/10.3390/su14063311.
[480] M. Aanestad, A. Kankanhalli, L. Maruping, M.-S. Pang, S. Ram, Digital technologies and social justice, MISQ 2 (2021) 1–8.
[481] S. Svetsky, O. Moravcik, Some barriers regarding the sustainability of digital technology for long-term teaching, in: K. Arai, R. Bhatia, S. Kapoor (Eds.), Advances in Intelligent Systems and Computing, Springer, Cham, 2019, pp. 950–961, http://dx.doi.org/10.1007/978-3-030-02686-8_71.
[482] M. van der Velden, Digitalisation and the UN Sustainable Development Goals: What role for design, Interact. Des. Arch. 16 (2018) 0–174, http://dx.doi.org/10.55612/s-5002-037-008.
[483] Y. Xin, R.U. Khan, V. Dagar, F. Qian, Do international resources configure SMEs' sustainable performance in the digital era? Evidence from Pakistan, Resour. Policy 80 (2023) 103169, http://dx.doi.org/10.1016/j.resourpol.2022.103169.
[484] Xue, Liang, Boulton, Information technology governance in information technology investment decision processes: The impact of investment characteristics, external environment, and internal context, MIS Q. 32 (2008) 67, http://dx.doi.org/10.2307/25148829.
[485] S. Lall, Promoting technology development: The role of technology transfer and indigenous effort, Third World Q 14 (1993) 95–108, http://dx.doi.org/10.1080/01436599308420315.
[486] J.M. Altayeb, H. Eleyan, N.D. Wishah, A.E. Elmahmoum, A.J. Khalil, B.S. Abu-nasser, S.S. Abu-naser, AI-driven innovations in agriculture : Transforming farming practices and outcomes, Int. J. Acad. Appl. Res. 8 (2024) 1–6.
[487] S. Zeb, N. Fnu, N. Abbasi, M. Fahad, AI in healthcare: Revolutionizing diagnosis and therapy, Int. J. Multidiscip. Sci. Arts 3 (2024) 118–128, http://dx.doi.org/10.47709/ijmdsa.v3i3.4546.
[488] A.M.K. Sherani, M. Khan, M.U. Qayyum, H.K. Hussain, Synergizing AI and healthcare: Pioneering advances in cancer medicine for personalized treatment, Int. J. Multidiscip. Sci. Arts 3 (2024) 270–277, http://dx.doi.org/10.47709/ijmdsa.v3i01.3562.
[489] S. Sarker, S. Henningsson, T. Jensen, J. Hedman, The use of blockchain as a resource for combating corruption in global shipping: An interpretive case study, J. Manage. Inf. Syst. 38 (2021) 338–373, http://dx.doi.org/10.1080/07421222.2021.1912919.
[490] A. Kaplan, Cryptocurrency and corruption: Auditing with blockchain, in: T. Aksoy, U. Hacioglu (Eds.), Auditing Ecosystem and Strategic Accounting in the Digital Era, Springer, Cham, 2021, pp. 325–338, http://dx.doi.org/10.1007/978-3-030-72628-7_15.
[491] D. Cagigas, J. Clifton, D. Diaz-Fuentes, M. Fernandez-Gutierrez, Blockchain for public services: A systematic literature review, IEEE Access 9 (2021) 13904–13921, http://dx.doi.org/10.1109/ACCESS.2021.3052019.
[492] S. Myeong, Y. Jung, Administrative reforms in the fourth industrial revolution: The case of blockchain use, Sustainability 11 (2019) 3971, http://dx.doi.org/10.3390/su11143971.
[493] C. Mahapatra, A. Moharana, V. Leung, Energy management in smart cities based on Internet of Things: Peak demand reduction and energy savings, Sensors 17 (2017) 2812, http://dx.doi.org/10.3390/s17122812.
[494] H. Mortaji, S.H. Ow, M. Moghavvemi, H.A.F. Almurib, Load shedding and smart-direct load control using Internet of Things in smart grid demand response management, IEEE Trans. Ind. Appl. 53 (2017) 5155–5163, http://dx.doi.org/10.1109/TIA.2017.2740832.
[495] M. Saleem, M. Shakir, M. Usman, M. Bajwa, N. Shabbir, P. Shams Ghahfarokhi, K. Daniel, Integrating smart energy management system with internet of things and cloud computing for efficient demand side management in smart grids, Energies 16 (2023) 4835, http://dx.doi.org/10.3390/en16124835.
[496] A.A. Musa, S.I. Malami, F. Alanazi, W. Ounaies, M. Alshammari, S.I. Haruna, Sustainable traffic management for smart cities using Internet-of-Things-oriented intelligent transportation systems (ITS): Challenges and recommendations, Sustainability 15 (2023) 9859, http://dx.doi.org/10.3390/su15139859.
[497] S.M.R. Islam, D. Kwak, M.H. Kabir, M. Hossain, K.-S. Kwak, The Internet of Things for health care: A comprehensive survey, IEEE Access 3 (2015) 678–708, http://dx.doi.org/10.1109/ACCESS.2015.2437951.
[498] Y. Yin, Y. Zeng, X. Chen, Y. Fan, The internet of things in healthcare: An overview, J. Ind. Inf. Integr. 1 (2016) 3–13, http://dx.doi.org/10.1016/j.jii.2016.03.004.
[499] H. Ding, J. Tian, W. Yu, D.I. Wilson, B.R. Young, X. Cui, X. Xin, Z. Wang, W. Li, The application of artificial intelligence and big data in the food industry, Foods 12 (2023) 4511, http://dx.doi.org/10.3390/foods12244511.
[500] A. Rejeb, J.G. Keogh, K. Rejeb, Big data in the food supply chain: a literature review, J. Data Inf. Manag. 4 (2022) 33–47, http://dx.doi.org/10.1007/s42488-021-00064-0.
[501] J. Díaz-Arancibia, J. Hochstetter-Diez, A. Bustamante-Mora, S. Sepúlveda-Cuevas, I. Albayay, J. Arango-López, Navigating digital transformation and technology adoption: A literature review from small and medium-sized enterprises in developing countries, Sustainability 16 (2024) 5946, http://dx.doi.org/10.3390/su16145946.
[502] M. Sharma, R. Gupta, P. Acharya, K. Jain, Systems approach to cloud computing adoption in an emerging economy, Int. J. Emerg. Mark 18 (2023) 3283–3308, http://dx.doi.org/10.1108/IJOEM-04-2021-0501.
[503] Z. Qadir, F. Ullah, H.S. Munawar, F. Al-Turjman, Addressing disasters in smart cities through UAVs path planning and 5G communications: A systematic review, Comput. Commun. 168 (2021) 114–135, http://dx.doi.org/10.1016/j.comcom.2021.01.003.
[504] S.A.R. Naqvi, S.A. Hassan, H. Pervaiz, Q. Ni, Drone-aided communication as a key enabler for 5G and resilient public safety networks, IEEE Commun. Mag. 56 (2018) 36–42, http://dx.doi.org/10.1109/MCOM.2017.1700451.
[505] B. Junge, T. Alabi, K. Sonder, S. Marcus, R. Abaidoo, D. Chikoye, K. Stahr, Use of remote sensing and GIS for improved natural resources management: case study from different agroecological zones of West Africa, Int. J. Remote Sens. 31 (2010) 6115–6141, http://dx.doi.org/10.1080/01431160903376415.
[506] J.K. Mani, A.O. Varghese, Remote sensing and GIS in agriculture and forest resource monitoring, in: G. Reddy, S. Singh (Eds.), Geospatial Technologies in Land Resources Mapping, Monitoring and Management, Springer, Cham, 2018, pp. 377–400, http://dx.doi.org/10.1007/978-3-319-78711-4_19.
[507] R.P. Fisk, A.S. Gallan, A.M. Joubert, J. Beekhuyzen, L. Cheung, R. Russell-Bennett, Healing the digital divide with digital inclusion: Enabling human capabilities, J. Serv. Res. 26 (2023) 542–559, http://dx.doi.org/10.1177/10946705221140148.
[508] A. Molina, The digital divide: The need for a global e-inclusion movement, Technol. Anal. Strat. Manag. 15 (2003) 137–152, http://dx.doi.org/10.1080/0953732032000046105.
[509] M. Wessels, P. van den Brink, T. Verburgh, B. Cadet, T. van Ruijven, Understanding incentives for cybersecurity investments: Development and application of a typology, Digit. Bus. 1 (2021) 100014, http://dx.doi.org/10.1016/j.digbus.2021.100014.
[510] N. Rawindaran, A. Jayal, E. Prakash, C. Hewage, Perspective of small and medium enterprise (sme's) and their relationship with government in overcoming cybersecurity challenges and barriers in Wales, Int. J. Inf. Manag. Data Insights 3 (2023) 100191, http://dx.doi.org/10.1016/j.jjimei.2023.100191.
[511] G. Alibekova, T. Medeni, A. Panzabekova, D. Mussayeva, Digital transformation enablers and barriers in the economy of Kazakhstan, J. Asian Financ. Econ. Bus. 7 (2020) 565–575, http://dx.doi.org/10.13106/jafeb.2020.vol7.no7.565.
[512] M. Toussaint, S. Krima, H. Panetto, Industry 4.0 data security: A cybersecurity frameworks review, J. Ind. Inf. Integr. 39 (2024) 100604, http://dx.doi.org/10.1016/j.jii.2024.100604.
[513] J. James, Confronting the scarcity of digital skills among the poor in developing countries, Dev. Policy Rev. 39 (2021) 324–339, http://dx.doi.org/10.1111/dpr.12479.
[514] K. Ameen, G.E. Gorman, Information and digital literacy: a stumbling block to development?, Libr. Manag. 30 (2009) 99–112, http://dx.doi.org/10.1108/01435120910927565.
[515] P.P. Nedungadi, R. Menon, G. Gutjahr, L. Erickson, R. Raman, Towards an inclusive digital literacy framework for digital India, Educ. + Train. 60 (2018) 516–528, http://dx.doi.org/10.1108/ET-03-2018-0061.